Dissertation

Spatial modeling of charging infrastructure for the traffic flow-based charging demand of battery-electric vehicle fleets

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Abstract

TODO: think about the title? maybe origin-destination-flow based? The objective of the thesis is to advance the analytical understanding of the demand-driven geographic allocation of charging infrastructure of future electrified vehicle fleets in different road transport segments. The core contribution to the scientific literature is the extension of charging infrastructure capacity planning to consider diverse drivers of charging demand. This includes the long-term battery-electric vehicle (BEV) adoption, consumer-specific utilization patterns and inter-regional traffic flows. This is addressed through four research papers. Three of the four methodologies applied are optimizations, one a simulation-based approach. The central feature of the analytical methods is the use of traffic flow data, leveraging geographic relations of traffic flow to model the spatial distribution of charging demand.

For passenger cars, we first address the planning of fast charging for long-distance passenger car travel at the national level under different decarbonization scenarios for the transport sector. We observe that the highest reduction in required charging infrastructure capacities is achieved in scenarios with high charging efficiencies, while the driving range of BEVs has no significant impact due to the dense network topology of the analysed highway network. Furthermore, we analyze public charging infrastructure for shorter-distance traffic and BEV adoption within and between three regions. Here, the analysis shows income-class differences in charger utilization preferences which are further shaped by the local spatial density of charging stations. The local expansion of rapid public charging infrastructure has a moderate impact on BEV adoption in neighboring regions.

For commercial fleets, the analysis focuses on the spatial heterogeneity of charging loads and flexibility potential. The sensitivity of charging load allocation to spatially varying electricity prices and network charges is analyzed along international road transport routes. Furthermore, this study identifies segments in an international transport corridor where rail infrastructure is cost-effective in contributing to the decarbonization of road freight and reducing the local concentration of charging loads. The temporal flexibility of commercial fleets is analyzed, considering light-duty vehicles, buses, and heavy-duty vehicles at the federal and state levels. The spatial heterogeneity results from geographical variations in regional fleet sizes and in national and international truck traffic flows. Moreover, high temporal flexibility in charging demand allocation is evident on weekends, which, in cases of cost-optimal charging, may result in substantial peak loads in the electricity system.

A central insight of this thesis is that planning of charging infrastructure—accounting for both adoption patterns and cross-regional vehicle movements — is necessary to support the electrification of particularly long-distance BEV application for passenger cars and trucks alike.

Motivation | Problem Statement		Research Question	Scientific publication
passenger car fleets			
• What are required fast-charging capacities along Austria's highway network for long-distance travel with BEV passenger cars and where are these located under different decarbonization scenarios? • Future charging infrastructure demand affected by different decarbonization scenarios for the transport sector (electrification share, technological improvments, shift to other travel modes,. ...)	**RQ1**	What are required fast-charging capacities along Austria's highway network for battery-electric passenger cars under different decarbonization scenarios and where are these located ?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2022). Minimum-Cost Fast-Charging Infrastructure Planning for Electric Vehicles along the Austrian High-Level Road Network. Energies, 15(6), 2147. https://doi.org/10.3390/en15062147
• Battery-electric vehicles are currently luxory item that are used in higher-income level classes • Unexplored impact of different public charging infrastructure roll-out strategies in two dimensions: on BEV adoption in different income levels and on neighbouring regions (spillover effect)	**RQ2**	To what extent do the speed and spatial distribution of public infrastructure expansion influence BEV adoption rates among various income groups and across neighboring regions?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2025). Public charger expansion: Impacts across income classes and beyond regional borders [Manuscript submitted for publication]. *Transportation Research Part D: Transport and Environment*.
commercial fleets			
• Cost differences in the installation of charging infrastructure and charging by country and electricity market zone indicate potential for "charging tourism" • Unknown magnitudes of potential charging infrastructure capacity allocations being affected by these differences	**RQ3**	What magnitude of charging infrastructure capacities is affected by geographically varying network fees and electricity prices in the cost-optimal planning of charging infrastructure allocation for long-haul battery-electric trucks?	Golab, A., Bakker, S., Zwickl-Bernhard, S., & Auer, H. (2025). Spatial flexible charging load allocation for inter-zonal long-haul truck electrification [Manuscript submitted for publication]. *Energy Reports*.
• Future impact of charging demand of battery-electric commercial fleet and potential for demand-side flexibility • Quantification of large-scale charging demand of battery-electric fleet at hourly temporal and NUTS-2 level under the consideration of transport flows within the countries and transit traffic in Austria	**RQ4**	What is the expected charging demand of the commercial BEV fleet spatio-temporally distributed in Austria in 2040?	Golab, A., Loschan, C., Zwickl-Bernhard, S., & Auer, H. (2025). The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. *Energy*, *316*, Article 134385. https://doi.org/10.1016/j.energy.2025.134385

Figure 1.: we have here a repetition of RQ1 in problem statement; needs change! Overview of most important building blocks of the thesis.

Graphical Abstract

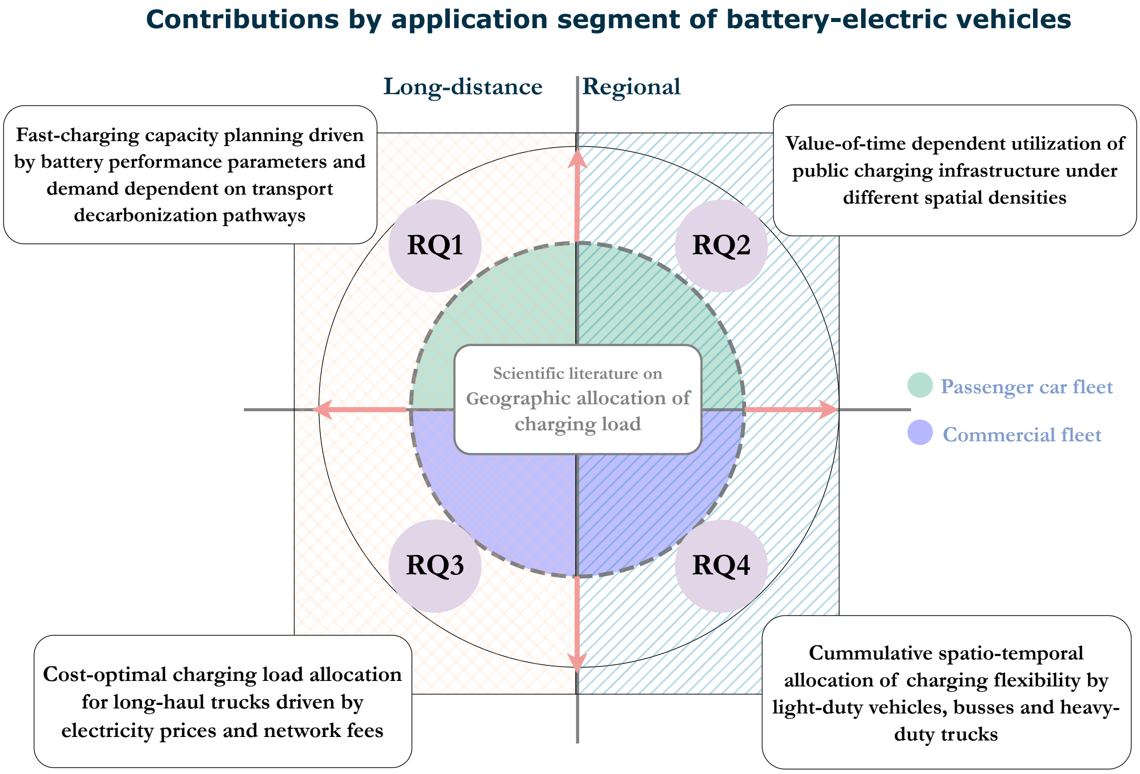

Figure 2.: Contributions of Papers 1—4, each reflecting one core research question (RQ1—4): Each research question extends the scientific literature for a different application segment of battery-electric vehicles. The application segments differ in long-distance vs. regional transport as well as between passenger cars and commercial fleet applications. This figure illustrates the relative allocation of the research questions and therefore the scope of this thesis.

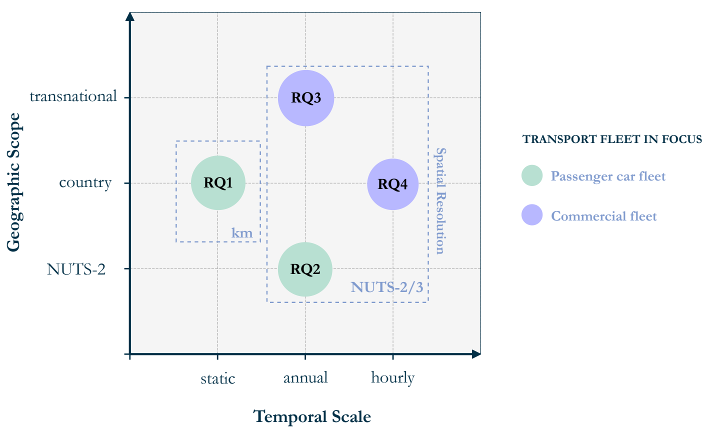

Figure 3.: Overview on the temporal scale, and geographic scope and scale as well as addressed application segments of battery-electric vehicles of the methods designed to answer RQ1—4.

Contents

Graphical abstracts

1. Introduction

1.1. Background: Electrification of road transport sector

Decarbonizing the transport sector is an urgent priority for achieving the European Green Deal goals [1]. In 2023, the transport sector accounted for 29% of total European greenhouse gas emissions, with road transport alone responsible for around 73% of these transport-related emissions [2]. In response, European countries have committed to reducing greenhouse gas emissions from the transport sector by 90% by 2050 [3]. To achieve this significant reduction over the next two and a half decades, zero-emission drivetrain technologies need to be rapidly adopted on a large scale. At the forefront of these technologies stands the battery-electric drive-train.

Direct electrification of road transport is widely regarded as the cost-optimal pathway for reducing fossil fuel demand in the sector [4]. This is primarily driven by the superior energy efficiency of electric drivetrains compared to internal combustion engines and the declining costs of electricity from renewable energy sources [5]. Moreover, the battery-electric drivetrain has achieved significant technological and market maturity over the last decade, which is marked by substantial advances in charging speed, driving range, and continuously decreasing battery costs [6]. The shift towards electrification is already evident through market shares across all road transport segments: In 2023, battery electric vehicles accounted for nearly 15% of all new car sales, while 40% of newly bought buses were electric and 10% of light and medium-haul commercial vehicles were battery-electric [7]. Yet these growing market shares have only begun to transform the existing fleet: as of 2023, just 1.8% of the EU passenger car fleet, 0.1% of trucks, and 2.5% of buses were electric [8]. This gap between electrified fleet numbers and total fleet composition underscores both the progress made and the scale of transformation still required.

The trajectory towards electrified road transport fleets is guided by two key policy frameworks at the EU level. A foundational document is the EU Sustainable and Smart Mobility Strategy, published in 2020 [9], which presents a comprehensive vision for decarbonizing all transport modes and sets the milestone of a 90% reduction in transport-related greenhouse gas emissions by 2050. For road transport, the strategy identifies two primary pathways: the uptake of zero-emission drivetrain technologies, primarily battery-electric vehicles, and the shift of transport demand towards lower-carbon modes such as rail.

The legislative translation of this strategy is the Fit for 55 Package, adopted in 2021 [10], which targets a reduction of net greenhouse gas emissions by at least 55% by 2030 relative to 1990 levels. Among its most significant provisions for road transport is the requirement of 100% CO_2 emission reduction for new passenger cars from 2035. Central to the infrastructure dimension of this package is the Alternative Fuels Infrastructure Regulation (AFIR; Regulation EU 2023/1804)

[11], which sets binding targets for the expansion of charging infrastructure for battery-electric vehicles. AFIR operates along two dimensions: distance-based targets, which mandate minimum charging capacity at regular intervals along the Trans-European Transport Network (TEN-T) [12], and national targets, which require minimum publicly accessible charging capacity proportional to the number of registered battery-electric vehicles in each member state.

Charging infrastructure is a fundamental prerequisite for road transport electrification as its availability directly influences vehicle adoption by alleviating range anxiety among passenger car drivers [13]. A key variable in the planning is the geographic placement which shapes the convenience and cost-effectiveness of charging for both private and commercial users [14]. Further, siting charging stations at locations of high utilization supports cost-effective infrastructure investments [15]. However, determining *where* charging capacity is needed and in what quantities remains a non-trivial problem. Capacity sizing must be balanced against infrastructure utilization; the integration with the electricity system adds another layer of complexity; and limited empirical experience—due to currently low adoption numbers—constrains the understanding of how these factors interact [16].

1.2. Problem statement: The geographic allocation of charging infrastructure

A central factor underlying the complexity of geographic charging infrastructure planning is the *mobility* of vehicles. Unlike stationary energy consumers—buildings, industrial sites—whose electricity demand is bound to a fixed grid connection point, vehicles consume energy at varying locations along their travel routes. A vehicle traveling from Vienna to Munich, for example, may recharge at any of several stations along the corridor, and this choice depends on remaining battery range, infrastructure availability, cost, and personal preferences. Battery-electric vehicles are therefore *spatially flexible* in their charging load allocation: they can charge at different locations along their travel route rather than being bound to a single point. This property is what makes planning for charging infrastructure fundamentally different from that of other energy-consuming sectors, as the resulting charging demand is not geographically predetermined. The challenge is further amplified at large geographic scales, where vehicles routinely traverse regional and national boundaries, as is the case along major European transport corridors.

This leads to the central problem addressed in this work: the geographic allocation of charging demand—and, consequently, the sizing and siting of infrastructure capacities—for aggregate battery-electric vehicle fleets at large geographic scales, spanning regional, national, and international transport networks.

This problem is relevant to three decision-making perspectives: First, transport policy makers who design regulatory frameworks, set targets for charging infrastructure deployment such as the AFIR, and initiate tendering processes for charging infrastructure. Second, transport and charging infrastructure planners, for example road network operators, who must translate these targets into concrete investment and siting decisions—determining where and how much charging capacity to deploy, including tendering for charging point operators and investments in local grid connection capacity. Third, electricity grid and system operators who require long-term estimates of future charging loads to plan grid reinforcement and to integrate demand-side flexibilities for system balancing and congestion management.

The central problem of geographic charging capacity allocation can be decomposed into three interconnected dimensions:

- **(i)** The identification of locations and required capacities for charging infrastructure,
- **(ii)** the influence of spatial configuration on charging demand and driver behavior, and
- **(iii)** the integration of charging operations with the electricity system.

The thesis addresses in particular questions related to **(i)** and **(ii)** and touches upon **(iii)**. Contributions 1 and 3 focus on dimension **(i)**, addressing the identification of charging infrastructure locations and capacities for passenger cars on highway networks and for long-haul trucks along international corridors, respectively. Contribution 2 addresses dimension **(ii)**, examining how the spatial configuration of public charging infrastructure influences BEV adoption across income groups and neighboring regions. Contribution 4 spans dimensions **(i)** and **(iii)**, estimating the spatio-temporal distribution of commercial fleet charging demand and its potential for integration with the electricity system.

The methodological foundation for addressing these dimensions is a traffic-flow-based representation of charging demand, derived from origin-destination data coupled with network topology. Origin-destination data indicate the volume of passengers or freight transported between two regions, from which traffic flow by location can be deduced while also capturing the geographic extent of fleet movements. This representation is suited to address the central problem of geographic charging capacity allocation as it captures the spatially flexible charging load allocation of large traveling fleets. Crucially, it inherently accounts for interregional and transit traffic — vehicles that pass through or across regions and national borders — which fleet-projection-based approaches, anchored to locally registered vehicles, typically do not account for. The following section presents how this approach is applied across four research contributions.

Motivation | Problem Statement		Research Question	Scientific publication
passenger car fleets			
• Long-distance BEV travel depends on en-route fast-charging, yet planning involves high uncertainty across decarbonization pathways (fleet size, driving range, modal shift) • The challenge of ensuring sufficient geographic coverage while avoiding overcapacity from misaligned assumptions	**RQ1**	What are the required fast-charging capacities along Austria's highway network for battery-electric passenger cars in 2030 under different decarbonization scenarios, and where are these allocated?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2022). Minimum-cost fast-charging infrastructure planning for electric vehicles along the Austrian high-level road network. Energies, 15(6), 2147. https://doi.org/10.3390/en15062147
• Broad BEV adoption requires a shift from home to public charging, while adoption needs to considered beyond the early adopters that are characterized by high income • Cross-regional spillover effects: local charging infrastructure expansion influences BEV adoption in neighboring regions	**RQ2**	• To what extent does the spatial density of public charging infrastructure influence BEV adoption across consumers with different income levels? • How does a local rapid charging infrastructure expansion affect BEV adoption in neighboring regions with slower expansion?	Golab, A., Zwickl-Bernhard, S., & Auer, H. (2026). Public charger expansion: Impacts across income classes and beyond regional borders. Transportation Research Part D: Transport and Environment. In press.
commercial fleets			
• Charging costs for battery-electric trucks vary geographically due to differences in electricity prices and network fees across countries • Unknown magnitudes of potential charging infrastructure capacity allocations being affected by these differences	**RQ3**	What magnitude of charging infrastructure capacities is affected by geographically varying network fees and electricity prices in the cost-optimal planning of charging infrastructure allocation for long-haul battery-electric trucks?	Golab, A., Bakker, S., Zwickl-Bernhard, S., & Auer, H. (2026). Spatial distribution of charging patterns of inter-zonal long-haul battery-electric truck fleets. Energy. Submitted, March 2026.
• Estimating future charging demand from multiple commercial vehicle segments (light-duty trucks, heavy-duty trucks, buses) requires segment-specific modeling approaches due to their distinct operational patterns and geographic reach • Fleet-projection-based methods miss transit freight traffic — vehicles passing through a country — which can account for a substantial share of charging demand	**RQ4**	What is the expected spatio-temporal distribution of charging demand from the commercial battery-electric vehicle fleet in Austria in 2040?	Golab, A., Loschan, C., Zwickl-Bernhard, S., & Auer, H. (2025). The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. Energy, 316, 134385. https://doi.org/10.1016/j.energy.2025.134385

Figure 1.1.: Overview of the four research questions (RQ1–4): each row summarizes the motivation, research question, and corresponding scientific publication. Contributions are grouped by vehicle fleet type — passenger car fleets (RQ1, RQ2) and commercial fleets (RQ3, RQ4). Adapt figure to updated RQ3 formulation.

1.3. The thesis' research questions

Each of the following research questions addresses a specific gap in the literature on charging infrastructure planning, as detailed in Chapter 2. The thesis formulates four research questions (RQ1–4), each within a scientific journal paper (Contributions 1–4). Figure 1.1 provides an overview.

Contribution 1 Golab, A., Zwickl-Bernhard, S., & Auer, H. (2022). Minimum-cost fast-charging infrastructure planning for electric vehicles along the Austrian high-level road network. *Energies*, *15*(6), 2147. https://doi.org/10.3390/en15062147

Contribution 2 Golab, A., Zwickl-Bernhard, S., & Auer, H. (2026). Public charger expansion: Impacts across income classes and beyond regional borders. *Transportation Research Part D: Transport and Environment*. In press.

Contribution 3 Golab, A., Bakker, S., Zwickl-Bernhard, S., & Auer, H. (2026). Spatial distribution of charging patterns of inter-zonal long-haul battery-electric truck fleets. *Energy*. Submitted, March 2026.

Contribution 4 Golab, A., Loschan, C., Zwickl-Bernhard, S., & Auer, H. (2025). The value of flexibility of commercial electric vehicle fleets in the redispatch of congested transmission grids. *Energy*, *316*, 134385. https://doi.org/10.1016/j.energy.2025.134385

Figure 1.2 illustrates the contextualization of the contributions in the existing literature. The research questions address gaps across different BEV application segments, each characterized by distinct requirements for charging infrastructure allocation. Contributions 1 and 2 address the passenger car segment, while Contributions 3 and 4 focus on commercial vehicle fleets. Contributions 1 and 3 focus on charging infrastructure for long-distance transport along highway and international corridor networks. Contribution 2 centers on regional charging infrastructure and its role in enabling broader BEV adoption. Contribution 4 spans both long-distance and regional transport, addressing multiple commercial vehicle segments and their distinct charging patterns.

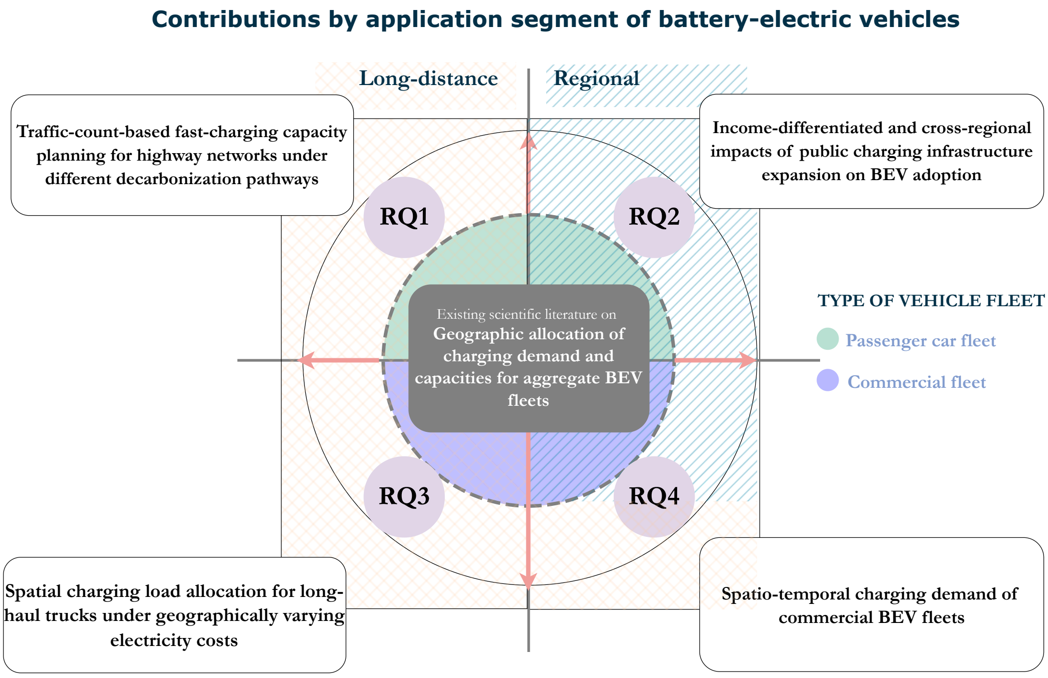

Figure 1.2.: Contributions 1–4 by application segment of battery-electric vehicles. The four research questions (RQ1–4) are positioned along two axes: long-distance vs. regional transport (horizontal) and passenger car vs. commercial fleet (vertical). The central area represents the existing scientific literature on the geographic allocation of charging demand and capacities for aggregate BEV fleets, which the contributions extend.

1.3.1. Contribution 1 (RQ1)

Contribution 1 [17] addresses the spatial allocation of fast-charging infrastructure for battery-electric passenger cars along highway networks. Long-distance travel by car depends on the availability of en-route charging, yet planning such infrastructure involves considerable uncertainty regarding future demand [16]. This demand is shaped by the technical characteristics of the electrified fleet — notably battery capacity, driving range, and charging power — and by the broader transport sector decarbonization pathway, including the rate of BEV adoption, modal shift towards rail and public transport, and changes in overall transport demand [18]. The challenge lies in ensuring sufficient geographic coverage to enable long-distance electric mobility while avoiding overcapacity that results from overly optimistic or misaligned assumptions. The research question addressed is:

Research question 1 (RQ1): What are the required fast-charging capacities along Austria's highway network for battery-electric passenger cars in 2030 under different decarbonization scenarios, and where are these allocated?

To address this, a charging station allocation problem for highway networks is formulated that combines a node-based optimization structure with flow-based demand representation. Charging demand is defined at discrete nodes corresponding to specific service areas along the network, while the flow of vehicles between nodes captures the spatial flexibility in demand allocation based on traffic direction and driving range. This formulation leverages granular data on service area positions and traffic flow counts along the Austrian highway network. Fast-charging capacities are planned for 2030 under different scenarios of BEV fleet shares, road traffic load, driving range, and charging power levels. Extensive sensitivity analyses are conducted to identify which parameters most significantly affect charging capacity requirements, thereby addressing the uncertainties arising from different transport sector decarbonization pathways.

1.3.2. Contribution 2 (RQ2)

Contribution 2 [19] focuses on public charging infrastructure, which plays a key role in enabling the adoption of passenger BEVs at a large scale. Early adopters of BEVs are predominantly higher-income households with access to home charging [20]. Achieving the long-term goal of broad BEV adoption across all socio-economic backgrounds [14], however, requires a shift in reliance from home charging toward publicly accessible infrastructure. Lower-income households, who are more likely to lack private charging access and to travel longer distances to reach affordable charging facilities [21], are particularly dependent on the availability of public charging infrastructure.

Along with this large-scale adoption, the degrees of freedom in the geographic allocation of charging activity increase with growing driving range [22]. Empirical studies have shown that local public charging infrastructure generates cross-border impacts on BEV adoption in neighboring regions, effects that are likely to be amplified as growing driving ranges allow vehicles to charge in more distant locations [23], [24].

Bringing these aspects together in the context of public charging infrastructure expansion at the regional level, the second contribution addresses the following overarching research question:

Research question 2 (RQ2): How does public charging infrastructure expansion influence the adoption of passenger BEVs across income groups and neighboring regions?

This is decomposed into two sub-questions:

RQ2.a: To what extent does the spatial density of public charging infrastructure influence battery-electric vehicle adoption across consumers with different income levels?

RQ2.b: How does a local rapid charging infrastructure expansion affect the adoption of battery-electric vehicles in neighboring regions with slower charging infrastructure expansion?

To address these questions, a system-cost optimization of passenger car fleet decarbonization is formulated. Intangible costs that vary by income class are introduced, capturing differential tolerances for both detouring time to recharge and charging duration. The key methodological contribution is modeling detour time as an endogenous decision variable, explicitly linking public charging infrastructure expansion to increased spatial charger density and, consequently, to reduced consumer travel time. To capture cross-regional interactions, the analysis is conducted at the regional level (NUTS-3[1]) using traffic flow and socio-economic data from the Basque Country, Spain. This approach accounts for commuter traffic patterns and the potential to flexibly allocate charging activity between trip origins and destinations, thereby quantifying income-class-dependent preferences for charger utilization and cross-border impacts.

1.3.3. Contribution 3 (RQ3)

Contribution 3 [26] focuses on long-haul international trucking. For fossil-fueled trucks, country-level differences in diesel taxation have significantly influenced the geographic allocation of fueling activity across borders [27]. As charging costs are location-dependent — varying with electricity-zone-dependent electricity prices and local network charges [28], [29] — this phenomenon will, with the adoption of battery-electric trucks, shift to the allocation of charging loads. This can lead to high concentrations of charging load at unfavorable locations in the electricity system, for example where insufficient grid capacity causes congestion. This leads to the following research question:

Research question 3 (RQ3): RQ3 formulation needs to be adapted What magnitude of charging infrastructure capacities is affected by geographically varying network charges and electricity prices in the cost-optimal planning of charging infrastructure allocation for long-haul battery-electric trucks?

To address this question, the impact of geographically varying charging costs on the cost-optimal allocation and the route-dependent adoption of battery-electric trucks is analyzed. International origin-destination truck flows are used as input for the cost optimization of fleet turnover, charging infrastructure development, and charging activities. In various electricity and technology pricing scenarios, the analysis shows how locally low charging costs along a corridor can lead to a high concentration of charging load, and, further, how local intervention in charging prices can prevent this.

1.3.4. Contribution 4 (RQ4)

With the electrification of the commercial fleet, large peak charging power levels are expected. The potential for shifting charging loads temporally to minimize required investments into grid capacity and balancing energy has been demonstrated by many studies [30]. However, quantifying this flexibility potential requires spatially-differentiated estimates of future charging demand that account not only for regional vehicle fleets but also for transit freight traffic—a critical consideration for European electricity systems where market zones align with national borders while

[1]Nomenclature of Territorial Units for Statistics [25].

substantial cross-border freight flows generate charging demand. Contribution 4 [31] addresses this challenge by estimating charging demand and flexibility potentials across multiple commercial vehicle segments (light-duty trucks, heavy-duty trucks, and buses) for Austria in 2040:

Research question 4 (RQ4): What is the expected spatio-temporal distribution of charging demand from the commercial battery electric vehicle fleet in Austria in 2040?

For heavy-duty trucks, the methodology adopts a traffic-flow-based approach, which—unlike fleet-projection-based methods—is able to capture charging demand from interregional and transit transport: vehicles operating across provincial boundaries and international freight passing to, from, and through the country. This approach distinguishes between en-route charging along transport corridors and depot-based charging at trip origins and destinations—a differentiation particularly important given that a substantial share of truck traffic on Austrian highways originates outside the country and would be missed by domestic fleet projections. In contrast, light-duty trucks and buses, which operate primarily within regional boundaries, are modeled based on provincial fleet characteristics and typical driving patterns. This differentiated modeling approach enables capturing the geographic heterogeneity in charging load magnitudes and temporal flexibility characteristics across Austria's regions.

1.4. Outline of the thesis

The remainder of this thesis is structured as follows. Chapter 2 reviews the relevant literature and identifies the gaps addressed by each contribution. Chapter 3 presents the methodological approaches developed for the four contributions. Chapter 4 presents the results of the respective case studies. Chapter 5 provides a discussion and synthesis of the findings across all contributions, including limitations. Chapter 6 draws conclusions and formulates implications for decision makers. Chapter 7 outlines directions for future research.

2. Review of State-of-the-Art & Progress Beyond

TODO include acronyms for BEV, OD The body of scientific literature on the electrification of road vehicles and charging infrastructure planning is extensive. In the following, selected papers are presented that help in understanding the state-of-the-art methodologies and most relevant findings, and that provide the context for the contributions of this thesis.

The literature review is split into two parts. The first part, Sections 2.1 and 2.2, focuses on the electrification of the road sector and the role of charging infrastructure in the adoption of private (Section 2.1) and commercial (2.2) battery-electric vehicles. The second part, Section 2.3, provides insights into the methodological approaches used to answer the research questions related to the demand-driven geographic charging capacity planning, which is the methodological foundation this thesis builds on. In Section 2.4, the core contributions of this work to the existing scientific literature are stated by article.

2.1. Electrification of the private passenger cars sector

2.1.1. Towards large-scale electrification of passenger car fleets

The pathways towards a zero-emissions passenger transport sector encompass three complementary strategies: reducing travel demand, shifting demand towards low-carbon modes, and transitioning to zero-emission technologies [5, 32, 33]. While the first two levers require substantial behavioral and structural changes, technological transformation in the private motorized passenger sector allows for preserving individual mobility patterns and flexibility of current car ownership [34].

For technological substitution, the transformation towards a passenger car fleet of battery-electric vehicles is established as the most cost-efficient way to decarbonize [5]. This is extensively demonstrated: Studies on total cost of ownership show the cost benefits of battery-electric passenger cars, stemming mainly from lower maintenance and fuel costs due to improved drivetrain efficiency [35]. Life-cycle analyses show significant reductions in carbon emissions from the use of battery-electric vehicles, best achieved with renewable electricity sources [36]. This sector-overarching cost-efficiency is also evident in studies that apply energy system models to quantify decarbonization pathways under various future policy scenarios [37]. The performance parameters of the battery-electric drive-train, particularly the driving range and charging speed, have significantly improved during the last decades, and this, together with future trajectories, solidifies the system-wide benefits of the large-scale adoption of battery-electric vehicles [38].

By the end of 2024, approximately 58 million battery-electric passenger cars were in use worldwide, compared to 1.47 billion fossil-fueled passenger cars, placing the battery-electric vehicle still in its early adoption phase [39]. This phase is predominantly driven by higher-income households with access to home charging [20], for whom the purchase cost barrier — historically significant [40] — is offset through subsidization schemes across many European countries [41] and continuously decreasing vehicle costs, with cost parity now achieved for several car segments [39]. Yet for lower-income households, adoption remains out of reach due to several persisting barriers: (1) limited availability in affordable car segments, with most BEVs concentrated in larger, premium categories [42], (2) declining purchase subsidies in many policy frameworks [39], and (3) reliance on the second-hand market, which remains underdeveloped for BEVs [43, 44].

Schiaroli and Fraccascia [45] empirically analyze socio-behavioral barriers and motivators to BEV adoption across three European countries, identifying range anxiety as one of the most prominent obstacles. Range anxiety encompasses the fear of insufficient charging infrastructure, the perception of limited driving range compared to internal combustion engine vehicles [46], and the inconvenience of charging times that substantially exceed conventional refueling [13], [47]. While these concerns have been partially mitigated over the past decade — through significant improvements in battery technology that have increased range and accelerated charging [48], and the fact that most travel routes fall well within a modern BEV's range [49] — charging infrastructure remains a critical factor. Baden [50] conduct a comprehensive review on range anxiety mitigation and emphasize that the targeted allocation of charging infrastructure is key, with capacities aligned to drivers' needs at convenient locations. Crucially, the authors argue that this allocation must be based on geographic and behavioral segmentation, as different EV usage patterns shape fundamentally different requirements for charging infrastructure.

2.1.2. Role of charging infrastructure roll-out for passenger cars

Charging infrastructure is categorized by the power level, the use case, and access type [51]. *Home* charging is typically at low voltage levels, providing charging directly to the car while parked at home. Home charging is currently the most popular method, as existing BEV owners are predominantly homeowners for whom it provides the most convenient and cost-efficient charging option [52]. Often, BEVs at home chargers are integrated directly into the home energy management system, and co-optimized with home PV electricity production, battery storage, and/or other home appliances [53, 54]. Similar to home charging, *workplace* charging provides charging at a regular daily parking place. However, workplace charging availability remains limited, as provision falls short due to unclear benefits to companies [55, 56].

Public charging provides the opportunity to recharge en route, thereby alleviating range anxiety. Most importantly for the current adoption stage, it enables BEV owners without access to home charging to recharge their vehicles [51, 57]. Functionally, public charging infrastructure resembles conventional fueling stations but is typically located at parking facilities due to longer charging times. The geographic placement of public charging stations aims to achieve both high spatial coverage and convenience in integrating charging into regular mobility patterns [58]. Convenience charging stations are often co-located with destinations for other activities, such as shopping [59], leading to their placement at so-called *points of interest*, e.g., shopping malls [60]. *Slow* charging offers an opportunity to recharge during longer stays and daily downtime, day or night, delivering comfort comparable to home or workplace charging. *Fast* charging, i.e., charging at higher power levels using DC technology, enables rapid recharging, e.g., during errands or en route [61]. From a business perspective, the challenge lies in balancing investment costs and competitive pricing against uncertainty in expected charging demand and its growth at individual sites [62]. Therefore, consumers currently face higher charging prices at public charging stations [28, 53].

Simultaneously, there is an access gap in home charging that correlates with income level [14, 63]. This makes lower-income groups especially dependent on public charging infrastructure and, consequently, exposed to higher charging prices than car owners with home chargers. However, charging point operators are more likely to allocate charging stations in high-income communities due to greater certainty about BEV adoption and charging demand [64, 65]. This creates a paradox: those most dependent on public charging have the least access to it. Therefore, equity in charging-station allocation is an important factor, alongside subsidization and other monetary incentives, to support rapid adoption across all income classes [14].

Understanding these equity implications requires examining how charging infrastructure influences BEV adoption more broadly. In scientific literature, the relationship between local charging infrastructure development and the uptake of BEVs remains unclear. There exists a wide range of choice models that relate the decision to purchase a BEV to socio-economic variables and reveal the influence of factors such as income level, home ownership, household size, a person's attitude towards renewable technologies, and the availability of charging infrastructure [66, 67]. Range anxiety is directly related to the absence of charging infrastructure, and poses a psychological barrier to the adoption of BEVs [13]. However, empirical studies show that the relationship between public charging infrastructure deployment and BEV uptake is more nuanced. Illmann and Kluge [68] find an evident but weak relationship between the deployment of public charging infrastructure and the uptake of BEVs in German communities. Another finding of their study is that the uptake is more affected by the charging speed provided at the charging stations, rather than the number of chargers themselves. Similar is found in the work of White et al. [69] who conduct an analysis for three U.S. metropolitan areas. In their work, no direct impact of range anxiety on the BEV adoption is found. The authors observe that higher spatial densities of public charging infrastructure increase the perceived social pressure to adopt BEVs. Globisch et al. [70] confirms through questionnaires that lower charging duration is valued over higher spatial density and lower fees, while Gebauer et al. [71] demonstrates that widespread DC fast-charging significantly improves potential users' perception of electromobility.

Public charging infrastructure deployment creates spatial spillover effects that extend beyond regional boundaries—a pattern consistent with the network nature of transport infrastructure more broadly [72, 73, 74]. Empirical evidence demonstrates these cross-regional impacts for battery-electric vehicle adoption: Qian et al. [23] find that fast-charging infrastructure in centrally located regions positively influences BEV adoption in neighboring areas, while Sheng et al. [24] show that in New Zealand public charging infrastructure in neighboring regions has a more significant impact on local BEV adoption than locally installed infrastructure itself. Both studies conclude that effective charging infrastructure planning requires spatially holistic approaches that account for cross-border effects rather than treating regions as isolated systems. However, these analyses are based on observed historical patterns; prospective assessments of how strategic infrastructure deployment decisions affect BEV adoption across regional boundaries have not been conducted.

2.2. Battery-electric vehicle adoption in commercial fleets

2.2.1. Electrification across commercial fleet segments

Currently, the electrification of the commercial fleet is in significantly earlier stages than the private passenger vehicle sector, though substantial progress has been made: 40% of buses and 10% of light and medium commercial vehicles sold in 2023 were battery-electric [75]. The core difference to the passenger sector is cost-efficiency as the primary driving force, rather than social aspects and personal freedom considerations in private passenger car ownership [76]. While the three fundamental decarbonization levers—reducing demand, shifting toward low-carbon modes, and replacing conventional drivetrains—apply to commercial transport as well [77], practical constraints limit their application. Modal shift from road to rail freight faces substantial barriers: cost-competitiveness only at longer ranges, bottlenecks in rail infrastructure capacity requiring high investment, and limited flexibility due to track-based operations and last-mile delivery challenges [78, 79]. Against this background, direct electrification of drivetrains has emerged as the primary pathway to decarbonize commercial road transport [80], [81].

The technology pathway toward electrification, however, differs markedly from the passenger car sector due to the greater diversity in vehicle segments, particularly regarding daily ranges and payload requirements [82, 83]. For light-duty commercial applications, battery-electric vehicles have become the clear solution for reducing fossil fuel consumption, mirroring developments in the passenger car sector [84]. For heavy-duty applications, particularly long-haul trucking, the technology landscape initially appeared less certain. Zero-emission options narrow to battery-electric vehicles and hydrogen fuel-cell electric vehicles, with early analyses suggesting that battery costs increase substantially for long-haul applications due to payload losses from heavy batteries needed to overcome long ranges while transporting heavy goods [85]. Noll et al. [83] analyze total cost of ownership across different truck segments in European countries, finding that while cost benefits for light- and medium-duty battery-electric drive-trains are clear, cost-competitiveness for long-haul heavy-duty applications varies substantially by country, driven primarily by operational expenditures, country-dependent tolls, and relative differences between diesel and electricity prices. Recent technological developments, however, have shifted this picture considerably. Plötz [81] argue that the core challenges of implementing battery-electric trucks for long-haul applications—particularly long charging times and limited range—have been largely overcome. Their analysis concludes that battery-electric technology will play the central role throughout all relevant road transport segments, including heavy-duty trucks, while hydrogen fuel-cell electric vehicles will remain in niche applications due to weak cost-effectiveness.

Despite this technological convergence toward battery-electric drivetrains, significant barriers to adoption remain. Scherrer et al. [86] identify two fundamental challenges for fleet operators based on empirical evidence from the German logistics sector: establishing adequate high-power charging infrastructure represents a major deployment hurdle, while operational planning must accommodate substantial flexibility in charging timing and location. This flexibility requirement stems from persistent uncertainties regarding the availability and reliability of public charging infrastructure—a constraint that forces fleet operators to maintain adaptive operational strategies rather than implementing fixed routing and scheduling patterns. These operational challenges create planning uncertainties that extend beyond vehicle acquisition decisions to encompass fundamental questions about charging infrastructure requirements and their integration with existing logistics operations [87, 88].

2.2.2. Charging infrastructure for battery-electric trucks

The transition to battery-electric vehicles in the commercial sector requires integrating time-consuming charging activities into operational schedules and logistics processes [89]. This integration challenge is compounded by the need to balance operational flexibility with cost-effectiveness: while the most economical approach involves low-power overnight charging at fleet depots [90], this strategy requires fleet operators to invest in grid connection reinforcement and on-site charging infrastructure [91], with costs potentially offset through on-site renewable generation and smart charging strategies [92, 93].

The diversity of commercial vehicle operations necessitates different charging infrastructure strategies. Al-Hanahi et al. [94] categorize charging infrastructure into three functional types based on operational requirements: (1) Depot charging—at fleet facilities, yards, and industrial sites—serves vehicles that return daily to a fixed location, enabling cost-effective overnight charging but requiring substantial upfront infrastructure investment. (2) Public charging infrastructure addresses vehicles with variable schedules requiring intermediate charging away from depots, supporting multi-shift operations and providing backup capacity for fleets with power-limited depot infrastructure. (3) Highway charging infrastructure supports long-haul operations through both fast charging during mandatory 45-minute driver breaks (currently at 350 kW, advancing toward 1 MW "megawatt charging") and slow charging during extended rest periods. The adequate driving ranges of modern battery-electric trucks enable strategic integration of these charging opportunities into logistics operations [95], though doing so requires careful spatial planning of infrastructure deployment along freight corridors. These different infrastructure types create distinct spatial planning challenges: while depot and public charging respond primarily to local demand patterns, highway charging infrastructure must align with long-distance freight flows along major corridors—including substantial international transit traffic that may not be captured in domestic fleet projections.

2.3. Modeling of charging demand, charging infrastructure capacities, and roll-out

2.3.1. Graph-based charging infrastructure allocation for highway networks

Charging infrastructure allocation problems are commonly formulated based on a graph representation with nodes representing locations of potential charging sites, and edges representing connecting road infrastructure. This approach enables high geographic resolution in modeling the spatial relationships between charging sites and demand locations [96, 97]. Metais et al. [96] differentiates these graph-based optimization tools by how the charging demand is derived from vehicle activity. The authors differentiate between node-based, flow-based[1], and tour-based approaches: Node-based approaches use a discrete expression of charging demand at nodes, which can be, for example, derived from local population density. In the flow-based application, the demand is expressed via vehicle flow numbers, often derived from origin-destination (OD) data. A concrete example is the flow-capturing approach by Kuby and Lim [98]. Charging stations are allocated along edges, and under cost minimization, the goal is to "capture" traffic flows, ensuring that in the optimized configuration, all vehicles have the opportunity to charge along their respective routes. The third category comprises tour-based approaches, which consider individual agents in the system and, therefore, model individual vehicle charging demand at high resolution. These concepts are suited to different charging applications and entail different types of input data and computational complexities. In general, the literature suggests that node-based approaches are well suited for urban areas, while flow-based approaches are better suited for highway networks [97, 99].

Building on the flow-based framework, several studies have developed optimization models for highway charging infrastructure. Wang et al. [100] presents a multistage equilibrium model utilizing origin-destination data to determine optimal highway charging infrastructure placement, targeting reductions in charging duration, waiting times, driver expenses, and infrastructure installation costs. In a flow-based framework, Jochem et al. [99] seeks to minimize the total number of charging points while determining their capacity requirements according to the traffic flow each station would serve. Similarly adopting a flow-based methodology, Yan [101] develops a two-stage genetic algorithm with multi-objective optimization to reduce both infrastructure construction expenses and charging costs incurred by drivers. Napoli et al. [102] introduces an iterative algorithm for station placement that establishes the spacing between consecutive charging facilities based on vehicle driving range capabilities. A sequential site selection approach is

[1] In the paper, it is referred to as *path*-based, but in the broader literature it is more commonly termed *flow*-based.

developed by Csiszár et al. [103], determining charging station locations according to traffic flow patterns and the population density of adjacent settlements.

More recent methodological developments have integrated on-site capacity planning into the optimization framework. Rose et al. [104] develops a node-capacitated flow-capturing approach based on origin-destination data that plans charging stations at discrete nodes while inherently planning on-site capacity based on past driven distances within the optimization, applying this to Germany. Building on this approach, Nugroho et al. [105] adopts the node-capacitated flow refueling location model to design fast-charging infrastructure for inter-city travel in the Greater Jakarta Area, analyzing different expansion scenarios for various electric vehicle ranges. Rehman et al. [106] extends this framework by proposing a capacitated flow refueling location model that jointly optimizes fast-charging station placement and on-site battery storage sizing.

Across these studies, vehicle driving range consistently serves as the primary constraint for determining the maximum permissible spacing between network charging stations. Csiszár et al. [103] examines how driving range influences infrastructure requirements, demonstrating that longer ranges enable a fixed charging network to serve a greater total number of vehicles. Kavianipour et al. [107] investigates the combined effects of enhanced driving range and charging power on infrastructure needs, revealing that technological advancements reduce infrastructure investment requirements as improved vehicle capabilities diminish the necessary number and capacity of charging points. Using mixed-integer linear programming, Wang et al. [108] evaluates optimal infrastructure deployment under budget constraints, identifying a saturation threshold in flow coverage—the proportion of BEVs successfully charged along their routes—beyond which further increases in driving range do not alter infrastructure requirements. Rehman et al. [106] finds the strongest impact of range variations between 100-300 km, with moderate effects beyond this threshold. Similarly, Nugroho et al. [105] observes in a sensitivity analysis across different BEV sizes and battery capacities that larger BEVs require only 6% fewer total stations in projections between 2025-2035, suggesting diminishing returns from range increases.

2.3.2. Modeling the impact of charging infrastructure expansion on passenger BEV adoption

Research examining optimal infrastructure expansion and its influence on BEV adoption can be classified into three literature streams: first, system dynamics modeling approaches that capture complex interactions among various system components [109, 110]; second, agent-based modeling frameworks featuring fine-grained spatial resolution in both charging station placement and driver mobility patterns [111, 112]; third, large-scale optimization methodologies mapping cost-efficient transport sector decarbonization trajectories, typically from a system-wide perspective [21]. Gao et al. [110] employs a system dynamics framework to project urban BEV adoption trajectories. The relationship between vehicle adoption and public charging deployment emerges through utility function optimization, incorporating variables such as average fueling station access distance, refueling duration, vehicle range, and total ownership expenses. An agent-based methodology presented by Wolbertus et al. [112] explores comparable research questions while also concentrating on urban BEV uptake. Unlike Gao et al. [110], this work models the adoption-infrastructure interaction at finer geographic granularity, specifically at the district scale. The BEV purchase decision is formulated as dependent on walking distance to the nearest charging point. Consumer segmentation by neighborhood reflects an assumption of consistent EV purchase propensity within geographic areas. Wolbertus et al. [112] compares single versus hub deployment strategies. The *single* approach distributes new charging stations broadly, whereas the *hub* strategy concentrates multiple charging points at centralized locations. Their findings indicate that distributed deployment positively influences BEV uptake, while concentrated hub placement demonstrates substantially weaker effects on adoption rates.

Regarding the third literature category, numerous studies address limitations inherent in total-cost-minimization frameworks [21, 113]. These limitations particularly concern consumer behavioral dimensions, including technology selection, modal preferences, and travel patterns. Common solutions involve coupling multiple models or augmenting individual model capabilities [114]. In their review paper, Venturini et al. [115] identify various methodologies for incorporating behavioral elements. For technology choice modeling, they highlight two particularly effective approaches: consumer segmentation to reflect realistic adoption dynamics, and integration of behavioral patterns alongside intangible (non-monetary) cost factors. Tattini et al. [21] implements a comparable methodology that considers income-differentiated consumer segments, assigns monetary valuations to mode- and technology-dependent time consumption, and constrains purchase decisions by household expenditure limitations. Luh et al. [114] segment consumers by residential type and incorporate diverse charging infrastructure categories. Their analysis examines how various charging access options affect BEV deployment and passenger transport emissions reduction. Results demonstrate that home charging availability critically influences adoption patterns. Public slow and fast charging become essential for widespread uptake when private charging access remains constrained. A subsequent publication by these authors [116] extends this analysis by incorporating travel time valuation, encompassing both recharging periods and detour time required to access charging facilities. This follow-up work emphasizes improvements in public transport utilization and the associated travel duration implications across different trip categories.

2.3.3. Consideration of different income levels in charging infrastructure planning

Methodological frameworks for charging infrastructure planning rarely differentiate consumers by income level, though several recent studies have begun addressing equity considerations. Loni and Asadi [117] propose a multi-objective model that minimizes charging site deployment and operation costs while maximizing both charging demand fulfillment and accessibility as a measure of social equity access. Batic et al. [118] develop a graph-neural network approach that deploys public charging infrastructure with explicit focus on establishing equitable network coverage, directly addressing accessibility gaps in low-income areas. Voinea et al. [119] propose an optimization-based framework that begins with spatio-temporal forecasting of adoption patterns across consumer segments defined by BEV model classes—specifically, more affordable short-range BEVs versus long-range premium models—which correlate with adopters' income levels. The proposed framework is applied to the spatial resolution of postal code level. Building on such segmentation approaches, Wang et al. [120] quantify charger access disparities derived from indicators including median household income, employment rate, and ethnicity, introducing an equity measure directly into their objective function to minimize access disparities between consumer groups within a multi-objective optimization. The charging infrastructure is planned explicitly for workplaces, while considering the spatial resolution of neighborhoods. Across these studies, income-level differentiation relies on spatial proxies that correlate neighborhood characteristics with income levels, rather than on behavioral parameters that capture income-dependent differences in charging preferences.

2.3.4. Overarching charging station deployment and battery-electric truck adoption at large-scale

Recent research advances dynamic infrastructure planning approaches that integrate fleet electrification trajectories with charging network development, without sacrificing spatial precision in site allocation. Karam et al. [78] examine this integration from a fleet operator's economic perspective within Denmark's borders. Their methodology employs network-based road representation combined with joint optimization of fleet operations costs and infrastructure investment decisions. The analysis tracks fleet conversion through 2040, determining when and where charging capacity additions become economically justified alongside vehicle replacements. Bakker et al. [121] broaden this scope considerably in their analysis of Norwegian freight transport evolution. Their framework addresses long-term infrastructure investment to support technological diversification and modal transitions in goods movement at the national scale. Operating at NUTS-2 geographic resolution, they model coordinated pathways for freight decarbonization encompassing both zero-emission vehicle adoption and modal shifts toward lower-carbon alternatives such as rail, alongside the corresponding expansion of transport networks and energy supply infrastructure. Their approach determines refueling and recharging capacity requirements directly from freight volumes served by each powertrain technology at each location, treating energy infrastructure sizing analogously to transport mode capacity planning. Building on this foundation, Pagnier et al. [122] integrate the Norwegian freight model from Bakker et al. [121] with a comprehensive energy system representation. Their coupled analysis reveals substantial interdependencies between heavy-duty electrification levels and charging infrastructure demands, demonstrating that spatially differentiated electricity pricing significantly influences optimal decarbonization strategies for the freight sector. However, these studies operate within national boundaries. The spatial allocation of charging demand along international corridors—where electricity costs and network charges vary across market zones—has not been addressed in an optimization framework.

2.3.5. Traffic-flow based modeling of charging profiles of commercial fleets for large-scale integration to the electricity system

To explore the potential benefits of electric vehicle flexibility provision—both for private passenger transport and commercial purposes—at large scales for the electricity system, various studies have estimated charging demand from fleet projections, driving patterns, and charging behavior assumptions, though differing substantially in their spatial resolution, fleet composition, and methodological frameworks. At the national scale, Li et al. [123] determine future charging loads for China by simulating daily charging behaviors based on regional fleet projections and driving patterns for private battery-electric vehicles, buses, taxis, and commercial light-duty vehicles, though notably excluding freight trucks. Similarly, Suski et al. [124] model charging demand for electric buses, cars, and two-wheelers in Malé, Maldives, incorporating charging flexibility into a capacity expansion framework, but also omitting heavy-duty commercial transport. While alternative methodologies exist—such as spatially explicit approaches that allocate charging demand based on observed vehicle movements [125] or traffic flow data [126]—these are primarily applied to passenger vehicle fleets at local or metropolitan scales. The spatial allocation of heavy-duty truck charging demand, which differs fundamentally from passenger vehicle patterns due to long-haul operations and corridor-specific infrastructure requirements, receives limited attention in the context of large-scale electricity system integration. In particular, fleet-projection-based approaches—which estimate charging demand from locally registered vehicle stocks—inherently miss the interregional and transit freight traffic that constitutes a substantial share of highway charging demand, especially in European transit countries. A traffic-flow-based approach, grounded in origin-destination data and observed traffic volumes, is needed to capture this demand component.

2.4. Contribution to the progress beyond state-of-the-art

Against this background, the novelties of this work are stated in the following, organized by research question.

Contributions to Research Question 1:

- A novel charging station allocation problem is formulated for highway networks that combines the node-based optimization structure with flow-based demand representation. This allows for direct leverage of openly available granular data on service area positions and densely sampled traffic flow counts along the highway network.

- Fast-charging infrastructure capacities are planned for the Austrian highway network for 2030 under different scenarios of BEV shares in the passenger car fleet, road traffic load, BEV driving range, and charging power levels. Extensive sensitivity analyses identify the parameters with the greatest impact on charging capacity requirements. The scenarios address uncertainties in demand resulting from the different transport sector decarbonization pathways.

Contributions to Research Question 2:

- A fundamentally different approach to representing income-group-specific requirements in passenger car charging capacity modeling is introduced. Rather than relying on spatial proxies, income groups are differentiated through values of time, reflecting behavioral heterogeneity regardless of residential location. This allows distributional consequences of charging infrastructure coverage decisions to be examined beyond neighborhood-level access disparities.

- The detour charging time, i.e., the extra walking or driving time to reach the nearest charging station, is used to link the impact of infrastructure roll-out to BEV adoption. In this way, exhaustive analysis based on high-resolution geographic data is avoided, while the densification of public charging infrastructure is still represented and connected to BEV adoption.

- The spillover effects between regional charging infrastructure expansion and BEV adoption are analyzed across neighboring regions within a techno-economic optimization framework. Previous studies of such cross-regional spillovers have relied exclusively on empirical data to explain observed patterns but have not assessed the prospective impact of strategic infrastructure deployment on the electrification of the passenger car fleet.

Contributions to Research Question 3:

- The *spatial* flexibility in the allocation of charging demand for the aggregated long-haul battery-electric truck fleet is examined at an international scale, enabling quantification of the effect of geographically varying charging costs — including electricity prices and network charges — on the cost-optimal spatial distribution of charging loads.

- The case study covers the Scandinavian-Mediterranean corridor, one of the most freight-intensive in the Trans-European Transport Network and, as the longest of its Core Network Corridors, characterized by extensive long-haul transport routes spanning from Northern to Southern Europe. Traversing multiple countries with distinct electricity prices and network charges, this corridor provides a well-suited case for examining cost-optimal transport decarbonization under both the transition to battery-electric trucks and potential modal shift to rail.

Contributions to Research Question 4:

- Origin-destination data are used to determine spatio-temporal patterns of charging demand for the future battery-electric truck fleet, extending beyond exclusively fleet-projection-based methods to account for charging demand from interregional and transit trucks.

- The study focuses on Austria and, thereby, addresses the lack of empirical spatio-temporal charging patterns for diverse commercial fleets at sub-national resolution for Central European countries. Austria's central geographic location and substantial commercial transit traffic make it representative of conditions across comparable countries in the region.

3. Methods

3.1. Overview

For each addressed research question, a tailor-made methodological approach was developed. Within this, three distinct models are formulated. Research questions 2 and 3 are addressed using the same model, which is then adapted to each research question. Model 1 is a static charging station planning tool for highway networks. Model 2, addressing RQ 2 and 3, is for long-term charging infrastructure roll-out. Model 3 is designed to generate hourly charging profiles for regional and interregional travel. Table 3.1 displays the core differences in method design. The main differences are in the functionalities and case studies which directly dictate temporal and geographic scopes and scales. Model 1 builds on the graph-based charging infrastructure planning methods [96]. The planning is for an exogenously defined charging demand for one time slice. The simplicity in temporal dynamics allows the high geographic resolution of the highway network, the demand, intersection, and service area nodes. In model 2, the scope is extended beyond charging capacity planning to include charging capacity planning alongside the adoption of BEVs. For this, year-by-year decisions are introduced, and geographic granularity is coarser. Model 4 is dedicated to the charger utilization at high temporal granularity.

Table 3.1.: Overview of methodological approaches employed in this thesis

Model	Paper/Study	Primary Functionality	Temporal Scope	Temporal Scale	Geographic Scope	Geographic Scale
Model 1	RQ1	Static capacity planning	Single year (2030)	Annual	Regional (Basque Country)	Municipal/grid cell
Model 2	RQ2 RQ3	BEV adoption + capacity expansion	Multi-year (2020-2050)	Annual	Federal State International transport corridor	NUTS-2/3
Model 3	RQ4	Profile modeling	Representative Week	Hourly	Austria	NUTS-2

Further details on the model design and main modeling assumptions are explained in the following subchapters.

3.2. *Static charging infrastructure planning*

MILP for fast-charging infrastructure sizing and allocation along highway networks (RQ1).

3.2.1. Problem description

new text: The model is designed to cost-optimally plan fast-charging infrastructure along highway networks. The panning encompasses the geographic allocation and the sizing of the charging capacities. The model aims to reflect optimal decisions from the perspective of an infrastructure planner seeking to determine a fast-charging infrastructure set-up for supporting long-distance travel with battery-electric passenger cars. This decision-making could, for example, apply to the highway network operator preparing tenders for charging service providers.

*new text:*The considered costs of charging infrastructure encompass two cost components: the onsite preparations to enable the support of large capacities provided by the charging points locally (c_X) and, on the other hand, the hardware and construction costs associated with the installation of a single charging point (c_Y). The planning refers exclusively to a defined network topology described by intersections, distances along highway segments, and positions of service areas. Potential sites for charging stations are exclusively service areas. *old text* The approach in this proposed methodology follows a node-based approach which has been reported to be mostly used in the allocation of charging infrastructure in urban areas where the demand at a node is defined through population density [96]. In this work, this model formulation is extended to the application to high-level road networks by reformulating the definition of demand at a node based on the energy consumed through BEVs driving along the road network, rather than population density, and further, regarding the traffic flow movement affecting the position of coverage of the demand.

new text In the following, the road connections between two junctions and between a junction and a network endpoint are referred to as *segments*, and the vertices are referred to as *nodes*. There are two types of nodes: some represent possible sites for charging station installation and lie along segments (*node type 1*), and others represent highway junctions or ending points of the network (*node type 2*). The nodes of type 1 can either be accessible from both driving directions ($k \in \{0, 1\}$) or from only one direction ($k = 0 \vee k = 1$). At each node, a charging demand exists which can be covered at the node itself or at other nodes within a defined range, $dist_{max}$. First, we proceed with a detailed description of the main contribution of this work, which is the mixed-integer linear program formulation of a node-based highway allocation method. Based on the demand at the nodes, the program allocates sites to install charging stations and sizes them by determining the optimal number of charging points. Then, the demand calculation used in this work is outlined.

*new text:*The charging infrastructure is planned in the consideration of the traffic flow and technological parameters of the vehicle's battery. The following key assumptions related to the representation of charging demand are made here: *old text:*

- The BEV fleet traveling along the highway network is treated as a homogeneous quantity, allowing to consider accumulated charging demand and translating this into the optimal sizing of a charging station. Based on this assumption, the technical parameters of an *average* BEV are assumed, such as average driving range ($\overline{dist}_{range,BEV}$), energy consumption ($\overline{d}_{spec,BEV}$) and charging capacity ($\overline{P}_{charge,BEV}$).

- All charging demands, $\sum_{n,k} \hat{D}_{n_k}$, result from the energy consumption of BEVs driving along the highway and need to be compensated for in total by charging stations built along the highway network.

- Highway charging infrastructure is primarily used by long-distance drivers as BEV owners mostly charge at home or at work.

- A fast-charging infrastructure along a highway network is designed based on peak demands, including seasonal peak demand ($\hat{f}_{l_k}$) and hourly peak demand, during a day (γ_h).

3.2.2. Model features and assumptions

The underlying methodology of this work is presented in Figure 3.1. It is based on a graph representation of the highway network. The approach in this proposed methodology follows a node-based approach which has been reported to be mostly used in the allocation of charging infrastructure in urban areas where the demand at a node is defined through population density [96]. In this work, this model formulation is extended to the application to high-level road networks by reformulating the definition of demand at a node based on the energy consumed through BEVs driving along the road network, rather than population density, and further, regarding traffic flow movement effecting the position of coverage of the demand.

3.2.3. Input-Output relations

TODO: insert the input-output relations here Figure 3.1 displays

3.2.4. Mathematical formulation

The objective function of the optimization model is to minimize the total charging infrastructure investment costs:

$$\min_{X_l, Y_{l_k}, \hat{E}_{l_k}^{charged,n_k}, \hat{D}_{l_k}^{in,n_k}, \hat{D}_{l_k}^{out,n_k}} c_X * \sum_{l} X_l + c_Y * \sum_{l_k} Y_{l_k} \tag{3.1}$$

The charging infrastructure is designed in such a way that all demand $\sum_{n,k} \hat{D}_{n_k}$ in the network is covered. For each node the following energy balance is defined:

$$\hat{E}_{l_k}^{charged,n_k} - \hat{D}_{l_k}^{demand,n_k} = \hat{D}_{l_k}^{out,n_k} - \hat{D}_{l_k}^{in,n_k} \quad : \forall n_k, l_k \in A_{n_k} \tag{3.2}$$

Set A_{n_k} encompasses all nodes l_k which are within the distance of $dist_{max}$[1] to node n_k. Within

[1] $dist_{max} = 4/5 * 0.6 * \bar{d}_{range,BEV}$: The factor 4/5 accounts for reduced driving range resulting from increased energy demand at highway speeds [52], [102]and 0.6 for charging processes starting at a state of charge of 20%

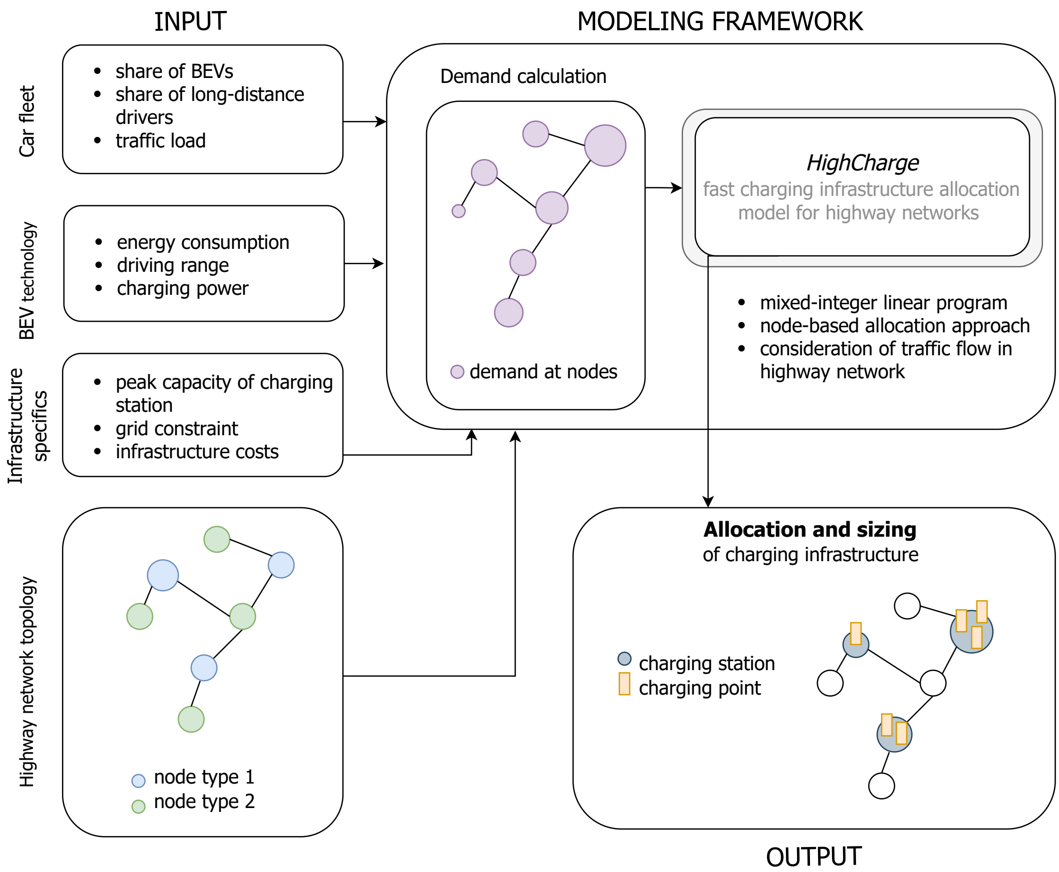

Figure 3.1.: **Overview of methodology applied in this work.** Input parameters encompass demand calculation and optimization determining the allocation and sizing of charging infrastructure.

this set radius around a node n_k, the energy demand stemming from node n_k, $\hat{D}_{n_k}$ has to be covered. Between the nodes l_k, which are part of A_{n_k}, the demand is shifted and for this, the following holds for adjacent nodes l_k and l'_k:

$$\hat{D}_{l_k}^{out,n_k} * r_{l_k,l'_k} = \hat{D}_{l'_k}^{in,n_k} \quad : \forall n_k, \{l_k, l'_k\} \in A_{n_k} \tag{3.3}$$

For parameter r_{l_k,l'_k}, the following applies:

$$r_{l_k,l'_k} = \begin{cases} 1, & \text{if } s_{l_k} == s_{l'_k} \\ < 1, & \text{otherwise} \end{cases} \tag{3.4}$$

$$\sum_{l_k} r_{l_k,l'_k} = 1 \tag{3.5}$$

up to 80% [127].

This rationing parameter is, therefore, only lower than 1, if node l is a junction point and in this case, the $\hat{D}_{l_k}^{output,n_k}$ needs to be split to simulate traffic flow partitioning at a junction point. For example, if, e.g., three segments meet and it is assumed that traffic flow is split equally between these segments at the junction, then $r = 1/2$.

The demand coverage by a charging point and charging station is defined through:

$$\sum_{n_k} \hat{E}_{l_k}^{charged,n_k} \leq \sum_k Y_{l_k} * \overline{P}_{charge,BEV} \quad : \forall l_k \tag{3.6}$$

$$\frac{P_{max}}{\hat{P}_{CP}} * X_l \geq \sum_k Y_{l_k} \quad : \forall l \tag{3.7}$$

Equation 3.7 establishes the relation between the sizing of a charging station, i.e., the number of charging points, and the binary variable, whether a charging station is installed at node n_k.

3.3. *Long-term charging infrastructure roll-out:* LP for charging infrastructure planning under consideration of multiple technologies and modes

3.3.1. Problem description and assumptions

new text The model represents the decision making from a social planner's perspective and determines the cost-optimal pathway for transitioning vehicle fleets and deploying supporting infrastructure. The costs for the vehicle fleet (purchase, maintenance, driver's costs), the costs for fueling, and the charging infrastructure (purchase, installement and OM) are minimized over multiple years.

*new text*Important features of the model are: *old text:*

- **Network design and transport demand coverage:** The geographic relation between multiple neighboring regions is conceptualized based on a graph representation. Nodes represent regions and edges represent the connections between neighboring regions. The modeling framework is designed based on origin-destination data. The travel demand is given exogenously by a set of trips that are defined by their origin-destination pair, purpose, distance, and consumer group on an annual level. These trips can be covered using different vehicle types, drivetrain technologies, and fuels. For trips that cross multiple geographic locations, i.e., for inter-regional trips that cross at least one other region, there can be a definition of multiple possible travel paths between the origin-destination pair.

- **Consumer groups by different income-class levels and level of service measure:** We introduce consumer-specific parameters that allow differentiation between income class levels in two dimensions: the value of time (VoT) and the monetary budget. The VoT relates to the monetary value of travel time as perceived by the consumer. The travel time is the level of service (LoS) that is provided by a drive-train technology, measured in hours. The value of time rises with the income level of a consumer. The monetary budget refers to the budget of a consumer that is spent on the purchase of a new vehicle and is defined together with a time horizon, reflecting the frequency of the purchase of a new vehicle.

- **Vehicle stock modeling:** The vehicle stock is modeled based on the numbers of different vehicle types and technologies. The *generation* of a vehicle is a particularly important dimension here. It defines the year in which a vehicle was purchased on the market. Each generation is associated with different technological attributes which allows to capture the technological improvements in the battery size and charging speed of BEVs while also modeling the presence of vehicles with variable technological attributes within the fleet.

- **Sizing of different types of fueling infrastructure:** Fueling infrastructure is sized based on the geographic allocation, and different types of fueling infrastructure, which may vary by the provided fueling power, maximum utilization rate, and availability. The expansion of fueling infrastructure capacity follows a defined set of investment periods.

- **Spatially flexible allocation of fueling activity:** For a given trip, there exists a set of different opportunities to fuel or charge, e.g., at work, at a public fueling station, or at home. The decision on whether the fueling demand is covered is endogenous and can change annually

3.3.2. Model features and assumptions

Figure 3.2 provides a conceptual overview of the most relevant input and output parameters as well as the core functionalities of the method. The basis of the methodology follows the classic formulation of the Network Design Problem [128]. Table 3.2 provides an overview of the state-of-the-art methods that are integrated here, together with corresponding extensions for the present work. The core model is formulated as a linear program with the objective of minimizing the total costs related to the vehicle stock, fueling, and fueling infrastructure[2], and other transportation costs, encompassing investments, operation and maintenance costs. This model is extended to a mixed-integer linear program to endogenously include the interaction between charging infrastructure expansion and the consumers' detouring fueling time.

The core features and functionalities of this model are:

[2]To provide a general model formulation here in Sections **??** and **??**, the general terminology *fueling infrastructure* and *fuel* include also charging infrastructure and electricity as a fuel.

- **Network design and transport demand coverage:** The geographic relation between multiple neighboring regions is conceptualized based on a graph representation. Nodes represent regions and edges represent the connections between neighboring regions. The modeling framework is designed based on origin-destination data. The travel demand is given exogenously by a set of trips that are defined by their origin-destination pair, purpose, distance, and consumer group on an annual level. These trips can be covered using different vehicle types, drivetrain technologies, and fuels. For trips that cross multiple geographic locations, i.e., for inter-regional trips that cross at least one other region, there can be a definition of multiple possible travel paths between the origin-destination pair.

- **Consumer groups by different income-class levels and level of service measure:** We introduce consumer-specific parameters that allow differentiation between income class levels in two dimensions: the value of time (VoT) and the monetary budget. The VoT relates to the monetary value of travel time as perceived by the consumer. The travel time is the level of service (LoS) that is provided by a drive-train technology, measured in hours. The value of time rises with the income level of a consumer. The monetary budget refers to the budget of a consumer that is spent on the purchase of a new vehicle and is defined together with a time horizon, reflecting the frequency of the purchase of a new vehicle.

- **Vehicle stock modeling:** The vehicle stock is modeled based on the numbers of different vehicle types and technologies. The *generation* of a vehicle is a particularly important dimension here. It defines the year in which a vehicle was purchased on the market. Each generation is associated with different technological attributes which allows to capture the technological improvements in the battery size and charging speed of BEVs while also modeling the presence of vehicles with variable technological attributes within the fleet.

- **Sizing of different types of fueling infrastructure:** Fueling infrastructure is sized based on the geographic allocation, and different types of fueling infrastructure, which may vary by the provided fueling power, maximum utilization rate, and availability. The expansion of fueling infrastructure capacity follows a defined set of investment periods.

- **Spatially flexible allocation of fueling activity:** For a given trip, there exists a set of different opportunities to fuel or charge, e.g., at work, at a public fueling station, or at home. The decision on whether the fueling demand is covered is endogenous and can change annually.

- **Fueling detour time:** The detouring time that is needed to the closest fueling station is included as a decision variable that is tied to the expansion of the fueling infrastructure. This particular modeling aspect allows linking the investments in fueling infrastructure to an improved LoS.

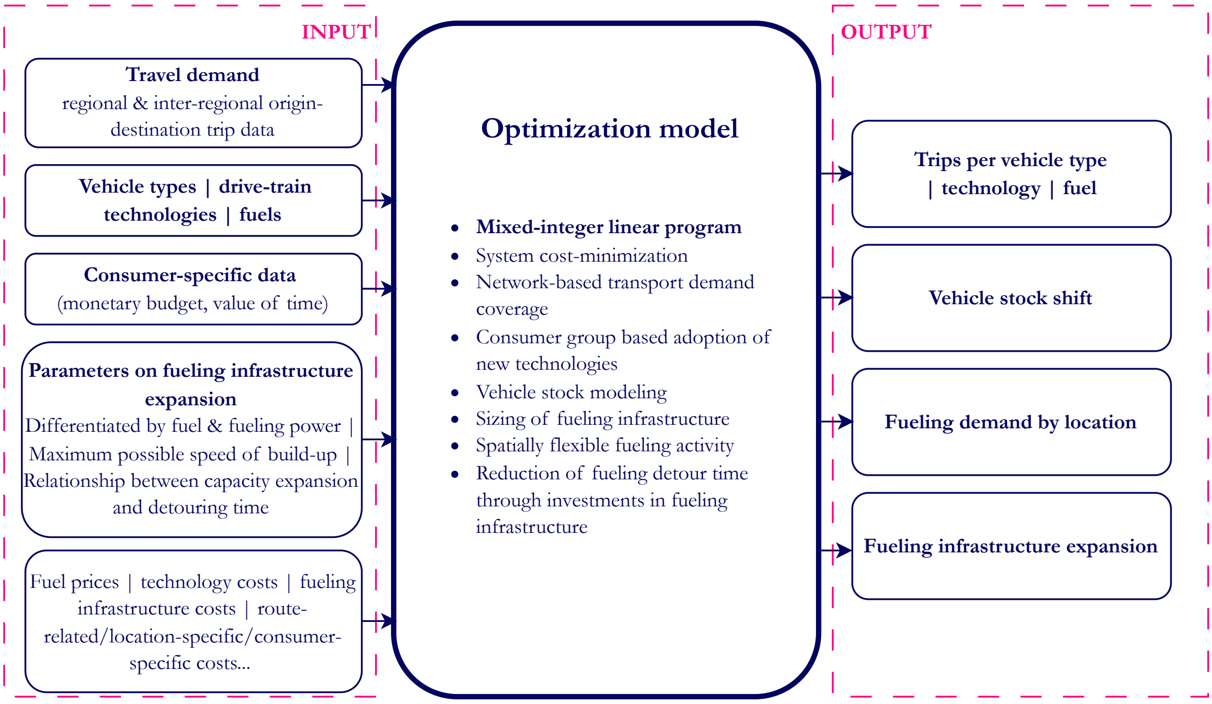

Figure 3.2.: Overview on the input and output parameters of the optimization model and its core functionalities

Table 3.2.: Overview of integrated model functionalities drawn from state-of-the-art methodology. *Fueling detour time* is written in cursive letters to imply that this model feature is not based in the existing state-of-the-art, and is, therefore, a novel extension. '-' implies that the included modeling feature is integrated without significant adaptations from the initial concepts. The listed sources are references to research papers that describe or apply these modeling approaches.

Modeling feature	State-of-the art references	Adoption/extension in this work
Network design and transport demand coverage	[128]	-
Consumer groups of different income-class levels and level of service measure	[21], [116]	-
Vehicle stock modeling	[129], [130]	-
Sizing of different types of fueling infrastructure	[121], [114]	-
Spatially flexible allocation of fueling activity	[126]	-
Fueling detour time		Consideration of the time required to reach closest charging station

3.3.3. Input-Output relations

3.3.4. Mathematical formulation

In the following, we elaborate on the mathematical formulation of the model functionalities that are extensions from state-of-the-art methodologies. The remaining constraints are to be found in the Appendix. Table 3.3 contains the relevant nomenclature used in the following equations.

Table 3.3.: Nomenclature (selection).

Sets and indices	Description
$y \in \mathcal{Y}$	year
$u \in \mathcal{U}$	consumer group
$p \in \mathcal{P}$	trip purpose
$r \in \mathcal{R}_p$	trip between origin-destination (OD) pair for a specified purpose
$k \in \mathcal{K}$	travel paths
$e \in \mathcal{E}$	geographic location
$v \in \mathcal{V}$	type of vehicle
$t \in \mathcal{T}_v$	drive-train technology fueled by a specific fuel
$g \in \mathcal{G}$	year that vehicle was introduced to market
$l \in \mathcal{L}$	types of fueling infrastructure (by capacity)
$i \in \mathcal{I}_l$	index for detour time reduction potential for type of fueling infrastructure l
$\mathcal{Y}^{exp}$	set of years of investment decisions
$\mathcal{K}_e$	set of paths that go through edge e
$\mathcal{L}^{nothome}$	fueling infrastructure that is not allocated at home of vehicle owner
$\mathcal{L}^{home}$	fueling infrastructure that is allocated at home of vehicle owner
$\mathcal{L}_{tk}$	types of infrastructure by drive-train technology and availability for travel path
$\mathcal{E}_{kl}$	geographic location along travel path and available fueling type
S^u_j	$= \{y_m \in Y \| j \in [m, m + \tau_u - 1]\}, m = 1, ..., \|Y\| - \tau_u + 1$ subsets of time periods for which a monetary budget is defined by consumer group

Decision variables		Unit	
$f_{yurkvtg}$	number of person-trips	-	$[0, \infty[$
$h_{yurkvtg}$	number of vehicles	-	$[0, \infty[$
$h^+_{yurkvtg}$	newly purchased vehicles	-	$[0, \infty[$
$h^-_{yurkvtg}$	number of vehicles leaving the vehicle stock	-	$[0, \infty[$
$s_{yurkvtgle}$	energy fueled	kWh	$[0, \infty[$
$n_{yurkvtgle}$	number of fueling vehicles		$[0, \infty[$
$b_{yurkvtle}$	fueling detour time to reach closest available public charging infrastructure	h	$[0, \infty[$
w_{yeli}	binary decision variable indicating whether detour reduction i is active	-	$\{0, 1\}$
$z_{yrkvtle,i}$	decision variable substituting the multiplication $b * w$	h	$[0, \infty[$
$q^{+,\text{fuel\_infr,home}}_{yrte}$	added fueling infrastructure at home	kW	$[0, \infty[$
$q^{+,\text{fuel\_infr}}_{ytle}$	added fueling infrastructure	kW	$[0, \infty[$
$q^{\Delta+,\text{fuel\_infr}}_{ytle}$	relative difference of fueling capacity expansion to fixed capacity size values cap	kW	$[0, \infty[$
budget^+_{yur}	budget overrun	€	$[0, \infty[$
budget^-_{yur}	budget underrun	€	$[0, \infty[$
Parameters			
LoS	level of service	h	$[0, \infty[$
VoT	value of time	€/h	$[0, \infty[$
γ_l	factor between annual peak hour and total annual demand	%	$[0, 100]$
$Q^{\text{fuel}_\text{i}\text{nfr,nothome}}_{urte}$	Initial fueling infrastructure at home	kW	$[0, \infty[$
$Q^{\text{fuel}_\text{i}\text{nfr,home}}_{urtle}$	Initial fueling infrastructure	kW	$[0, \infty[$
α_{iel}	reduction $i = 1, \dots, I$ of detour time	%	$[0, 100]$
cap_{iel}	fixed lower bounds for capacity which tie to a reduction of detour time of α_i	kWh	$[0, \infty[$
τ^u	number of years between consecutive vehicle purchase decisions, representing the vehicle replacement period.	a	$[0, \infty[$
β	relative threshold for budget under- or overrun	%	$[0, 100]$
B^{init}_{el}	initial detour time	h	$[0, \infty[$
L_k	length of path	km	$[0, \infty[$
C^{CAPEX}_{yvtg}	expenditure costs for vehicles	-	$[0, \infty[$
Budget_u	monetary budget of consumer group	€/trip	$[0, \infty[$

Objective function

The objective function comprises costs for the vehicle stock ($C^{\text{vehicle\_stock,total}}$), required fueling ($C^{\text{fueling,total}}$), costs for fueling infrastructure ($C^{\text{infrastructure,total}}$), and intangible costs ($C^{\text{intangiblecosts,total}}$) as well as penalty costs ($C^{\text{penaltycosts,total}}$) that are introduced here to penalize the overrun or the underrun of monetary budgets of consumers. The objective function is defined as follows:

$$\begin{aligned} \underset{f,h,h^+,h^-,s,n,b,w,z,q^+,\text{budget}^+,\text{budget}^-}{minimize} \quad z = C^{\text{vehicle\_stock,total}} + C^{\text{fueling,total}} \\ + C^{\text{infrastructure,total}} + C^{\text{intangiblecosts,total}} + C^{\text{penaltycosts,total}} \end{aligned} \tag{3.8}$$

The vehicle stock is sized to cover all car trips and the required fueling activity is allocated among the visited geographic locations of a car trip and different fueling infrastructure types. The latter may imply, for example, fueling infrastructure installed at home or work, or public fueling infrastructure.

Sizing of different types of fueling infrastructure

Fueling infrastructure is expanded for each investment period, $y' \in \mathcal{Y}^{exp}$, and sized based on an assumed factor which represents the ratio between the annual demand peak and the annual total demand, γ_l. We differentiate here between fueling infrastructure of which the availability is independent from the particular trip ($\mathcal{L}^{\text{nothome}}$) and those that are dependent on it as this fueling infrastructure is installed at home ($\mathcal{L}^{\text{home}}$).

$$\begin{aligned} Q^{\text{fuel\_infr}}_{tle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_y} q^{+,\text{fuel\_infr}}_{y'tle} \geq \gamma_l \sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{K}_e} \sum_{u \in \mathcal{U}} \sum_{g \in \mathcal{G}} s_{yurkvtgle} : \forall y, t, e, l \in \mathcal{L}^{\text{not home}} \\ Q^{\text{fuel\_infr,home}}_{urtle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_y} q^{+,\text{fuel\_infr,home}}_{y'urtle} \geq \gamma_l \sum_{k \in \mathcal{K}_e} \sum_{g \in \mathcal{G}} s_{yurkvtgle} : \forall y, u, t, r, e, l \in \mathcal{L}^{\text{home}} \end{aligned} \tag{3.9}$$

s refers to the annual fueling demand.

Table 3.4.: Nomenclature (extended).

Sets and indices	Description
$y \in \mathcal{Y}$	year
$u \in \mathcal{U}$	consumer group
$p \in \mathcal{P}$	trip purpose
$r \in \mathcal{R}_p$	trip between origin-destination (OD) pair for a specified purpose
$k \in \mathcal{K}$	travel paths
$e \in \mathcal{E}$	geographic location
$v \in \mathcal{V}$	type of vehicle
$t \in \mathcal{T}_v$	drive-train technology fueled by a specific fuel
$g \in \mathcal{G}$	year that vehicle was introduced to market
$l \in \mathcal{L}$	types of fueling infrastructure (by capacity)
$i \in \mathcal{I}_l$	index for detour time reduction potential for type of fueling infrastructure l

$\mathcal{Y}^{exp}$	set of years of investment decisions		
$\mathcal{K}_e$	set of paths that go through edge e		
$\mathcal{L}^{nothome}$	fueling infrastructure that is not allocated at home of vehicle owner		
$\mathcal{L}^{home}$	fueling infrastructure that is allocated at home of vehicle owner		
$\mathcal{L}_{tk}$	types of infrastructure by drive-train technology and availability for travel path		
$\mathcal{E}_{kl}$	geographic location along travel path and available fueling type		
S_j^u	$= \{y_m \in Y \| j \in [m, m+\tau_u - 1]\}, m = 1, ..., \|Y\| - \tau_u + 1$ subsets of time periods for which a monetary budget is defined by consumer group		
Decision variables		**Unit**	
$f_{yurkvtg}$	number of person-trips	-	$[0, \infty[$
$h_{yurkvtg}$	number of vehicles	-	$[0, \infty[$
$h^{+}_{yurkvtg}$	newly purchased vehicles	-	$[0, \infty[$
$h^{-}_{yurkvtg}$	number of vehicles leaving the vehicle stock	-	$[0, \infty[$
$s_{yurkvtgle}$	energy fueled	kWh	$[0, \infty[$
$n_{yurkvtgle}$	number of fueling vehicles		$[0, \infty[$
$b_{yurkvtle}$	fueling detour time to reach closest available public charging infrastructure	h	$[0, \infty[$
w_{yeli}	binary decision variable indicating whether detour reduction i is active	-	$\{0, 1\}$
$z_{yrkvtle,i}$	decision variable substituting the multiplication $b * w$	h	$[0, \infty[$
$q^{+,\text{fuel\_infr,home}}_{yrte}$	added fueling infrastructure at home	kW	$[0, \infty[$
$q^{+,\text{fuel\_infr}}_{ytle}$	added fueling infrastructure	kW	$[0, \infty[$
$q^{\Delta+,\text{fuel\_infr}}_{ytle}$	relative difference of fueling capacity expansion to fixed capacity size values *cap*	kW	$[0, \infty[$
budget^{+}_{yur}	budget overrun	€	$[0, \infty[$
budget^{-}_{yur}	budget underrun	€	$[0, \infty[$
Parameters			
LoS	level of service	h	$[0, \infty[$
VoT	value of time	€/h	$[0, \infty[$
γ_l	factor between annual peak hour and total annual demand	%	$[0, 100]$
$Q^{\text{fuel}_\text{i}\text{nfr,nothome}}_{urte}$	Initial fueling infrastructure at home	kW	$[0, \infty[$
$Q^{\text{fuel}_\text{i}\text{nfr,home}}_{urtle}$	Initial fueling infrastructure	kW	$[0, \infty[$
$\hat{Q}^{\max}_{yule}$	upper limit for charging infrastructure capacity by consumer group	kW	$[0, \infty[$
$\hat{Q}^{\max}_{yle}$	upper limit for charging infrastructure capacity	kW	$[0, \infty[$
α_{iel}	reduction $i = 1, \ldots, I$ of detour time	%	$[0, 100]$
cap_{eli}	fixed lower bounds for capacity which tie to a reduction of detour time of α_i	kWh	$[0, \infty[$
τ^u	number of years between consecutive vehicle purchase decisions, representing the vehicle replacement period	a	$[0, \infty[$
β	relative threshold for budget under- or overrun	%	$[0, 100]$
B^{init}_{el}	initial detour time	h	$[0, \infty[$
D^{spec}_{vtg}	specific energy consumption	kWh/km	$[0, \infty[$
W_{vtg}	load factor	persons/vehicle	$[0, \infty[$
$\text{Cap}^{\text{tank}}_{vtg}$	tank capacity	kWh	$[0, \infty[$
L^{annual}_{vtg}	annual range	km	$[0, \infty[$
L_k	length of path	km	$[0, \infty[$
C^{CAPEX}_{yvtg}	expenditure costs for vehicles	-	$[0, \infty[$
Budget_u	monetary budget of consumer group	€/trip	$[0, \infty[$

appendix Decision variable f expresses the number of trips covered by a specific vehicle type and technology-fuel pair of a specific generation. The sum over all possible travel paths, vehicle types, technology-fuel pairs and vehicle generations equals the exogenously given travel demand:

$$\sum_{k\in\mathcal{K}_r}\sum_{v\in V}\sum_{t\in\mathcal{T}_v}\sum_{g\in\mathcal{G}} f_{yurkvtg} = F_{yur} \quad : \forall y, u, r \tag{3.10}$$

appendix The right side of the equation expresses the coverage of the demand by different vehicle types and drive-train technologies of different generations.

appendix Based on the trips covered by a vehicle type of a technology, the vehicle stock is sized as follows:

$$h_{yurvtg} \geq \sum_{k\in\mathcal{K}_r} \frac{L_k}{W_{yvtg}L^{\text{annual}}_{vtg}} f_{yurkvtg} \quad : \forall y, u, r, k, v, t, g \tag{3.11}$$

$$\sum_p h_{yurvtg} = \sum_p h_{(y-1)rvtg} + \sum_p h^+_{yurvtg} - \sum_p h^-_{yurvtg} \quad : \forall y \in \mathcal{Y} \setminus \{0\}, u, r, v, t, g \tag{3.12}$$

$$\sum_p h^-_{yurvt(g=y-\text{Lifetime}^{max}_{vtg})} = \sum_p h_{yrvt(g=y-\text{Lifetime}_{vtg})} \quad : \forall y, u, r, v, t \tag{3.13}$$

Equation 3.11 expresses the sizing of the vehicle stock via a vehicle's annual range L^{annual} and load factor W. With equations 3.12 and 3.13, the vehicle stock is aging. Vehicles exit the vehicle stock when a maximum lifetime is reached.

Fueling demand is derived from the number of trips f and via the specific demand D^{spec} of the vehicle:

$$\sum_{l\in\mathcal{L}_{tk}}\sum_{e\in\mathcal{E}_{kl}} s_{yurkvtgle} \geq \sum_{e\in\mathcal{E}_k} \gamma_l \frac{D^{\text{spec}}_{gvt}}{W_{gvt}L_k} f_{yurkvtg} : \forall y, u, r, k, v, t, g \tag{3.14}$$

n indicates the number of fueling vehicles and is deducted from the fueled energy:

$$\sum_{v,t} \frac{1}{\text{Cap}^{\text{tank}}_{vtg}} s_{yurkvtgle} = n_{yurklge} \quad : \forall y, u, r, k, g, e, l \in \mathcal{L}^{vt} \tag{3.15}$$

The geographic allocation of the fueling demand is given by the summation over all geographic elements that are traveled by the vehicle. The summation over all geographic locations along the travel path and different types of fueling infrastructures l ensures that the allocation of fueled energy along the path is an endogeneous decision. The expansion of fueling capacities are also limited in its expansion by maximum values:

$$
\begin{aligned}
Q^{\text{fuel\_infr}}_{tle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_{y}} q^{+,\text{fuel\_infr}}_{ytle} &\leq \hat{Q}^{\text{fuel\_infr}}_{tle} &&: \forall y, t, l, e \\
\sum_{r \in \mathcal{R}_u} \left(Q^{\text{fuel\_infr,home}}_{urtle} + \sum_{y' \in \mathcal{Y}^{\text{exp}}_{y}} q^{+,\text{fuel\_infr,home}}_{yurtle} \right) &\leq \hat{Q}^{\text{fuel\_infr,home}}_{utle} &&: \forall y, u, t, l, e
\end{aligned}
\tag{3.16}
$$

3.3.5. MILP extension for regional application (RQ2)

3.3.5.1. Extended problem description and model features

3.3.5.2. Extended mathematical formulation

Consumer groups by different income-class levels and level of service measure

The intangible costs ($C^{\text{intangiblecosts,total}}$) represent non-monetary costs that are monetized in the objective function by weighting them with the consumer-group-dependent VoT. The non-monetary costs refer to in particular the total travel time associated with different vehicle types and drivetrain technologies, i.e., the level of service:

$$
C^{\text{intangiblecosts,total}} = \sum_{y,u,r,k,v,t,g} \text{VoT}_u * \left(\text{LoS}^{f}_{ykvt} * f_{yurkvtg} + \sum_{l} b_{yurkvtle} \right) \tag{3.17}
$$

f is the number of person-trips. The parameter LoS is the level of service, i.e, total travel time, associated with a vehicle type and drivetrain technology for a trip:

$$
\text{LoS}^{f}_{ykvt} = \frac{L_k}{\text{speed}_{vt}} + \text{fueling\_time}_{ykvt} \quad : \forall y, k, v, t \tag{3.18}
$$

Decision variable b is the fueling detour time, i.e., the additional travel time.

Monetary budgets vary by consumer group. These are defined by the temporal frequency of car purchases, i.e., every τ years. The budget constrains expenditures for vehicles and fueling infrastructure installed at home, which is the left side of these constraints:

$$\sum_{y \in \mathcal{S}_j^u} \left(\sum_{vtg} C_{yvtg}^{\text{CAPEX}} * h_{yurvtg}^{+} + \sum_{tl} \sum_{y' \in \mathcal{Y}_{y'}^{\text{exp}}} q_{yurtle}^{+,\text{fuel\_infr,home}} \right) \\ \leq \beta * \tau_u * \text{Budget}_u * \sum_{kvtg} f_{yurkvtg} + \sum_{y \in \mathcal{S}_j^u} \text{budget}_{yur}^{+} \quad : \forall y, u, r, j \tag{3.19}$$

$$\sum_{y \in \mathcal{S}_j^u} \left(\sum_{vtg} C_{yvtg}^{\text{CAPEX}} * h_{yurvtg}^{+} + \sum_{tl} \sum_{y' \in \mathcal{Y}_{y'}^{\text{exp}}} q_{yurtle}^{+,\text{fuel\_infr,home}} \right) \\ \geq \beta * \tau_u * \text{Budget}_u * \sum_{kvtg} f_{yurkvtg} - \sum_{y \in \mathcal{S}_j^u} \text{budget}_{yur}^{-} \quad : \forall y, u, r, j \tag{3.20}$$

Equation 3.19 expresses the lower limit for the expenditures, which relate to the investments in new vehicles (left hand-side of the equation). Equation 3.20 imposes an upper limit on these expenditures. A threshold for the budget is included with the factor β. Budget underrun and overrun, budget_{yur}^{-} and budget_{yur}^{+}, are decision variables that are penalized in the objective function and leveraged with a factor in the term $C^{\text{paneltycosts,total}}$ (Equation 3.8).

We introduce lumpiness in the sizing of the infrastructure, which is expressed by:

$$Q_{tle}^{\text{fuel\_infr}} + \sum_{y' \in \mathcal{Y}_y^{\text{exp}}} q_{y'tle}^{+,\text{fuel\_infr}} = \sum_i w_{yiel} * cap_{iel} + q_{y'tle}^{\Delta+,\text{fuel\_infr}} : \forall y, t, e, l \in \mathcal{L}^{\text{not home}} \\ \sum_i w_{yiel} = 1 \quad : \forall y, e, l \in \mathcal{L}^{\text{not home}} \tag{3.21}$$

$q_{y'tle}^{\Delta+,\text{fuel\_infr}}$ is a continuous variable that expresses the relative difference in capacity to the fixed capacity values cap_{iel}. The lumpiness is introduced via the binary variable w and ties fueling infrastructure expansion to the reduction of detour time.

The fueling detour time is explicitly defined only for fueling types not allocated at home, for $l \in \mathcal{L}^{\text{not home}}$. Based on an assumed fueling detour time B^{init} that is initially needed to reach the nearest fueling station for refueling, we model the relative reductions of the detour time that directly result from the expansion of the fueling infrastructure capacities. The decision variable b represents the total annual detour time of all vehicles by trip for this vehicle type, drivetrain technology fueling using the type of fueling infrastructure l at the geographic location e. We express this in the following way:

$$b_{yiel} = \begin{cases} B^{init} * (1 - \alpha_{iel}) * \sum\limits_{u,r,v,g} \sum\limits_{k \in \mathcal{K}^e} \sum\limits_{t \in \mathcal{F}^e} n_{yurkvtge} & \text{if } w_{yiel} = 1 \\ \\ 0 & \text{if } w_{yiel} = 0 \end{cases} \tag{3.22}$$

α_i^- represents the ith reduction of the detour time in percentage at fueling capacity cap_i. For better understanding of the computation of this equation, we provide the following example: At location e, there are 1000 vehicles charges (expressed by $\sum_{u,r,v,g} \sum_{k \in \mathcal{K}^e} \sum_{t \in \mathcal{F}^e} n_{yurkvtge}$). Due to the total installed public charging capacity at location e, the reduction $\alpha_{(i=7)e}$ is active which reduces the initial detouring time, B^{init} by 80%. B^{init} equals 60 minutes. Based on this, the total detouring time at this location is $b = 60\text{min} * (1 - 0.8) * 1000 = 120,000min$.

3.3.6. Extension for long-distance transport (RQ3)

add from paper 3

3.3.6.1. Extended problem description and model features

3.3.6.2. Extended mathematical formulation

Travel time

State of charge

3.4. *Bottom-up charging profile modeling:* Traffic-flow-data-based charging profile modeling and flexibility potential estimation (RQ4)

3.4.1. Problem description

demand and flexibility potential approximation for electricity market modeling inserting here the modeling steps ; also integrating the definition of flexibility potential (this one graph); refer here to the model of Loschan/EDISON The applied methodology consists of two distinct steps. The aim in the first step is the modeling of the future charging demand of the commercial BEV fleet together with an estimation of the demand-side flexibility potential. This flexibility option is then integrated into the electricity market model.To analyze the effect of commercial fleet charging on redispatch costs, we focus on the usage of flexibility in the dispatch market and redispatch measures.

Figure 3.3 gives an overview of these building blocks of the methodology and its most important features. In the modeling of charging profiles and the electricity market processes, the temporal resolution is hourly. The optimization of the day-ahead market clearing which has the objective of social welfare maximization, and the optimization of redispatch measures with the objective of minimizing the associated costs. Both models have the functionality of coordinating the aggregated charging processes of the commercial BEVfleet. Therefore, this modeling approach is capable of simulating the centrally aggregated coordination of charging processes and its participation in the dispatch market and using its flexibility for dispatch measures. It is important to note that, with this approach, the demand-response of charging is not modeled as the optimization of the charging processes is directly integrated in the dispatch and redispatch optimizations.

Throughout the methodology, different types of geographic information are referred to. We introduce here three graphs to support the description of the methodology ($\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3\}$; see Figure 3.4). These mirror the three models outlined in Figure 3.3. Graph $\mathcal{G}_1 = (\mathcal{N}_1, \mathcal{E}_1)$ represents a network of regions and route connections of the commercial vehicle fleet. At the visited nodes, different charging opportunities exist. $\mathcal{G}_2 = (\mathcal{N}_2, \mathcal{E}_2)$ functions as the basis for optimizing the dispatch, presenting market price zones. $\mathcal{G}_3$ is the representation of the transmission grid. These graphs overlap and values transferred between the layers are aggregated or disaggregated based on their geographic allocation.

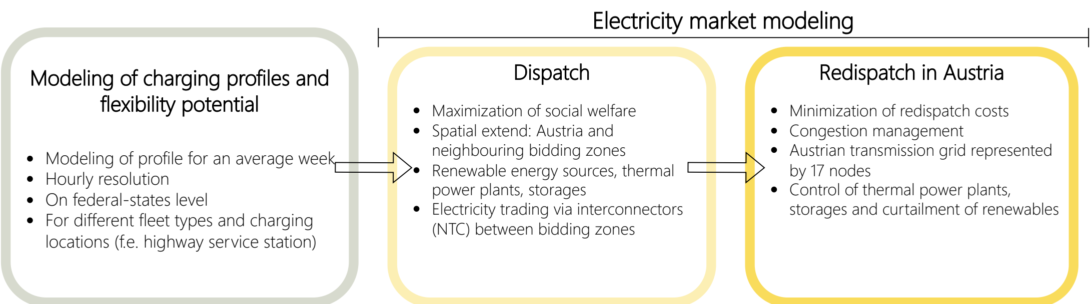

Figure 3.3.: Overview of the steps of the methodology

3.4.1.1. Definition of the flexibility potential

Figure 3.5 describes the definition of the flexibility potential in the charging of vehicles graphically as a function of time and the state of charge: A vehicle is connected to a charging station for a defined period during which a certain amount of electricity must be charged. In the present example, the battery needs to be recharged until its maximum capacity, $\mathrm{SOC}^{\mathrm{BEVmax}}$. An initial charging profile is given based on an assumption of the charging strategy of the vehicle. In the

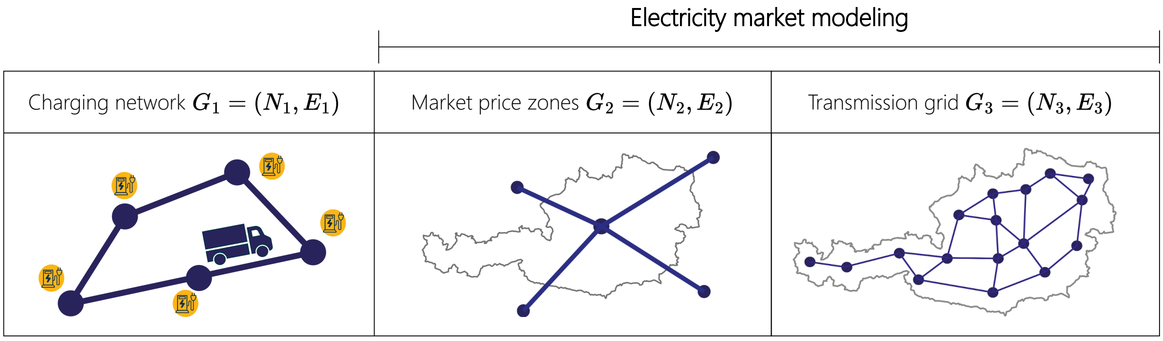

Figure 3.4.: The three layers of geographic information used throughout the methodology.

illustrated case, this charging strategy is charging at the lowest possible continuous power level ($\mathrm{CAP}^{\mathrm{BEVconst}}$), distributing the charged energy evenly during the plug-in period (indicated with the dark red line). A maximum possible charging power, $\mathrm{CAP}^{\mathrm{BEVmax}}$, defines the inclination of the green lines and, therefore, the time span of the fastest possible charging process and the latest possible point in time at which the charging process must start in order to fully recharge. The orange line illustrates an example of a load curve with a variable power level, with the power level at the beginning and at the end being significantly higher than during the mid-part of the plug-in period. The area bounded by the green oblique lines indicates the extent of possible charging curves to reach $\mathrm{SOC}^{\mathrm{BEVmax}}$ during the plug-in time period, which translates to the flexibility potential of the charging process. For example, in fast-charging processes where the charging happens at $\mathrm{CAP}^{\mathrm{BEVmax}}$ and the plug-in period is closer to the point of "fastest achievable charging completion", flexibility is more limited.

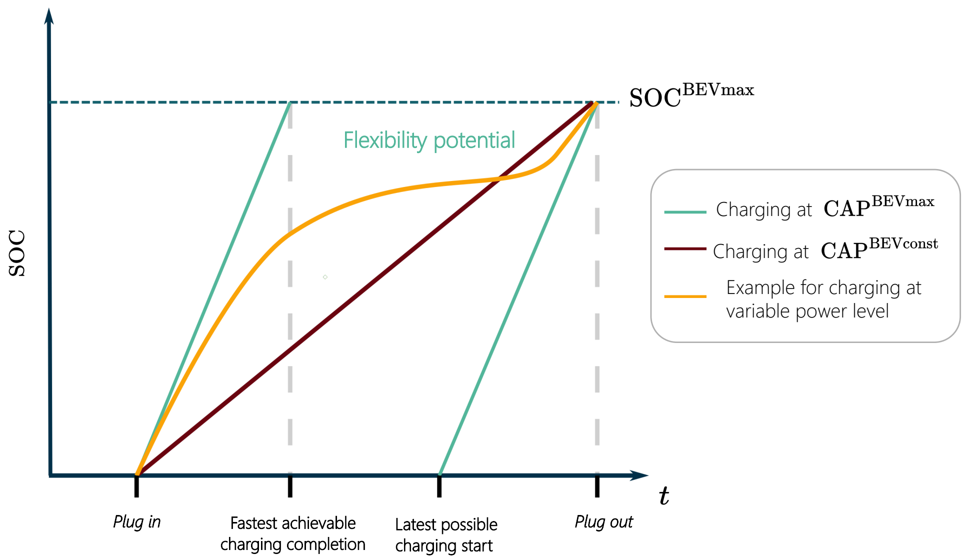

Figure 3.5.: Graphic illustration of the flexibility of a charging process

3.4.2. Local transport

For the case of local transport, the charging demand of a vehicle is assigned to exclusively one node in $\mathcal{G}_1$. We aggregate all vehicles of a specific type to a fleet $f \in \mathcal{F}^{\text{local}}$ and the energy demand for node n is estimated as follows:

$$D_{f,n}^{\text{BEV}} = \overline{D_f^{\text{spec,BEV}}} \cdot dist_f \cdot k_{f,n} \quad \forall f \in \mathcal{F}^{\text{local}}, n \in \mathcal{N}_1 \tag{3.23}$$

$$\sum_{l \in \mathcal{L}^f} D_{f,n,l}^{\text{BEV}} = D_{f,n}^{\text{BEV}} \quad \forall f \in \mathcal{F}^{\text{local}}, n \in \mathcal{N}_1 \tag{3.24}$$

The charging demand during the observed time period is deducted from the distance $dist_f$ that is driven by a vehicle of fleet f, its energy consumption $\overline{D_f^{spec,\text{BEV}}}$ and the amount of the vehicles of fleet type f operating at node n $k_{f,n}$ is. The charging demand is assigned to different charging locations $\mathcal{L}^f$ that are available to the fleet. These are, for example, depots at node n. Each charging location type is characterized by an installed maximum charging power $\text{CAP}_{f,n,l}^{\text{BEVmax}}$. The particular distribution of the charging demand to the charging location is conducted based on the assumed charging strategy for this fleet.

3.4.3. Interregional

For interregional transport, vehicles are assumed to travel between regions and charge at more than one node. In this case, a set of routes is given for the fleet $R^f = \{R_1, R_2, \dots R_I\}, i = 1, 2, \dots I$. The routes are all allocated in graph $\mathcal{G}_1$. The charging demand is determined and assigned to nodes $n \in \mathcal{N}_1$ for each route separately and later aggregated at each node for all considered types of charging locations. Each route R_i is traveled by a part of the fleet, $f_i \in \mathcal{F}^{\text{inter}}$. Charging locations are either allocated along the route $\mathcal{L}_f^{\text{enroute}}$ or at origin and destination nodes $\mathcal{L}^{f,\text{OD}}$. In accordance with this, the nodes contained in a route are categorized into two sets: $\mathcal{N}_i^{\text{enroute}}$ and $\mathcal{N}_i^{\text{OD}}$.

$$\begin{aligned} D_{f_i}^{\text{BEV}} &= dist_i \cdot \overline{D^{\text{spec,BEV}}}_f \cdot k_i \\ &= \sum_{n \in \mathcal{N}_i^{\text{enroute}}} \sum_{l \in \mathcal{L}_f^{\text{enroute}}} D_{f_i,n,l}^{\text{BEV}} + \sum_{n \in \mathcal{N}_i^{\text{OD}}} \sum_{l \in \mathcal{L}_f^{\text{OD}}} D_{f_i,n,l}^{\text{BEV}} \\ &\forall i = 1, 2, \dots I, f \in \mathcal{F}^{\text{inter}} \end{aligned} \tag{3.25}$$

k_i defines the number of vehicles traveling on the route R_i in the observed period. The charging demand for this route is distributed between the locations positioned along the route and at the origin or destination nodes. The particular assignment of the magnitude of charging demand to the nodes, $D_{f_i,n,l}^{BEV}$ depends further on the assumed charging strategy.

Nodal aggregation and temporal distribution We aggregate the charging demand for each fleet type and the respective locations on node-level:

$$D^{\mathrm{BEV}}_{f,n,l} = \begin{cases} D^{\mathrm{BEV}}_{f,n,l} & \text{if } f \in \mathcal{F}^{\mathrm{local}} \\ \sum_i D^{\mathrm{BEV}}_{f_i,n,l} & \text{if } f \in \mathcal{F}^{\mathrm{inter}} \end{cases} \tag{3.26}$$

The plug-in periods are during periods of operational downtime of the vehicles, during operation times only when needed. Independently of obtaining the absolute magnitude for each fleet, the charging demand is assigned to timestep $t \in \mathcal{T}$:

$$D^{\mathrm{BEV}}_{f,n,l,t} = g(D^{\mathrm{BEV}}_{f,n,l}, t) \tag{3.27}$$

$g(\cdot)$ indicates here the function assigning the amount of charging load to each time, resulting in a charging demand defined for each time step, $D^{BEV}_{f,n,l,t}$.

3.4.4. Estimation of charging capacities

3.4.5. Integration to electricity system model

Hier die Implementierung mit dem EDISON

4. Results of case studies

4.1. Fast-charging infrastructure network capacities for passenger cars along Austria's highway network in 2030 (RQ1)

4.1.1. Case study: Austrian highway network 2030

The proposed modeling framework is applied to Austria‘s highway network by modeling a cost-optimal charging infrastructure expansion for 2030 under different future scenarios. In this study, year 2030 has been chosen here mainly due to the two following reasons: first is that the aim of the analysis of this paper is to illustrate short-term infrastructure requirements (short term in terms of infrastructure planning), and second is that it is a significant year in decarbonization plans for the transport sector as the Paris Agreement explicitly states a 20% electrification of global road transport by 2030 [131].

To extrapolate future scenarios, model input parameters were set, which describe the current status of Austria in 2021. Table 4.1 presents the selected values. Until the end of December 2021, $76,539$ electric vehicles have been registered in Austria [132], which make up for 1.5 % of all the registered passenger cars in Austria. In order to determine traffic load on highways during peak hour and the share of long-distance drivers, the most recently acquired data (Austrian-wide) on mobility patterns was used [133]. Similar to the work of Jochem et al. [134], it was assumed that all car trips taking longer than 45 min and are longer than $25km$ or car trips longer than $50km$ are most probable to include driving on highways or motorways. Furthermore, trips were classified as long-distance if the drive was at least $100km$ long, following the common definition of *long-distance* travel [135]. Based on the starting times of these trips, it was evaluated that during the hour of peak demand on Austrian highways 12% of the daily traffic takes place and 24% of the traffic are long-distance travelers.

Traffic counts were obtained from ASFINAG [136], which provide averaged weekly traffic counts for up to 276 positions along Austrian‘s highways and motorways. Their data encompasses counts for vehicles of two categories, namely: vehicles weighing up to 3.5 tons and those weighing greater than 3.5 tons. As no differentiation is made between light-duty vehicles and passenger cars in the category of < 3.5 tons, it was assumed that all vehicles of this category are passenger cars, as light-duty transport will also be transitioned to electromobility [137]. Data for the year 2019 was chosen to be representative for 2021 as the complete data set for 2021 has not been published as of the time of the conduction of this analysis, and the 2020 dataset cannot be used due to being effected by the month-long lockdowns during the COVID pandemic.

The technical parameters $\overline{d}_{spec,BEV}$, $\overline{dist}_{range,BEV}$, and $\overline{P}_{charge,BEV}$ were set in such a way that they would represent an average Austrian BEV. For this, the technological attributes of the respective top 10 sold cars during the years 2019, 2020 and 2021 were used to evaluate the average values (see Appendix **??** for details on this).

Infrastructure cost component c_X is particularly difficult to set here as these onsite preparation costs may significantly vary based on the location of the site. Indications for the approximation of this value were drawn from [134], [62] and [138]. The installation costs of one charging point, c_Y, was assumed to be €$60,000$ for the hardware of a charging pole with $\hat{P}_{CP} = 150kW$ — which currently enables the fastest charging of passenger cars; an added construction costs to be €$7,000$ [138, 139]. Furthermore, it was assumed that the maximum installed capacity at a charging station is $12MW$.

The shape of the Austrian highway and motorway network was mapped based on geographic data by OpenStreetMap contributors [140]. The positions of the service areas were gathered based on the information drawn from ASFINAG [141] and complemented with geographic information from OpenStreetMap contributors [140]. The retrieved highway network used throughout the analysis has an overall length of $220km$ and consists of 55 segments. Moreover, there are 249 nodes in total, 139 of which represent service areas and 110 junctions. Further details on data preparation are found in the Appendix **??**.

Table 4.1.: Model input parameters reflecting the status quo in Austria.

Base case	
Input parameter	Value
BEV share ϵ	1.5%
Share of traffic load during peak hour γ_h	12%
Share of long-distance drivers μ	24%
Share of overall traffic load compared with the survey year of traffic counts α	100%
traffic count data $\hat{f}_{ik}$	maximum recorded daily traffic counts 2019
energy consumption of an average BEV in the car fleet $\overline{d}_{spec,BEV}$	$0.24kWh/km$
driving range of an average BEV in the car fleet $\overline{dist}_{range,BEV}$	$340km$
charging capacity of an average BEV $\overline{P}_{charge,BEV}$	$81kW$
peak capacity of a charging point $\hat{P}_{CP}$	$150kW$
maximum installed charging capacity at a charging station P_{max}	$12MW$
investment costs for installation a charging station c_X	€ $40,000$
investment costs for installation of one charging point $c_{Y,150kW}$	€ $67,000$

4.1.1.1. Future scenarios

Four quantitative scenarios for Austrian highway charging infrastructure expansion are outlined. These were developed in the course of the Horizon 2020 research project openEntrance [142]. The scenarios outline pathways to climate change mitigation by reaching the $1.5°C$ or $2.0°C$ targets and are referred to as the *Societal Commitment* (SC), *Techno-Friendly* (TF), *Directed Transition* (DT) and *Gradual Development* (GD) scenarios. The former three pathways aim to mitigate climate change by keeping the temperature rise to a maximum of $1.5°C$ as specified in the Paris Agreement, the latter to $2.0°C$. Within these scenarios, the extent of societal engagement, implementations of technological novelties and the strong presence of political interventions in climate change mitigation vary [see also 143, 144]. These scenarios were developed to outline mitigation paths for different economic sectors in Europe and are used here to align the analysis described in this paper into the large-scale context of decarbonization. Based on this, different developments of the transport sector relevant to the present work are projected:

- **Societal Commitment ($\boldsymbol{SC}$):** Within this scenario, politics are strongly intervening which is met by wide-spread societal acceptance, triggering behavioral changes in the face of awareness of the necessity of climate change mitigation. While this scenario is characterized by a reduction in energy demand due to behavioral changes, societal engagement supporting circular economy, and new market solutions, it is assumed that no significant technological breakthroughs appear. This translates in the transport sector to an increased modal shift to sharing concepts and public transport, which causes a significant decrease in individual passenger road transport.

- **Techno-Friendly ($\boldsymbol{TF}$):** This setting combines the appearance of major technological breakthroughs and strong societal engagement, which results in an increased top down push effect in the application of new technologies that improve energy efficiency. Simultaneously, similarly as in the SC scenario, there is a strong social commitment driving an increased modal shift away from individual passenger car transport.

- **Directed Transition ($\boldsymbol{DT}$):** Similarly as in the TF scenario, there is a strong active policy push supporting new technology options. While there are major technological developments, the social commitment to adopting such developments is missing. This results in the moderate growth of BEV share throughout the years and a decreased modal shift, but registered BEVs of the Austrian car fleet still show similar technological improvements as in the TF scenario.

- **Gradual Development ($\boldsymbol{GD}$):** This scenario represents the projection of less ambitious climate change mitigation goals. It embodies the exertion of all three dimensions, namely, social engagement, technological breakthroughs, and significant political interventions, only a weaker extent of each. Therefore, while BEV penetration will grow to some degree, and technological improvements will appear, no changes in mobility patterns are expected for this pathway.

Based on these scenario descriptions, we embed projections on changes in the modal split and developments in the BEV technology in European climate mitigation pathways for 2030. Parameters ϵ and α are specified based on the climate change mitigation goals for Austria‘s passenger transport sector described in the governmental document “Austria’s Mobility Master Plan 2030” (*AMMP*) [137]. One of these key goals is to reduce motorized private transport by 31% between the years 2018 and 2040 to meet the $1.5°C$ target set by the Paris Agreement. In the SC scenario where strong societal and political engagement act together, this goal will be met by 2030. The AMMP further specifies that by 2030, 100% of newly registered passenger vehicles must be exclusively powered by an electric engine. In the SC and TF scenarios, it is assumed that the share of new registrations of electric vehicles will gradually increase until 100% in 2030 due to the strong awareness of the necessity of electromobility by the society. For the DT and GD scenarios, the steady growth rate as experienced in 2020-2021 is projected until 2030, which will result in an EV share of 27%[1].

Table 4.2 presents the projected values. The presence of major technological breakthroughs is expressed by a significant development in the driving range of BEVs and increasing charging power, which is in line with the current major focus in BEV technology research on battery technology with the goal of allowing faster charging and longer trips [145, 146]. According to the study by Thielmann et al. [146], there is a great potential in battery storage research suggesting that technological breakthroughs in the next years could lead to the sale of BEVs with a driving range of up to $1000km$. Based on this, the sale of BEVs with an average driving range of up to $800km$ is projected for 2030 in the TF and DT scenarios, whereas in the SC scenario, this range is assumed to be $450km$, being slightly larger than the ranges of BEVs currently on the market (see Table **??** in Appendix **??**). To reflect a weak extent of technical developments in the GD scenario, the increase in the average driving range of up to $600km$ is projected for 2030.

To compare the results of the scenarios, a similar peak power for charging points is assumed for all four scenarios. The study by Andersen et al. [145] projected $\hat{P}_{CP} = 350kW$ to be the predominant peak charging power by 2030. The costs for the hardware of a charging pole are estimated to be €120,000, as well as an additional construction cost of €7,000 [147, 148]. To reflect different improvements in charging efficiency, technological breakthroughs are assumed to lead to charging power levels of up to $\overline{P}_{charge,BEV} = 315kW$, given that the efficiency will increase up to 90%, as for charging poles with peak capacity of $50kW$ nowadays [149]. Today, one of the passenger car models with the highest charging power is Porsche Taycan with $\overline{P}_{charge,BEV} = 197kW$. For the SC and GD scenarios, this is assumed to be adopted by all BEV vehicles in 2030.

Other model input parameters are assumed to be similar as in 2021, using the values from Table 4.1. Table 4.3 summarizes the overall projected values describing the average Austrian car fleet in 2030 which were extrapolated based on the assumption that these parameters would gradually change in a linear way until 2030 on the basis of the values for 2021 (see also Appendix **??**).

[1] This was calculated under the assumption that the size of the Austrian car fleet will not grow further from today.

Table 4.2.: Car fleet parameters and average technical parameters of an average BEV on the market in 2030 for different scenarios (BEV share ϵ, share in road traffic α, average driving range $\overline{dist}_{range,BEV}$, average charging power $\overline{P}_{charge,BEV}$).

Model parameters	Projections for 2030 under different scenarios			
	Social Commitment	*Techno-Friendly*	*Directed Transition*	*Gradual Development*
$\epsilon(\%)$	33	33	27	27
$\alpha(\%)$	69	83	83	100
$\overline{dist}_{range,BEV}(km)$	450	800	600	800
$\overline{P}_{charge,BEV}(kW)$	200	315	315	200

Table 4.3.: Model parameter values projected for the scenarios of climate change mitigation, set for year 2030 (BEV share ϵ, share in road traffic α, average driving range $\overline{dist}_{range,BEV}$, average charging power $\overline{P}_{charge,BEV}$, peak charging power of a charging point $\hat{P}_{CP}$, costs of installment of one charging point c_Y).

Model parameters	Input parameters for scenarios 2030			
	Societal Commitment	*Techno-Friendly*	*Directed Transition*	*Gradual Development*
ϵ (%)	33	33	27	27
α (%)	69	83	83	100
$\overline{dist}_{range,BEV}$ (km)	420	670	660	520
$\overline{P}_{charge,BEV}(kW)$	166	248	243	164
$\hat{P}_{CP}(kW)$	350	350	350	350
$c_{Y,350kW}$ (€)	127,000	127,000	127,000	127,000

4.1.2. Expansion of Fast-Charging Infrastructure along Austrian Highway Network under Different Scenarios

In this section, the most relevant results are presented. It is divided into two parts: First, we elaborate on the expansion requirements for the existing fast-charging infrastructure along Austria's highway network under different future scenarios for 2030. Details on the results obtained for the DT scenario, for which the costs of fast-charging infrastructure expansion are the lowest, are presented. Subsequently, the results of the modeled expansion given four different scenarios, SC, TF, DT and GD, are compared based on parameters describing the costs and the required charging infrastructure expansion. To gain better insight into how the observed differences between the scenarios come to place, selected input parameters of the scenario result with the highest infrastructure expansion costs for 2030, which is the GD scenario, are altered. In the second part of this section, the focus shifts from the required infrastructure expansion for 2030 to the analysis of changing infrastructure requirements in the face of technological development and increasing demand. First, the effect of change in the driving range of an average BEV is observed and, second, that of increasing demand by the rising share of BEVs in the Austrian car fleet.

4.1.3. Expansion of fast-charging infrastructure along Austrian highway network under different scenarios for 2030

Under the consideration of existing charging points with the peak capacity of $\hat{P}_{CP} = 350kW$, which is assumed to be the predominant charging capacity of fast-charging infrastructure for 2030, and the assumption that the investment of on-site preparation (c_X) has been made for all existing charging stations with charging points allowing fast-charging, the expansion of the current charging infrastructure along the Austrian highway network was modeled under different climate mitigation scenarios.

4.1.3.1. Austria's fast-charging infrastructure for 2030 under the Directed Transition scenario

Table 4.4 and Figure 4.1 display the modeling results for the expansion of the current $350kW$ charging infrastructure along highways in Austria for 2030 under the DT scenario. In Figure 4.1, the charging infrastructure expansion is expressed through additionally needed capacities. The expansions at existing charging points are indicated in dark green, and required capacities at newly installed charging points are indicated in orange . The charging stations in the lighter shades of green and orange indicate the required expansion of $350-4900kW$, i.e., by 1-14 charging points, and charging stations in the darker shades installations of additional $5200-10500kW$, i.e., 15 - 28 additional charging points. The total infrastructure expansion costs are estimated to be €54 Mio., which are translated to €/kW 369 and €39 per registered BEV in the Austrian car fleet in 2030. While currently, 40 charging points with $\hat{P}_{CP} = 350kW$, i.e., $14MW$, are installed along Austrian highways, an additional $+146MW$ of charging capacity are required until 2030. Further, additional 24 charging stations are needed, of which 10 are positioned within the dense part of the highway network in the East of Austria, in the vicinity of Vienna. The results also indicate the need for the installation of new charging stations near the Austrian border in order to cover all demand within the boarders, which mostly results from the model formulation enforcing all coverage of demand within the network as described in Section **??**. Aside from this, there exist three charging stations that do not need any infrastructure expansion.

Table 4.4.: Cost parameters and charging infrastructure attributes resulting from the expansion of the existing $\hat{P}_{CP} = 350kW$ charging infrastructure under the *Directed Transition* (DT) scenario.

	Nb. of charging stations	Total capacity	Specific capacity costs	Specific costs per BEV	Total expansion costs
DT scenario 2030	54	$160MW$	€/kW 369	€/BEV 39	€ 54 Mio.

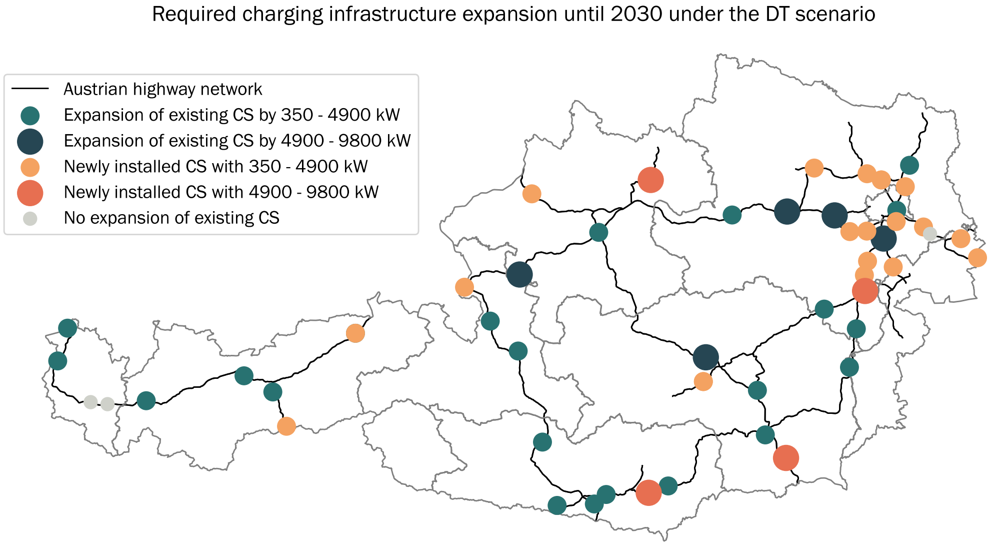

Figure 4.1.: **Required expansion of fast-charging infrastructure along Austrian highways in 2030.** Sizing of capacities that need to be added at currently existing and nonexisting charging stations (CS) given the *Directed Transition* (DT) scenario.

4.1.3.2. Comparison of the results from different future scenarios

Table 4.5 presents a comparison of key parameters describing the required infrastructure expansion under different future scenarios for Austria in 2030. The following observations are made:

- The expansion under the DT scenario ,for which the input parameters were set based on the assumption of a strong presence of political incentives pushing technological developments, results in the lowest costs of charging infrastructure expansion (€54 Mio.).

- There is a relative difference of up to +84% between the scenario causing the minimum expansion costs (DT) and the highest costs arising in the GD scenario, within which weaker climate change mitigation measures are assumed.

- The number of charging stations remains in a similar range for all scenarios, varying between 54 and 57. The specific infrastructure expansion costs per kW remain also very stable at around €/kW 368.

- The specific costs per BEV range between 39 and 72 and are the lowest in the TF and DT scenarios. The common trait of these two scenarios is the presence of technological breakthroughs leading to higher driving ranges and charging capacity of BEVs.

Table 4.5.: Comparison of the results for expansion of fast-charging infrastructure expansion along Austrian highways under different scenarios for 2030. The lowest specific costs per BEV and lowest total infrastructure expansion costs are highlighted.

Model output	Scenarios 2030			
	Societal Commitment	*Techno-Friendly*	*Directed Transition*	*Gradual Development*
Nb. charging stations	54	53	54	56
Total capacity (MW)	238	192	160	285
Specific capacity costs (€/kW)	368	368	369	367
Specific costs per BEV (€/BEV)	49	39	39	72
Total infrastructure expansion costs (€)	85 Mio.	68 Mio.	54 Mio.	100 Mio.
Rel. change of costs to DT scenario	+57%	+26%	-	+85%

4.1.3.3. Cost-reduction potentials in the Gradual Development scenario

To shed light into why the infrastructure expansion costs in the GD scenario are up to +85% higher than in the other scenarios and how the high specific costs per BEV of €/BEV 72 come to place, selected input parameters are altered and the effects on the costs are observed. These alterations reflect developments which are absent in the GD scenario but present in the others and which essentially lead to a cost reduction in charging infrastructure expansion investments. The following projections are drawn from the other scenarios and based on these, input parameters are altered in the GD scenario: a medium decrease in road traffic by -17% until 2030 as in the TF and DT scenarios, a major decrease by -31% as in the SC scenario, technological breakthroughs as in the TF and DT scenarios altering the driving range of an average BEV sold in 2030 to be $800km$, and having an average charging capacity of $315kW$ at a charging point with $\overline{P}_{CP} = 350kW$. Further details on these alterations are displayed in Table 4.6.

Figure 4.2 and Table 4.7 display the resulting cost reductions and changes in the required fast-charging infrastructure in response to the respective parameter changes. The lowest infrastructure expansion costs are achieved through the increase of BEV charging power which causes a large decrease in the required capacity as the efficiency of charging increases. Similarly, high cost reduction is accomplished by decreasing the overall traffic load which essentially reduces the demand for charging infrastructure. No cost reduction results from the increase in BEV driving range. Figure 4.2 illustrates these changes in costs in response to the specific parameter alterations visually.

Table 4.6.: Description of input parameter alterations reducing costs in the *Gradual Development* (*GD*) scenario. The parameter alterations are based on the other three scenarios: *Societal Commitment* (*SC*), *Techno-Friendly* (*TF*), *Directed Transition* (*DT*).

Parameter change	**Description (reference scenario)**	**Altered input parameter**	**Value in GD scenar**
Medium decrease in road traffic	The overall road traffic load is subject to a reduction of -17% (DT, TF).	α	100%
Major decrease in road traffic	The overall road traffic load is subject to a reduction of -31% (SC).	α	100%
Increase in driving range	The driving range of BEVs being sold in 2030 is increased to $1000km$ (DT, TF).	$\overline{dist}_{range,BEV}$	$520km$
Increase in charging power	The average charging capacity of BEVs sold in 2030 is projected to be $315kW$ (DT, TF).	$\overline{P}_{charge,BEV}$	$164kW$

Table 4.7.: Changes in fast-charging infrastructure and associated investment costs in the face of different cost-reduction measures as described in Table 4.6. The highest cost reductions are highlighted.

		Cost-reduction measures			
	GD scenario 2030	Medium decrease in road traffic	Major decrease in road traffic	Increase in driving range	Increase in charging pow
Nb. of charging stations	54	54	54	55	54
Total capacity (MW)	285	238	197	286	139
Total expansion costs (€)	100 Mio.	82 Mio.	68 Mio.	100 Mio.	66 Mio.
Rel. change	-	-18%	-32%	-0%	-34%

4.1.4. Sensitivity Analyses on the Requirements for Fast-Charging Infrastructure

In the following, the mere infrastructure requirements in 2030 without regards to the charging infrastructure currently in place are analyzed. This is done to better understand how requirements for the fast-charging infrastructure change as a function of technological BEV parameters and in response to growing demand. First, the effect of the average driving range of an average BEV in the Austrian car fleet is observed in the context of the TF scenario during which major developments in battery technology are expected. Second, the change in modeled charging infrastructure depending on the BEV share in the car fleet within the SC scenario, which reflects high societal commitment to BEV uptake, is demonstrated.

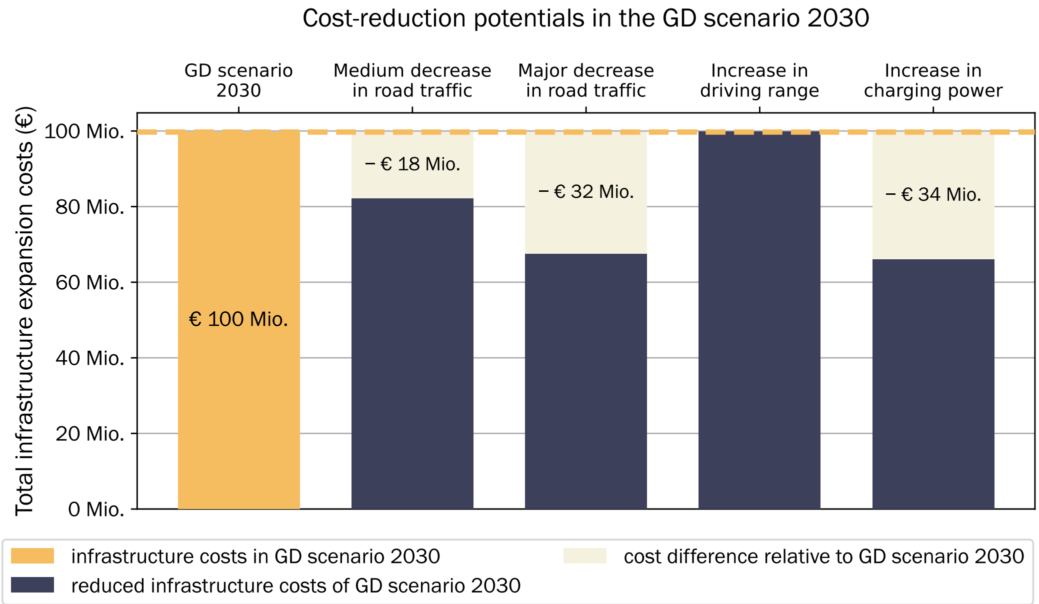

Figure 4.2.: **Cost reduction potentials in the *Directed Transition* (*DT*) scenario.** (Detailed descriptions of the measures are in Table 4.6.)

4.1.4.1. Increasing driving range in the Techno-Friendly scenario

The driving range is altered between 200 and 1400km. Figure 4.3 presents the changing distribution of the numbers of the charging points at the charging stations and total number of charging stations in response to gradually increasing driving range in the TF scenario. In this figure, the blue boxplots display the distribution of charging points (CP) along the charging stations (CS). The gray dashed line indicates the maximum possible number of charging points at a charging station which is 34. This is given by the maximum installed capacity of 12MW at a charging station. The dark red connected points indicate the total number of charging stations in the modeled charging infrastructure. Table 4.8 presents the change in the key parameters for the driving ranges of 200, 800 and 1400km and gives an impression on how the key parameters of the modeled fast-charging infrastructure change in the course of the conducted sensitivity analysis. Given the assumed development in the TF scenario, the average driving range of BEVs in the Austrian car fleet is projected to be 670km by 2030 and reach approximately 1000km by around 2040.

With the increasing range, energy demand can be shifted further away from the node where it originates, which results in wider solution space during the optimization. Overall, the infrastructure costs stay stable between €71 Mio. and €72 Mio. throughout this parameter alteration. The total number of charging stations decreases from 55 at 200km to a minimum of 38 at 1100km. The installed capacity stays constant. Starting at the range of 300km, fully occupied charging stations occur throughout the sensitivity analysis. Overall the highest costs of investments are at 200km and drop to €71 Mio at the range of 500km.

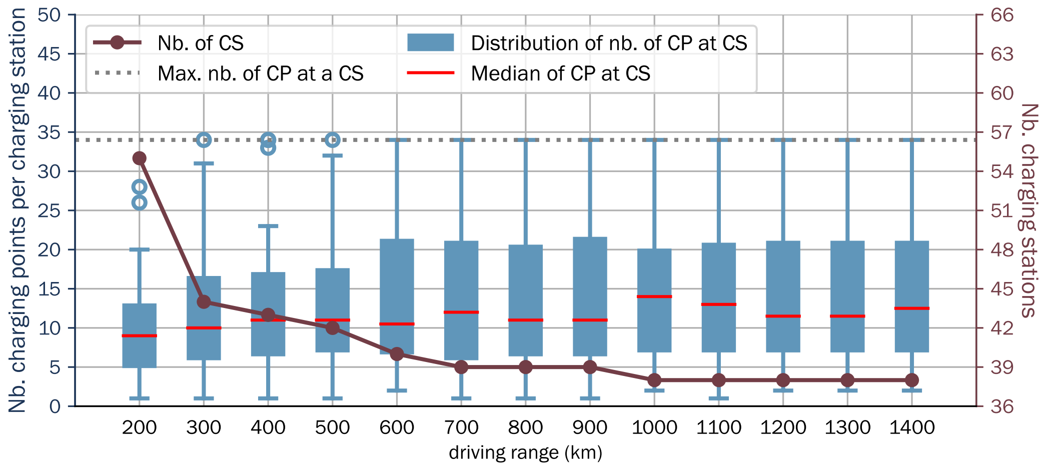

Figure 4.3.: **Sensitivity analysis on driving range.** Impact of the increase of the driving range of BEVs on the distribution of the number of charging points (CP) at charging stations and the total amount of charging stations (CS).

During this analysis, the model was run for each of the ranges between 200 and 1400km every 100km and, therefore, in total, for 13 times. During all of these model runs, at 34 of all service areas, charging station installations occurred in each of the model runs. The top sub-figure in Figure 4.4 illustrates these charging stations in black. The bottom illustration shows which of these are currently existing charging stations (in bright green) and which are not (in red). Overall, 11 of these 34 permanent charging stations are existing charging stations.

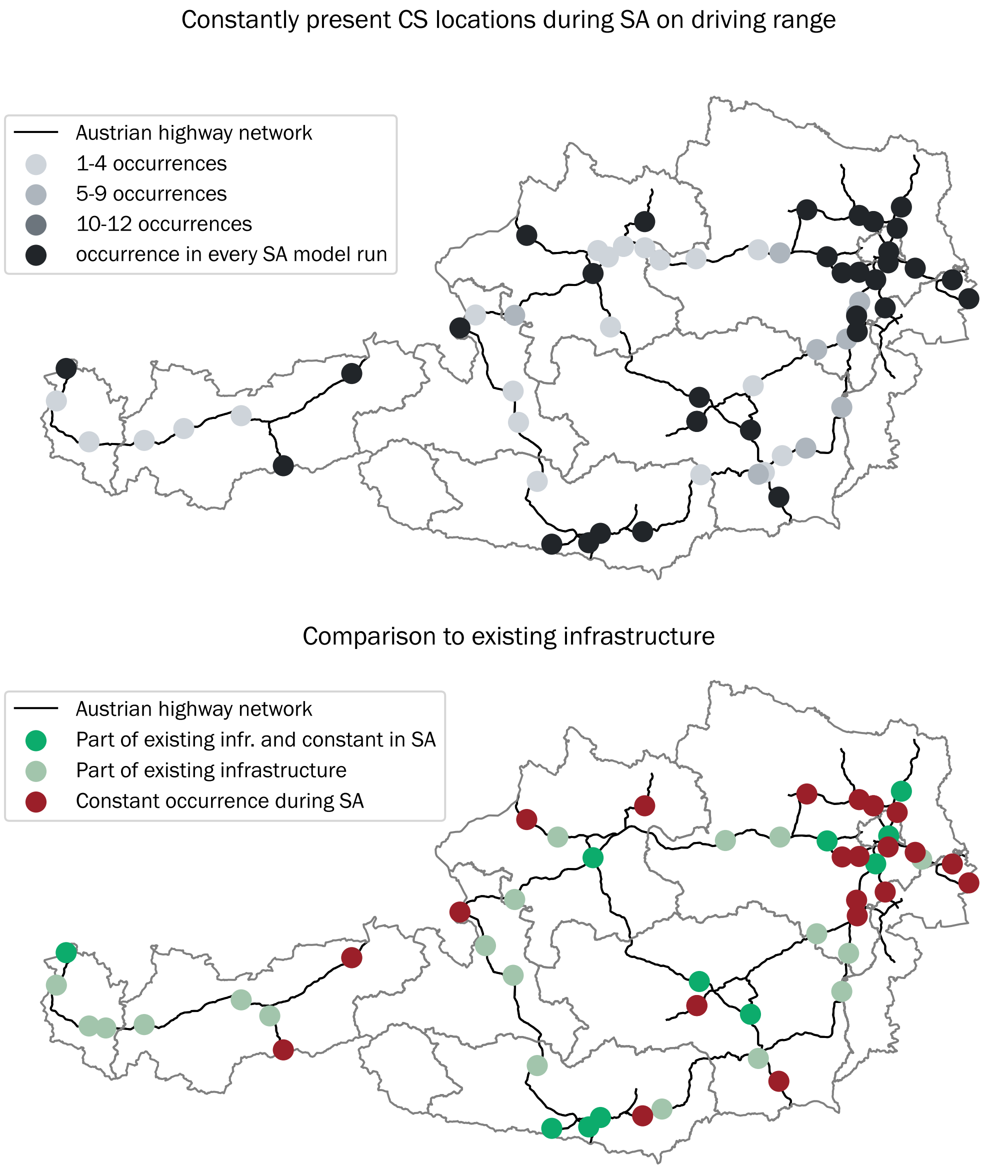

Figure 4.4.: **Frequent charging point allocations. Top:** Visualization of charging stations (CS) which occur during the model runs of the sensitivity analysis (SA) on BEV driving range. **Bottom:** Visualization of CS which are constantly present during the SA and are part of the existing infrastructure.

4.1.4.2. Increasing share of BEVs in the Societal Commitment scenario

Figure 4.5 presents the results of the sensitivity analysis on the share of BEVs in the Austrian car fleet under the SC scenario for 2030. The top-left sub-figure presents the total installed capacity of the fast-charging infrastructure as a function of rising share of BEVs. The top-right sub-figure displays the change in the number of charging stations. Boxplots illustrating the distribution of the number of charging points at charging stations are presented in the bottom-left sub-figure. The timeline in this figure is projected based on the assumption that as of 2030, all registered cars will be BEVs and a BEV has a lifetime of 10 years which would lead to an Austrian car fleet that is 100 % electric by 2040. Table 4.8 presents the values of parameters describing the key features of modeled infrastructure for the different shares of BEVs, 10%, 50% and 100%.

The required capacity for fast-charging is rising linearly with the share of BEVs. The number of charging stations is significantly increasing between 40% to 100%, indicating the requirement for a densification of the fast-charging infrastructure. This effect of densification can be attributed to the increasing amount of fully occupied charging stations which is visible in the bottom left sub-figure of Figure 4.5. The overall cost increase by a factor of about 10 between 10% and 100% share of BEVs, so does the required capacity.

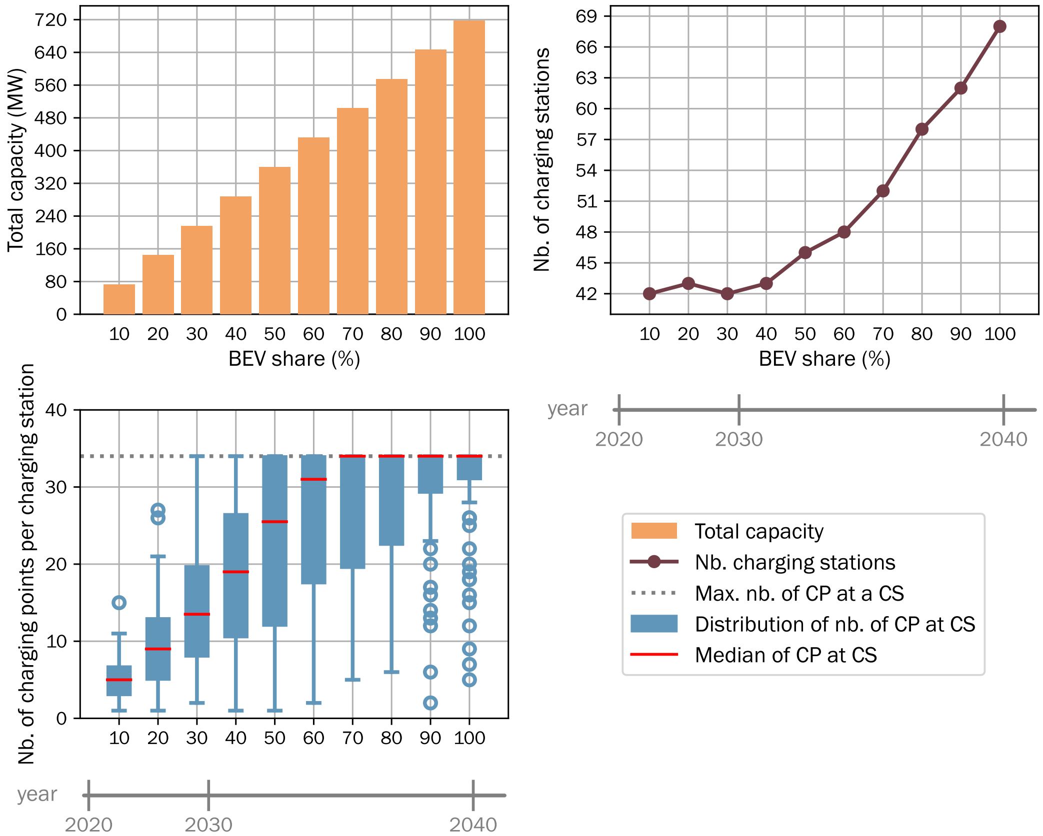

Figure 4.5.: **Sensitivity analysis of the increasing BEV share. Top-left:** Total required capacity of the fast-charging stations of required charging infrastructure. **Top-right:** Number of charging stations (CS). **Bottom-left:** Distribution of the number of charging points (CP) at the charging stations.

Table 4.8.: Key parameters describing the results of the sensitivity analyses on the driving range in the *Techno-Friendly* (*TF*) scenario and share of BEVs in the *Societal Commitment* (*SC*) scenario.

	driving range $\overline{dist}_{range,BEV}$ (TF)			**share of BEV ϵ (SC)**		
	$200km$	$800km$	$1400km$	10%	50%	100%
Nb. of charging stations	55	39	38	42	46	68
Total capacity (MW)	192	192	192	73	360	718
Total investment costs (€)	72 Mio.	71 Mio.	71 Mio.	28 Mio.	132 Mio.	263 Mio.

4.2. Impact of charging infrastructure roll-out strategies on BEV adoption in the Basque Community, Spain in the long-term (RQ2)

4.2.1. Case study: Passenger car fleet of the Basque Country, Spain

4.2.1.1. Basque Country

The geographic extent of this analysis of this paper is the autonomous community Basque Country (in Spanish: *País Vasco*, NUTS code: ES21). This region is located in the North of Spain. The current passenger car vehicle stock is in the magnitude of 1.08 Mio. (2022) [150]. In 2023, the electrification rate of the passenger car fleet was at 0.05% [151]. The Basque Country is classified as a NUTS-2 region in the EV NUTS classification scheme [25] and consists of three NUTS-3 regions: Araba/Álava[2] (ES211), Guipuzcoa (ES212), Bizkaia (ES213). Figure 4.7 displays the relative geographic allocation of the regions and of the capital cities. For RQ1, we represent the NUTS-2 region as one node. For RQ2, we derive a graph-based conceptualization with three nodes representing each of the NUTS-3 regions and three edges representing the transport connections between neighboring regions.

Origin-destination data We use the modeled passenger car trips by ETISplus database [152] which holds information about: origin, destination zones (NUTS-3 level [25]), trip purposes, as well as route lengths and shortest-path route information for trips between all regions. A subset including NUTS-3 regions of the Basque Country is extracted. We further exclude all trips that go beyond the border of the case study area. This corresponds to the selection of 99% of all trips originating from there. Based on this, we set the system boundaries of the analysis equal to the boundaries of the Basque Country.

The ETISplus data were calibrated for transportation demand in 2010. To consider the growth

[2]For the readability of the article, we refer to the region by its Basque language, Araba.

in passenger car trips since then, we use the relative growth of the passenger car fleet since then [153]. The subsequent growth of the travel demand until 2050 is extrapolated based on historic GDP growth since 2020 [154].

Trips are categorized by the following purposes: *Business*, *Private*, *Vacation*, and *Commuting*. Trip purposes of the category *Business* are taken into account in the sizing of fueling infrastructure, but are not assigned to explicit consumer groups of different income levels. *Private* trips are defined as non-business trips with a duration of up to four days, while *Vacation* refers to trips that take more than four days. *Commuting* includes daily trips for working and studying purposes. Figure 4.6 displays the initial data trip count by purpose for local trips, i.e. within a NUTS-3 region, and inter-regional trips between NUTS-3 regions within the Basque Country.

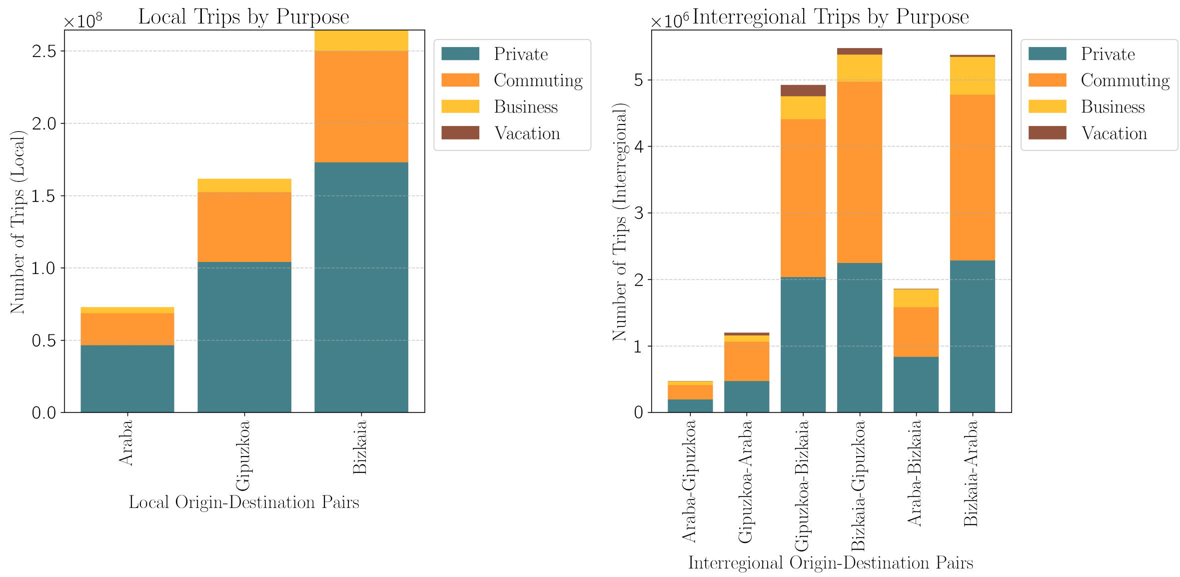

Figure 4.6.: Passenger car trips by purpose: Within a NUTS-3 region (*left*) and between NUTS-3 regions (*right*)

In the underlying data set by ETISplus, route lengths are given for the connections between NUTS-3 regions but not for trips within a region. To compensate for this missing information, we perform the assignment of route lengths (see Table 4.9).

Table 4.9.: Path lengths assigned to trips within a region (same origin and destination on NUTS-3 level)

		Average trip length (km)		
Region type	Share of total trips within a region	Private	Commuting	Business
Urban	21%	7.4	10.4	13.0
Rural	79%	9.7	17.1	23.2

Introduction of income-class specific parameters The key assumptions in the introduction of the income-class specific parameters is that the central difference in travel patterns by income class is the motorization rate, by monetary budget and home ownership [155, 156]. The income classes are characterized by having different monetary budgets for the purchase of vehicles and different VoT values. We consider five different income classes: *Very high income*, *High income*, *Medium low income*, *Low income* and *Very low income* class, mirroring the income class classification of EUROSTAT via five quintiles [157]. The most important preprocessing steps related to the income classes include the distribution of trips among them, the determination of monetary budgets and limitations for home-based charging infrastructure. We, therefore, characterize the five consumer groups by the motorization rate, VoT, average monetary budgets for new vehicles, the frequency of vehicle purchase, and the share of home ownership. Assumptions for these are summarized in Table 4.10.

Mattioli et al. [158] define a relation between motorization rate and purchasing power on NUTS-2 region level, derived from EUROSTAT data. We use this as a proxy to determine a relationship between the purchasing power standard (PPS) of the income class and the motorization rate, together with the average motorization rate in the Basque Country. This relationship is: Motorization rate $= 0.02 \times \text{PPS} + 79$. The VoT parameter is reported to vary significantly based on the monetary budgets. We use the value used in the case study by Tattini et al. [21] who apply these values in the context of a Danish case study. Due to the lack of published data for VoT values calibrated for different income classes in Spain or the Basque Country, we assume similar values for the present case study as found for Denmark and adapt these based on relative purchase power numbers, applying the factor $\times(92/139)$ [159].

Table 4.10.: Overview on assumptions between the income classes

Income class	*Purchase Power Standard*	*Motorization rate (nb. of veh/1000)*	*Value of Time (€/h)*	*Average monetary budget over 10 years (€)*	*Purchase frequency (a)*	*Assumed home ownership*
Very low	10081	338	6.60	4160	10	30%
Low	19685	425	12.77	9282	8	55%
Medium low	27932	500	18.93	13910	6	85%
High	37910	599	25.10	18824	4	100%
Very high	61086	800	31.27	26545	2	100%

A statistic for the UK details expenditures for purchases [160] [161], displaying an approximately linear relationship between level of income and expenditure for purchases, among with how much is spend on the primary market. Based on this, we determine the monetary budgets for a new vehicle from the primary market for the *Very high*, *High* and *Medium High* income class, while we use the vehicle purchase price from the secondary market for the budget quantification of income class levels *Low* and *Very low*. Home ownership is extrapolated from the average Spanish home ownership rate reported in [162].

4.2.1.2. Drivetrain technologies and fuels

When it comes to the selection of the drivetrain technologies, we simplify it to two options: fossil-fueled vehicles and plug-in battery-electric vehicles. In contrast to this choice, studies in the literature have included larger technology portfolios by considering different electric car technologies, such as hydrogen fuel-cell or hybrid plug-in electric. The reason for the limitation to two technologies is twofold: Plug-in battery-electric vehicles have been largely proven to be cost-wise and environmentally the most viable, with a promising outlook on further cost decreases [163]. Therefore, most policies focus on supporting this technology. We reduce the fuels used to *fossil fuel*, including diesel and petrol, and electricity. Cost parameters for electricity are drawn from marginal costs for the energy system optimization for Spain performed with GENeSYS-MOD [164]. The average fuel price is based on [165]. Costs of the drive-train technologies are drawn from a detailed modeling of the manufacturing of all components by [163]. We reduce the complexity of different car sizes to one representative vehicle type as related parameters to the choice of the car size - for example, household size, housing type [166] - are not part of the modeling. Therefore, different sizes are excluded to avoid introducing a modeling bias due to insufficient complexity of the model. Table 4.11 summarizes all assumptions on cost parameters and sources. The cost assumptions are based on the work by Grube et al. [163] who have evaluated a wide range of different drive-train technologies for passenger cars and different passenger car sizes, including small, medium, and SUV. Here, we derive the costs for an *average*-sized vehicle by averaging the costs using weights that correspond to the magnitudes of the sizes in the current vehicle stock, assuming that this distribution would be constant throughout the optimization horizon. The data on different vehicle sizes represented in the statistic for different engine sizes in the Spanish vehicle stock are retrieved from [167]. The costs for the maintenance are increased at a rate of 3% per year for the first three years, and then by 10% per year.

To also allow for purchases from the second-hand vehicle market, purchasing options for vehicles of older generation are allowed for. For each technology, we introduce three groups of older vehicles with decreased investment costs but higher costs for maintenance and operation. The technological performance of vehicles available on the second-hand market is a function of their counterpart on the first-hand market and their average age. These three age-groups are: *1-5*, *6-10* and *11+ years*. The maximum lifetime of a vehicle is set to 25 years. An initial vehicle stock with vehicles of different ages is assumed based on data from [167]. The cost for vehicles on the second-hand market are calculated assuming a 65% drop in resale price after the first year and after this, an annual 10% cost decrease. The maintenance costs follow the annual increase in number as described above.

4.2.1.3. Types of charging infrastructure

For fossil-fueled vehicles, we assume that the fueling infrastructure with sufficient capacities has been established. Therefore, for combustion-engine passenger cars, no expansion of the fueling infrastructure is required.

Table 4.11.: Overview of a selection of assumed parametrization for passenger cars of battery-electric vehicles (BEV) and fossil-fueled internal-combustion engine vehicles (ICEV).

	2025	2035	2045	Reference
BEV (first-hand market)				
CAPEX (€)	18,000	16,100	15,200	[163]
fuel costs (€/kWh)	0.05	0.03	0.03	[164]
Specific consumption (kWh/100km)	16.9	13.4	12.7	[163]
Battery capacity (kWh)	56	69	70	[163]
ICEV (first-hand market)				
CAPEX (€)	12,000	12,800	13,200	[163]
fuel costs (€/kWh)*	0.08	0.12	0.18	[163]
Specific consumption (kWh/100km)	58.1	44.9	36.9	[163]
carbon price (€/tCO2)	52	113	250	[168]
Charging infrastructure				
Level II: CAPEX (€/kW)		108		[28]
DCFC: CAPEX (€/kW)		350		[28]
OPEX		30% of CAPEX		

*without carbon pricing

For charging infrastructure, we introduce four different types: Home (Level II), Workplace (Level II), Public slow (Level II), and Public fast (DCFC) (based on [51]). Tables 4.12 and 4.13 summarize relevant assumptions. The reduction in fueling detour time is relevant for public infrastructure. In the case of fast charging (DCFC), the fueling time is considered in addition to the level of service (similar as in the work by Luh et al. [116]), while for the other types of infrastructure, we assume that there the charging is during typical down times of the vehicle and therefore does not impact the level of service. Initial home charging capacity (40 MW in 2020) reflects the early adopter profile of BEV owners, who were predominantly homeowners with dedicated parking and charging infrastructure. This assumption is consistent with empirical findings that early BEV adopters had home charging access rates exceeding 90% [169].

Table 4.12.: Overview on the introduced types of charging infrastructure and most important assumptions.

Type of charging infrastructure	Peak power level	Location	Max. utilization rate	Fueling detour time	Extra fueling time
Home (Level II)	11 kW	home	2%		
Workplace (Level II)	11 kW	workplace	20%		
Public slow (Level II)	22 kW	public	25%	×	
Public fast (DCFC)	50 kW +	public	25%	×	×

Table 4.13.: Assumptions for the sizing of initial capacities and upper limits for the expansion of charging infrastructure.

Type of charging infrastructure	Sizing of initial charging infrastructure	Limitation for capacity installation
Home (Level II)	based on current observations in the share of charging processes at home chargers (88%, [169])	Home charging infrastructure can be installed in 30% of owned homes.
Workplace (Level II)	based on current observations in the share of charging processes at workplace chargers (8.8%, [169])	based on today's charging demand coverage at work and scaled by current total number of passenger cars in the given region.
Public slow (Level II)	current BEV-to-charging point ratio for public slow chargers [170, 171]	No upper limit
Public fast (DCFC)	current BEV-to-charging point ratio for public fast chargers [170, 171]	No upper limit

. The maximum expansion speed of charging infrastructure is assumed to be 30% which is in line with reported expansion speed by [6].

To ensure that next to public slow charging, capacities for fast charging are installed to reduce range anxiety and enable recharge in cases of emergency (f.e. battery not sufficiently charged), we introduce a constraint imposing a minimum requirement for fast charging depending on the capacity installed for slow charging, which is set at the proportion of 2. This number is derived from the relative difference of the peak power level (the current value in Spain is 4.6 [170]) .

4.2.2. Charging infrastructure roll-out scenarios

Figure 4.7 illustrates the setup for analyzing different scenarios of the roll-out of public charging infrastructure. Illustrations 1 and 2 address the setup to answer RQ1: two expansion strategies are compared, the *Concentrated* and the *Distributed expansion*. For this, the case study area is considered as one NUTS-2 node, while for addressing RQ2, the area is modeled on NUTS-3 level with three nodes.

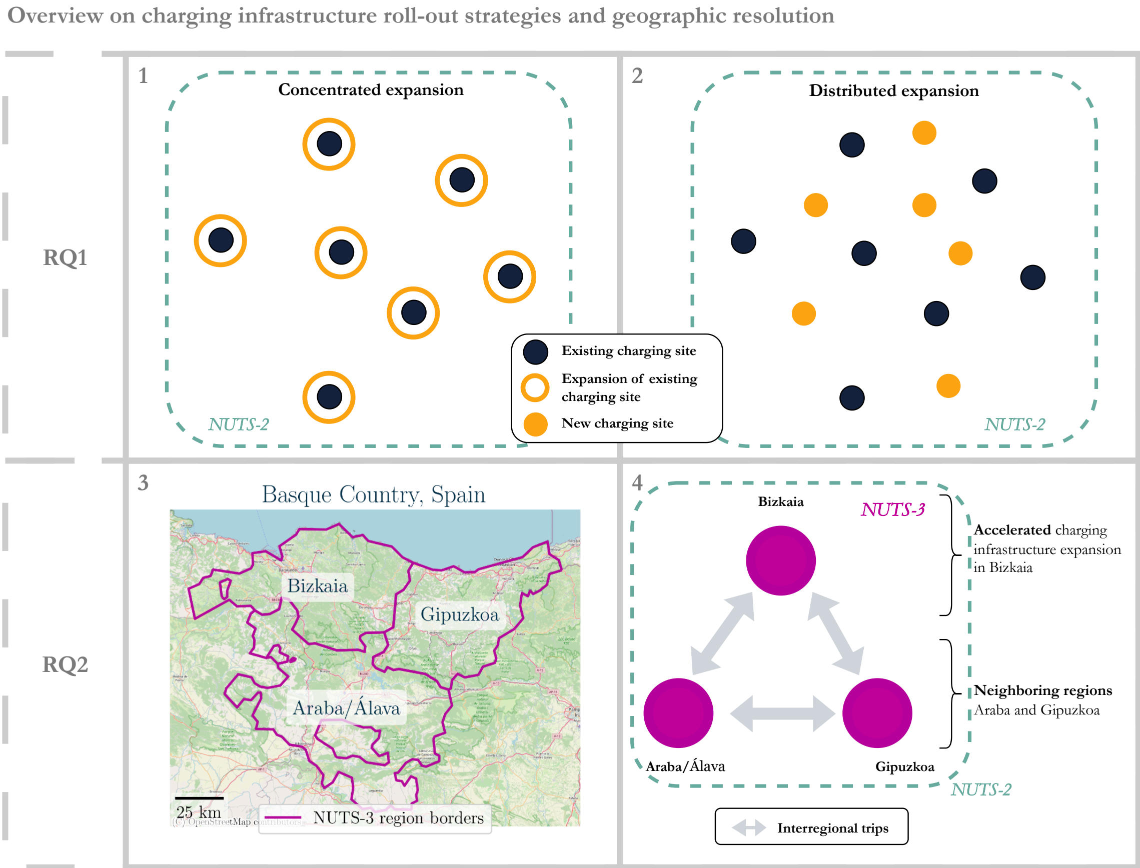

Figure 4.7.: Overview of the scenario setups for the analysis. The top row (illustrations 1 and 2) displays the contrasting approaches to the densification of public charging infrastructure, addressing research question 1 (RQ1). In the bottom row, illustration 3 displays a map of the geographic area of Basque Country by its three NUTS-3 regions, together with the corresponding capitals. In illustration 4, it is implied which pace of charging infrastructure roll-out is assumed for the NUTS-3 regions — addressing research question 2 (RQ2).

Spatial densification (RQ1)

The spatial densification of charging infrastructure is addressed using the mixed-integer extension of the optimization model. For this, the trip data is accumulated to the NUTS-2 level.

To quantify the interaction between charging infrastructure investments and the detouring time required to reach the closest public charging station, a function between the average detouring time and installed charging capacities needs to be defined:

$$\alpha = g\left(q^{fuel\_infr,+}\right) \tag{4.1}$$

α is the factor for the relative decrease of the fueling detour time and is a function of additionally built capacities. By formulating this function, we directly assume that the chargers are spatially evenly distributed throughout the regions' road network. By doing this, we neglect the different degrees of urbanization and the option of allocating charging sites at points of interest, which would decrease detour time in reality. We do this with the premise that for a share of 100% BEV penetration of the passenger car fleet, a spatially even distribution of charger availability is a necessity, to be equally accessible for all. Further, this assumption indicates that there are no malfunctions of charging points which could lead to extended detouring times. The function $g()$ represents this relationship and varies for different charging infrastructure expansion strategies, which are explained in the following:

- *Distributed expansion:* Charging sites with singular charging stations are installed with the aim of achieving a dense charging network. Only when a maximum number of charging sites is reached, the number of charging points at the charging stations is expanded.

- *Concentrated expansion:* The focus lies on expanding the number of stations at charging sites. A maximum is set for the average number of charging stations at charging sites. New stations are only installed when existing sites are expanded to their maximum.

Figure 4.7 illustrates these strategies (Illustrations 1 & 2): Filled circles of dark blue color indicate initial charging infrastructure sites; orange color indicates the allocation of the installation of new charging points. The two strategies are contrasting as the *Concentrated expansion* leads to significantly lower increases in the density of charging stations versus *Distributed expansion* rapidly leads to a dense charging infrastructure. This boils down to the installed charging points ($n^{\text{points per site}}$) per charging site:

$$n^{\text{sites}} = \left\lfloor \frac{\text{installed capacity}}{n^{\text{points per site}}} \right\rfloor \tag{4.2}$$

We simplify the shape of an area using the rectangular form and translate the detour distance to a closest available charging station using the following definition: addi info on budget penalty + home charger assumption

$$\begin{aligned} d^{\text{arieal}} &= \frac{1}{2}\sqrt{\frac{\text{area}}{n^{\text{sites}}}} \\ \text{detour distance} &= \mu \times d^{\text{arieal}} \end{aligned} \tag{4.3}$$

Factors μ is a value ≥ 1 and translated the areal distance to the driving distance in the street network. Finally, this is translated to detouring time, B:

$$B = \text{detour distance} \times \frac{1}{\text{driving speed}} \tag{4.4}$$

Three scenarios are defined for the analysis of spatial densification via different values for the parameter $n^{\text{points per site}}$, reflecting the *Distributed* roll-out scenario ($n^{\text{points per site}} = 2$), the *Concentrated* roll-out scenario ($n^{\text{points per site}} = 40$), and one between these two extreme cases with a balanced ten charging points per site ($n^{\text{points per site}} = 10$).

For the definition of suitable ranges in charging infrastructure expansion, we define a minimum detour time that is possible. This is set to five minutes[3]. No greater reductions of the initial detour time are possible. Figure 4.8 displays reduction curves for different cases of $n^{\text{points per site}}$ for both, fast and slow public charging.

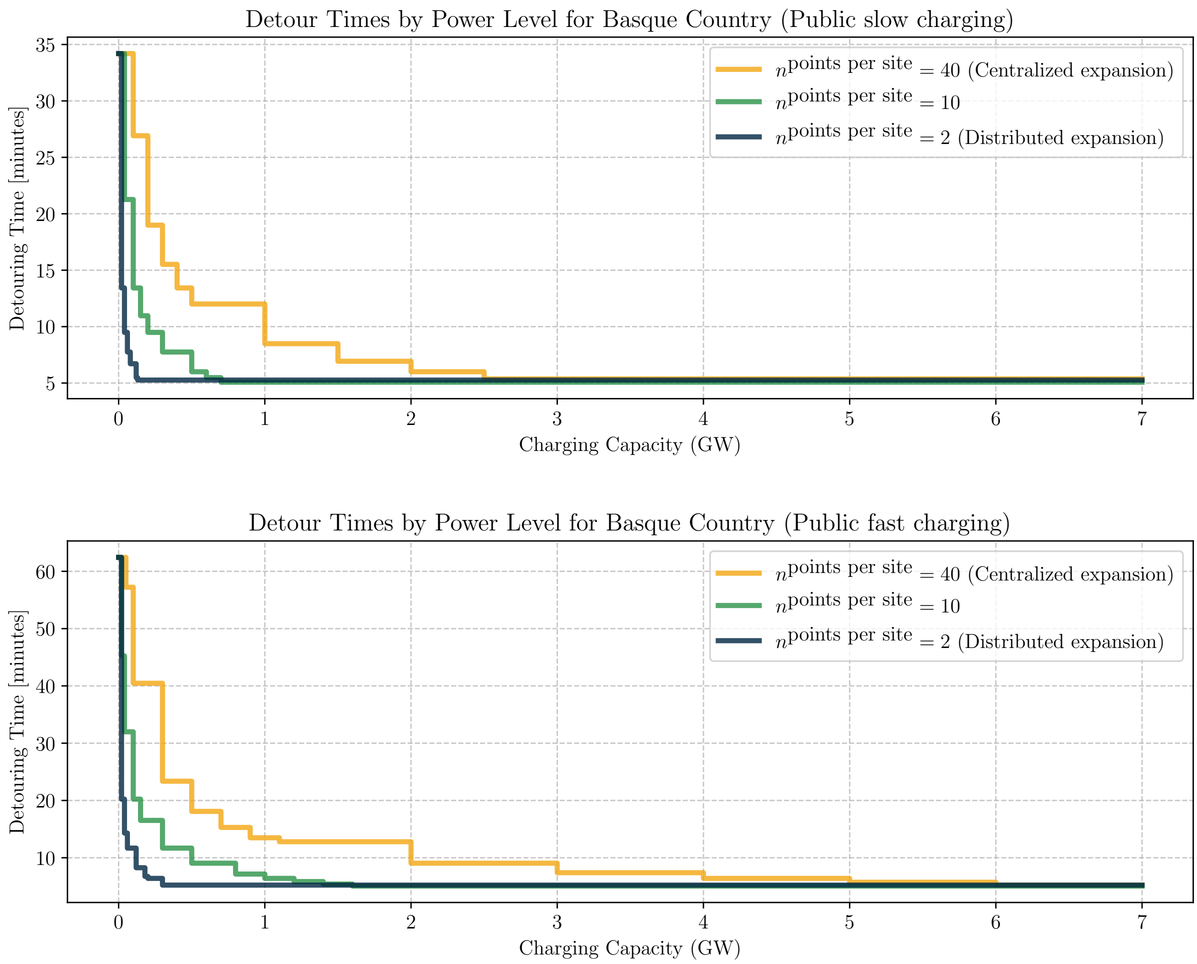

Figure 4.8.: Comparison of two vertically aligned figures.

[3]The five minute threshold is based on an educated guess by the authors. This is based on the assumption that lower times for detouring (time required to reach the charging station **and** back from the charging station) are unrealistic. Further, a lower limit is set to avoid very small values in the optimization which could lead to increased computational complexity in the solution of the MILP.

4.2.2.1. Speed of roll-out (RQ2).

The setup for the analysis of RQ2 is displayed in the bottom row of Figure 4.7: We consider BEV adoptions in the three NUTS-3 level regions of the Basque Country and focus on the accelerated charging infrastructure expansion in Bizkaia, while observing the neighboring regions, Araba and Gipuzkoa. For this analysis, we assume the charging infrastructure capacities exogenously by including hard constraints for the infrastructure capacities. Table 4.14 displays an overview of the designed scenarios for the analysis. *Baseline Roll-out* defines a reference scenario based on the disaggregated expansion pathway to the region. The *Central Acc. Roll-out* is designed to analyse the relative changes in electrification compared to this reference scenario caused by an increase of charging infrastructure by 30%. In the scenarios *Neighbor Slower Roll-out* and *Central Acc. + Neighbor Slowed Roll-out*, the neighboring regions have slower expansion than the reference, and the charging capacity roll-out pathway is reduced by 30%. These two scenarios differ again by first assuming the reference expansion pathway and in the second, the accelerated expansion with a charging capacity increased by 30% to the reference.

Table 4.14.: Scenarios for analysing RQ2. The cross indicates the assumed expansion pathway for the regions in each scenario.

		Expansion pathway		
Scenario name (RQ2)	**Region name**	-30%	Reference	+30%
Baseline Roll-out	Bizkaia		×	
	Gipuzkoa		×	
	Araba		×	
Central Acc. Roll-out	Bizkaia			×
	Gipuzkoa		×	
	Araba		×	
Neighbor Slowed Roll-out	Bizkaia		×	
	Gipuzkoa	×		
	Araba	×		
Central Acc. + Neighbor Slowed Roll-out	Bizkaia			×
	Gipuzkoa	×		
	Araba	×		

The reference expansion pathway is determined based on a model run, during which the constraint of 100% decarbonized passenger car fleet is imposed. The resulting total charging capacity values for the investment periods between 2020 and 2050 are included as hard constraints for the charging capacity expansion. Figure 4.9 displays the reference values by region.

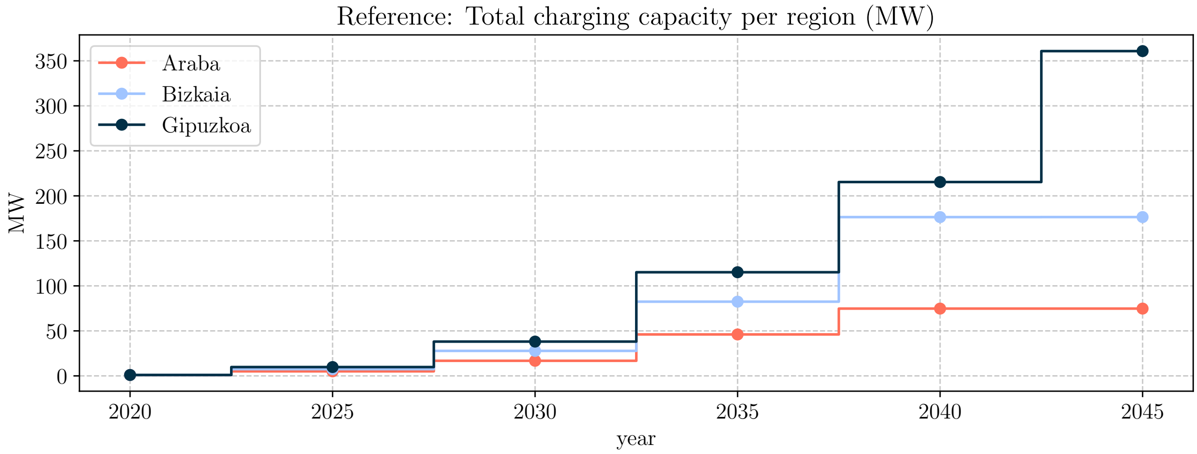

Figure 4.9.: Reference expansion pathway for scenario design of different charging capacity build-up scenarios.

4.2.3. Income-class dependent impact of spatial densification of public charging infrastructure

Figure 4.10 displays two snapshots of the development of the share of electrification in the passenger car fleet by consumer group: years 2030 and 2040. $n^{\text{points per site}}$ indicates how many charging points are installed in newly built charging stations as of 2020, implying the degree of spatial densification of the public charging infrastructure network. The development of technology turnover is different among the consumer groups. In 2030, the electrification in the vehicle fleet belonging to the *Very high income* and *High income* consumer groups is significantly higher, at the level of around 48.9-65.1%, than that of the vehicle fleet belonging to the other consumer groups. By 2040, the share of BEVs rises in all consumer groups to the level of 9.6-69.0%. It is important to note here that in absolute numbers, the BEVs in the fleet owned by the consumer group *High income* do not exceed the *Very high income* consumer group.

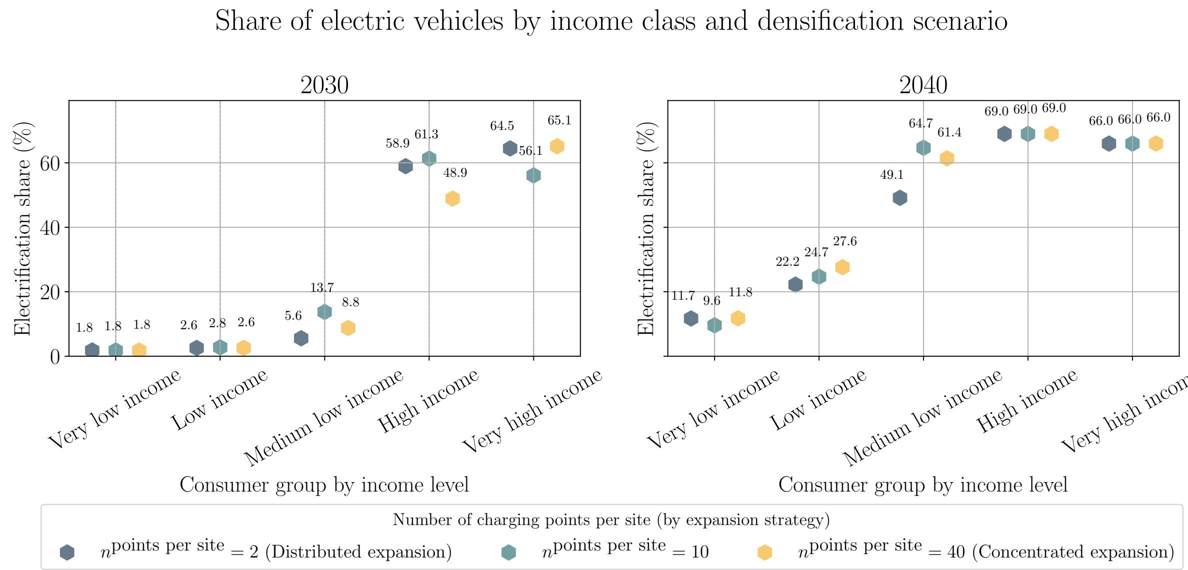

Figure 4.10.: Share of battery-electric vehicles in the passenger car fleet in 2030 and 2040 in consumer groups of different income levels for roll-out strategies of public charging infrastructure defined via the number of charging points per site ($n^{\text{points per site}}$).

The relative order of the number of BEVs in the fleet by consumer groups is not changed by the different roll-out strategies. However, significant differences in the adoption are visible between the different densification strategies: While the *Concentrated expansion* leads to a gap of 6.2% between the electrification share of the *Very high income* consumer group and the *High income* consumer group, this gap is lower in the scenario *Distributed expansion*. The electrification of the *Medium low income* group is accelerated in the more balanced scenario of ten points per site. Income groups *Very low income* and *Low income* are not significantly affected by 2030. By 2040, a significant gap of 5.4% between the *Distributed* and *Concentrated* expansion scenario arises. Overall, there is no visible correlation to be deduced between the number of points per site and the development of the electrification share in the fleets owned by the consumer groups of different income levels. Figure 4.11 displays differences in the development of the gap in the share of electrification between the *Distributed* and *Concentrated* scenarios. It shows that the highest overall impacts are on *High income* and *Medium low income*. By the end of the horizon, differences converge towards 0.0%, indicating that the $n^{\text{points per site}}$ parameter affects the speed of electrification but is not decisive for the resulting size of the decarbonized passenger car fleet.

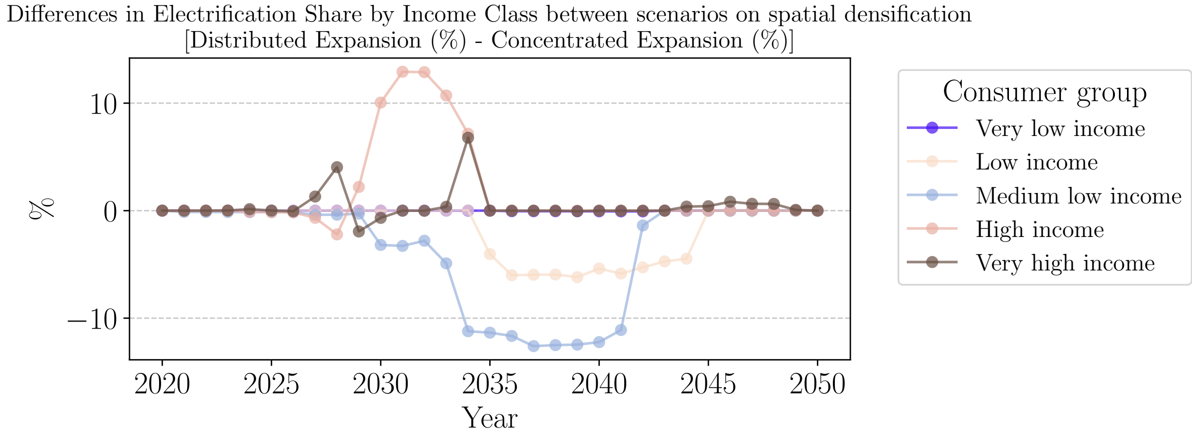

Figure 4.11.: Maximum differences in the electrification share between scenario $n^{\text{points per site}} = 2$ *(Distributed expansion)* and $n^{\text{points per site}} = 40$ *(Concentrated expansion)* by consumer groups over time.

To gain a better understanding of the variation in the electrification across consumer groups, we take a look at the three key output parameters that describe the charging infrastructure expansion decisions and their utilization by scenario. Figure 4.14 displays expanded charging infrastructure capacities together with the development of the utilization rates of public infrastructure for all scenarios. Figure 4.12 shows how the average detour time of public charging infrastructure is reduced by these capacity expansions. Figure 4.13 illustrates the charger utilization by income class, showing how much energy is charged by charger type and income class in years 2030, 2040, and 2050.

The following observations are the most relevant here:

- With an increase of the value for the parameter $n^{\text{points per site}}$, higher investments in public charging infrastructure are made earlier. In the scenario $n^{\text{points per site}} = 10$, public charging infrastructure capacity expands more rapidly in the investment periods 2030-2040 compared to the scenario $n^{\text{points per site}} = 2$ (*Distributed expansion*). In the scenario $n^{\text{points per site}} = 40$ (*Concentrated expansion*), the capacity of the public charging network in 2035 equals that of the 2040 level under lower values of the $n^{\text{points per site}}$ parameter (see Figure 4.14). In scenario $n^{\text{points per site}} = 2$ (*Distributed expansion*), substantial increases in the public charging infrastructure network occur during later years, 2040 and 2045. In all scenarios, these investment periods of substantial expansion coincide with decreases in detouring times for public charging infrastructure (see Figure 4.12).

- The accelerated expansion in the scenario $n^{\text{points per site}} = 10$, the increased scenarios in 2030 coincide with the sharp increase in adoption in the consumer group *Medium low income* (as displayed in Figure 4.10), indicating that the increased charging capacities particularly accelerate the electrification of the passenger car fleet owned by consumers of medium level income.

- Home and work charging capacities are consistent across all scenarios. While work charging is expanded early on to its maximum possible capacity values, home charging capacities are not further expanded, though they are used consistently by the consumer groups *Very high income* and *High income* as shown in Figure 4.13.

- The utilization rates of public charging infrastructure diverge substantially between the slow and fast charging types. For both charger types, maximum utilization — indicated by the red dashed line in graphs of the right column in Figure 4.14 — is not reached. In particular, the utilization rate of fast charging is very low (maxima are at 6-11%), indicating substantial investments in overcapacities.

- While, in all scenarios, the absolute numbers of BEVs and total energy consumption are similar by 2050, installed charging capacities and their utilization by income groups are different. Looking at the right column in Figure 4.13, there is a significant difference between the charged energy by income class and type between the scenario of $n^{\text{points per site}} = 2$ and the other two with charging sites of higher concentration. Particularly, there is a significant difference in the usage of fast charging by the consumer groups *Very low income* and *Low income*[4]. In the *Distributed expansion* scenario, higher amounts of public charging are available and used, while, in the *Very low income* group, fast charging is the dominating charging technology, leading to higher total utilization of fast charging.

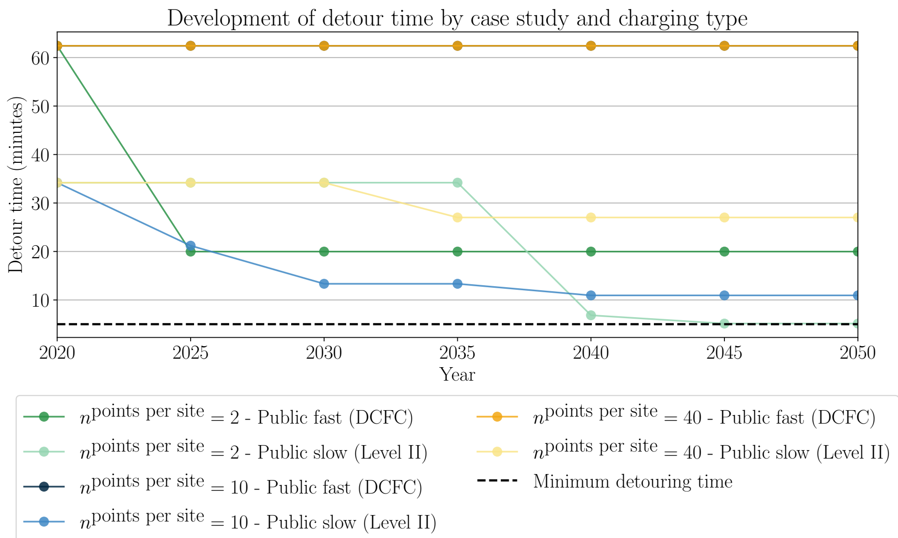

Figure 4.12.: Development of average detouring times for public fast and slow charging in Basque Country for different scenarios of spatial densification.

[4] We want to emphasize here for the interpretation of the results that we do not consider the different *pricing* of charging types here.

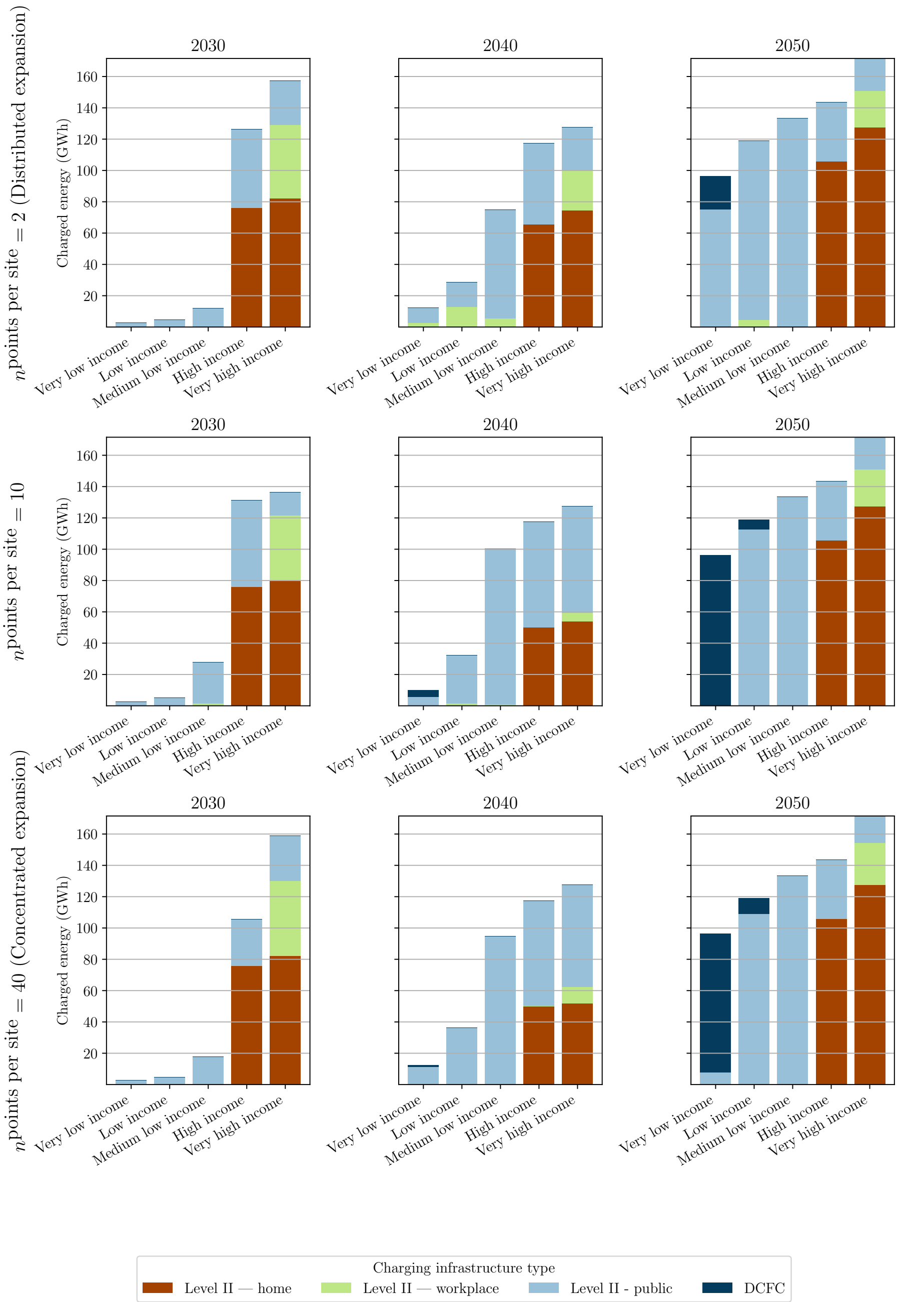

Figure 4.13.: Charged energy in 2030, 2040 and 2050 at different infrastructures

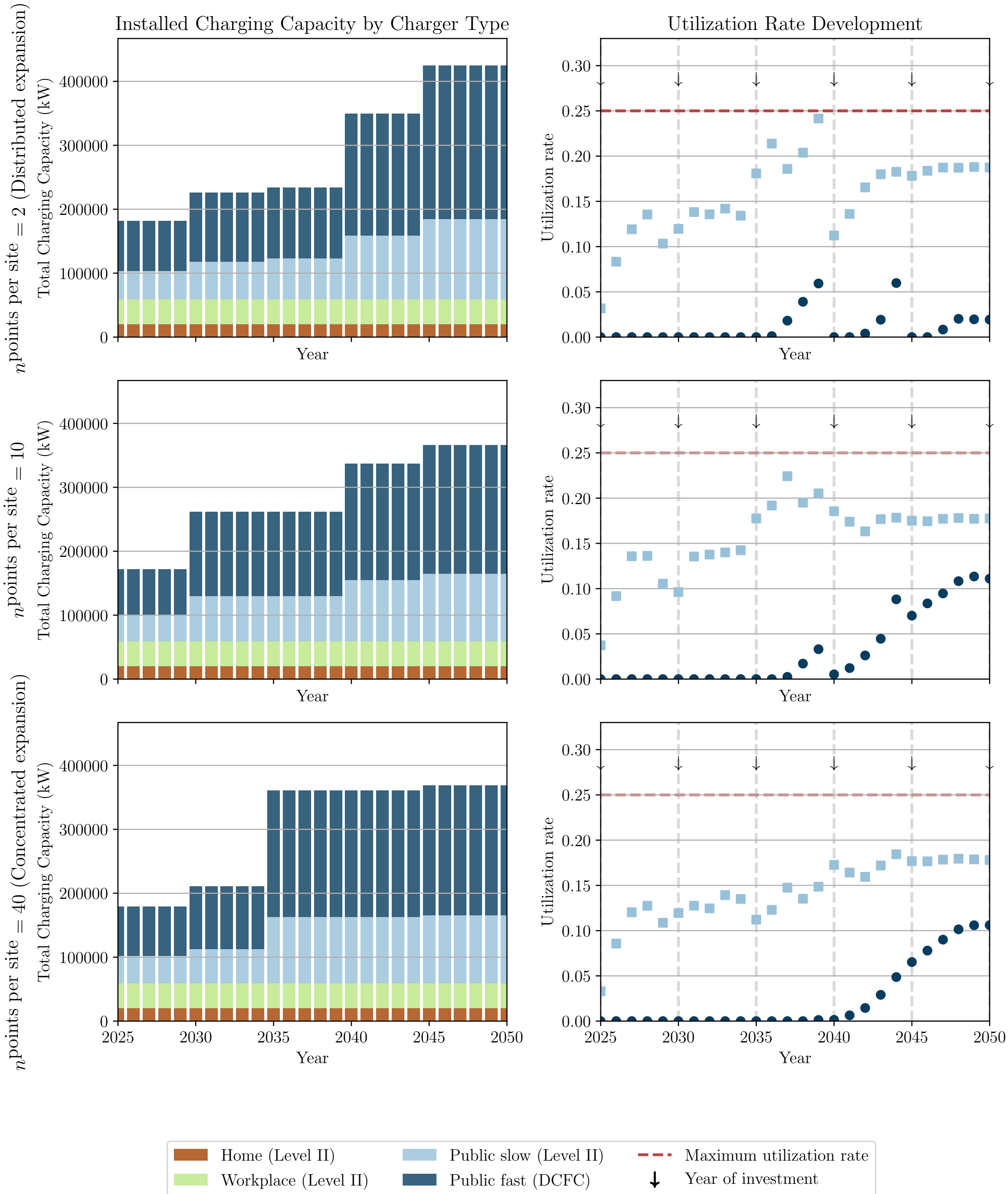

Figure 4.14.: Charging infrastructure expansion for Basque Country region in the scenario of *Concentrated expansion* ($n^{\text{points per site}} = 40$).

4.2.4. Spill-over effect of charging infrastructure planning for destination charging

Figure 4.15 displays the development of the adoption of BEVs by years 2030 and 2040. In particular, this is shown for the NUTS-3 regions Bizkaia, Araba, and Gipuzkoa, under the different scenarios in the speed of charging infrastructure expansion. In the *Baseline Roll-out* case, in 2030, the range of the electrification rate varies by region within 2.6-6.0%. By 2040, the region-dependent range becomes wider, 35.5-42.8%, with Araba having the highest rate of electrification and Bizkaia the lowest. With increased charging capacities built in Bizkaia (case *Central Acc. Roll-out*), the electrification share in Bizkaia increases slightly (0.9 % by 2030 and 3.3% by 2040), while the electrification rates of Araba and Gipuzkoa are also positively affected (increase by 0.6-1.3%). When the charging infrastructure capacity in Araba and Gipuzkoa (scenarios *Neighbor Slowed Roll-out* and *Central Acc. + Neighbor Slowed Roll-out*) are reduced, the values for the share of electrification in Araba and Gipuzkoa in 2040 are significantly lower than in *Baseline Roll-out*. Figure 4.16 displays the temporal development of the difference in electric vehicle numbers between the*Central Acc. Roll-out* and the *Baseline Roll-out*. A peak in the positive absolute differences occurs in 2045. The maximum relative difference is in 2044 and is 2.1% for Araba and 1.1% for Gipuzkoa. For some time periods, we observe negative spillover. In particular, there is a distinctive decrease in the BEV numbers in 2049.

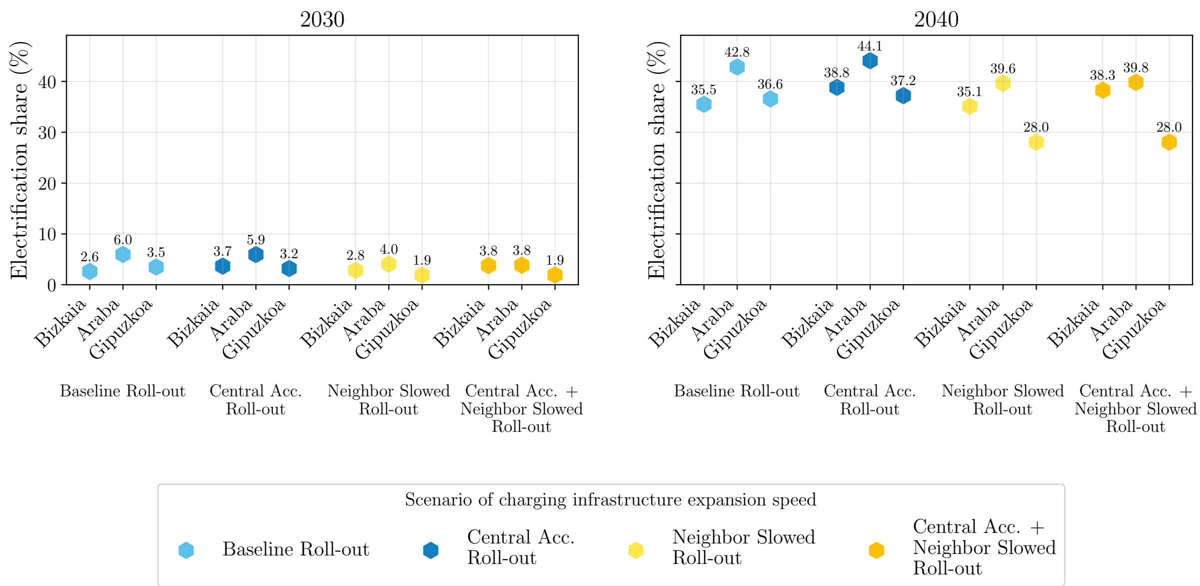

Figure 4.15.: Share of electrification of the passenger car fleet in the regions Bizkaia, Araba, and Gipuzkoa under different scenarios referring to the speed of charging infrastructure roll-out, for years 2030 and 2040.

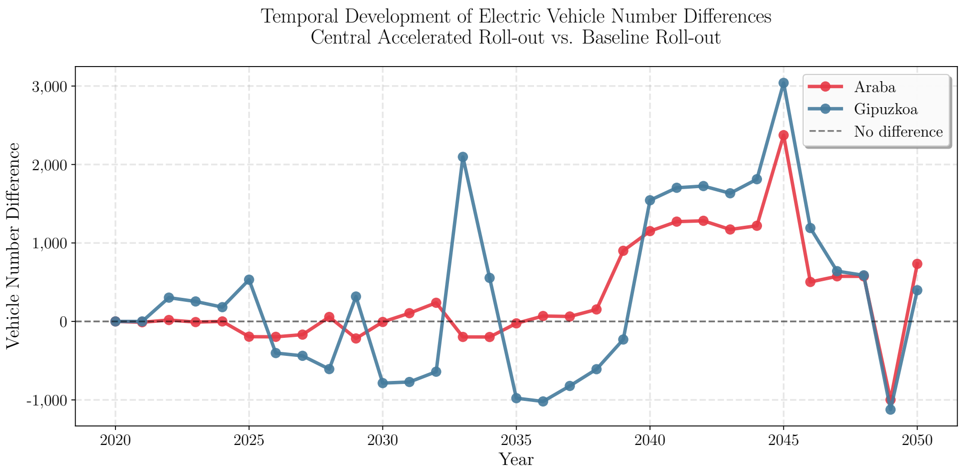

Figure 4.16.: Temporal development of the difference of adopted battery electric vehicles numbers in Araba and Gipuzkoa between the *Basic Roll-out* and *Central Acc. Roll-out.*

With higher expansion in Bizkaia in *Central Acc. + Neighbor Slowed Roll-out*, the electrification share in Bizkaia is increased by 2.1% by 2040. The BEV adoption shares in Araba and Gipuzkoa are here impacted significantly less, i.e., than in the reference capacity expansion. In 2030, a slight negative impact is observed.

Figure 4.17 gives an insight into the effect of increased charging capacity in Bizkaia on the electrification of the commuting traffic. The plots display the absolute numbers of BEVs used for commuting trips originating in Araba or Gipuzkoa and traveling to Bizkaia. We zoom in on three years: 2030, 2040, and 2050. The text in blue indicates the relative differences to the *Baseline Roll-out* scenario. The gray text indicates the relative difference to *Neighbor Slowed Roll-out.* The following four key observations are made here:

- The slowed roll-out in Araba and Gipuzkoa has little impact on the BEV adoption in the commuting fleet. Some negative impact is observed during 2040-2050 which is partly compensated with increased capacities in Bizkaia.
- In Araba, the electrification of the commuting fleet is mainly positively accelerated during the years 2030 and 2040 by the increased charging infrastructure deployment in Bizkaia. By 2050, this positive impact is limited to +0.6%.
- In Gipuzkoa, the impact of increased infrastructure expansion in Bizkaia remains not significant.
- In the case of reduction of charging infrastructure in Araba and Gipuzkoa, the increased charging infrastructure in Bizkaia does not fully compensate for the reduced capacities.

In Figure 4.18, we examine the allocation of charging processes for commuting trips between the origin and destination regions. We generally observe a high share of destination charging across all cases, years, and both regions. These shares increase under the accelerated rollout in Bizkaia. One exception is the development in Gipuzkoa between 2030 and 2040, during which there is a 43% increase in charging processes at the origin (between the Central Acc. Roll-out and Baseline Roll-out scenarios).

Figure 4.19 displays utilization rates for the public charging infrastructure in the Baseline Roll-out across all three regions. Here again, utilization rates of public slow charging are consistently higher than those of public fast charging. Overall, utilization rates are significantly higher when spatial density is not considered.

TODO: acknowledge the co-optimization with the Commercial sector in the appendix

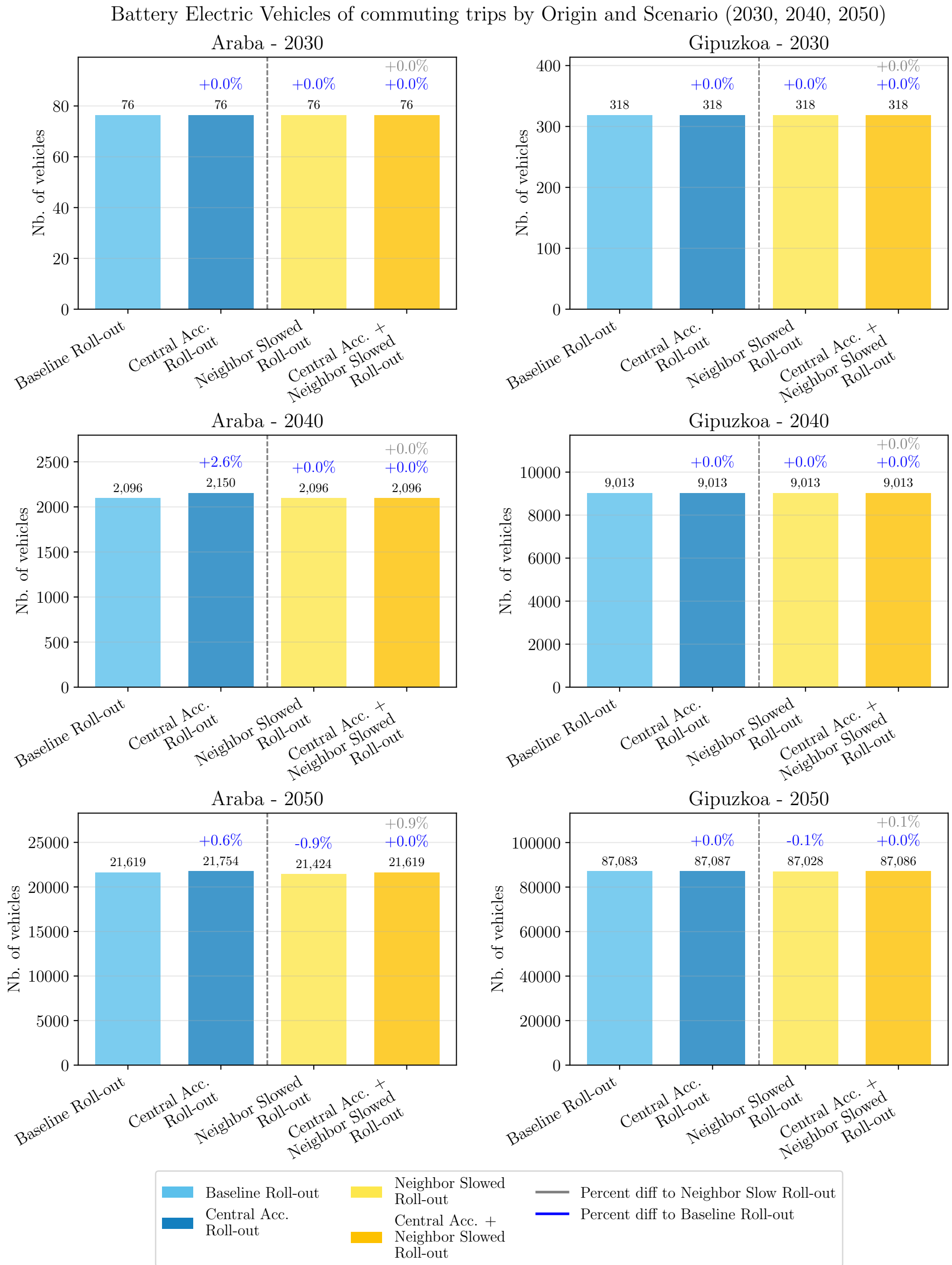

Figure 4.17.: Numbers of BEVs in the commuting trips between Araba or Gipuzkoa and Bizkaia under different scenarios of charging infrastructure roll-out scenarios for years 2030, 2040 and 2050.

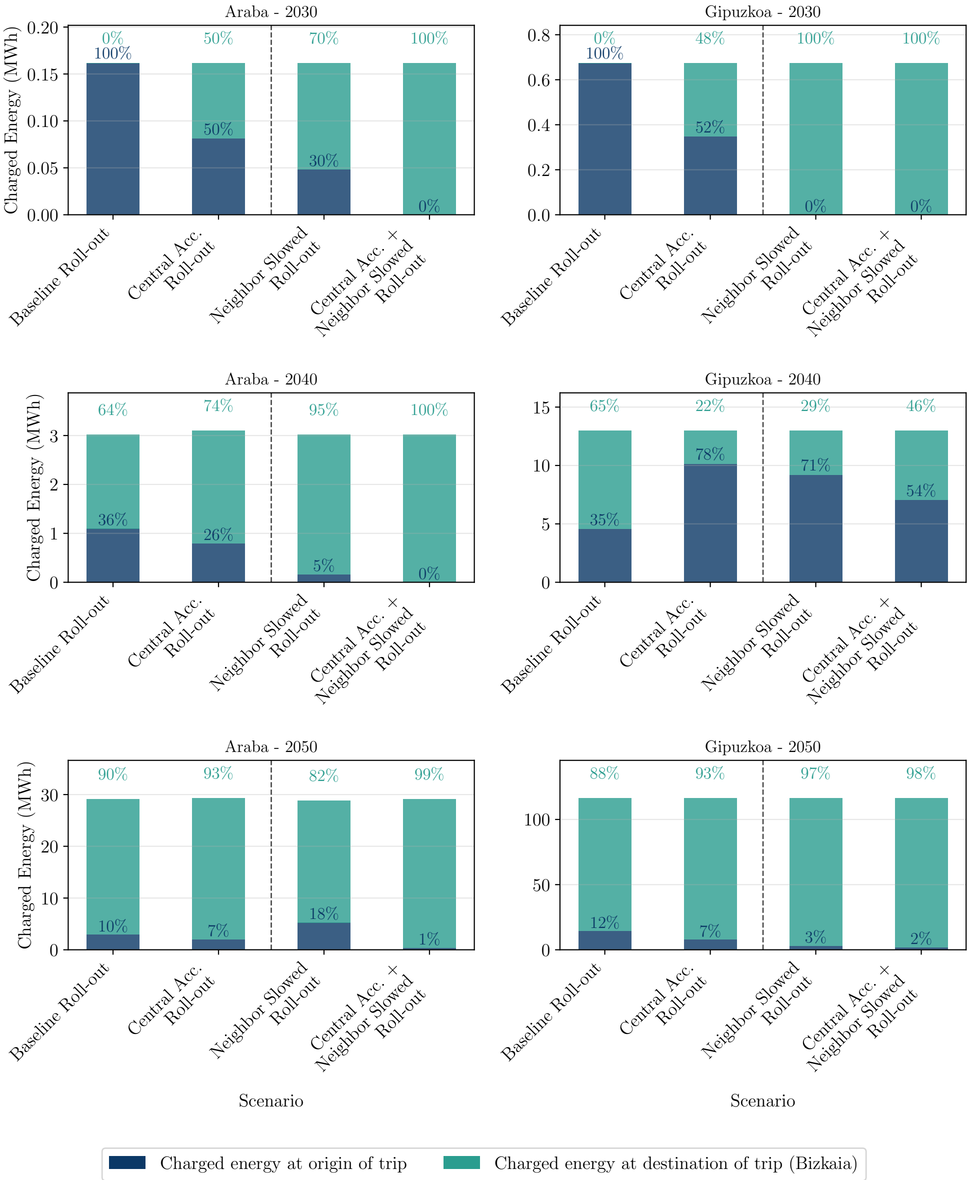

Figure 4.18.: Snapshots for 2030, 2040 and 2050: Allocation of charging processes between origin and destination of commuting trips originating in Araba or Gipuzkoa and going to Bizkaia with text indicating relative percentages of charging processes.

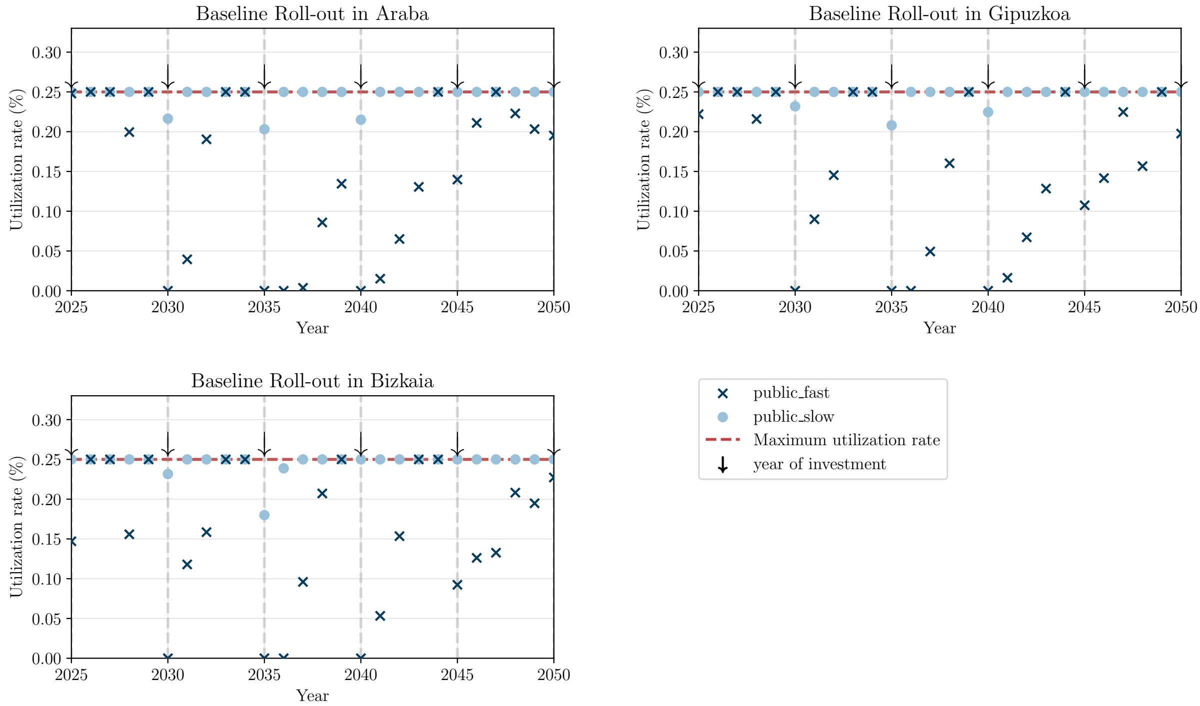

Figure 4.19.: Development of utilization rates for public charging infrastructure network in Bizkaia, Araba and Gipuzkoa.

4.3. Charging infrastructure for long-haul battery-electric trucks among the Scandinavian-Mediterranean corridor in the TEN-T (RQ3)

4.3.1. Case study: Road freight along Scandinavian-Mediterranean corridor

4.3.2. Cost-optimal adoption of battery-electric trucks and charging activity

4.3.3. Role of modal shift towards rail freight

4.4. Hourly charging demand profiles of the commercial fleet in Austria in 2040 (RQ4)

4.4.1. Case study: Austria's commercial fleet at Federal State level

The representation of the commercial fleet is focused on three fleet types: LDT, HDT, and buses. These are chosen based on their substantial fleet size and transport activity. Their electrification is therefore expected to play a significant role in the electricity system. Based on the mobility patterns of the different fleet types and expected charging strategies, future charging profiles are generated together with plug-in times and installed charging power. The charging profiles are generated for an average week in 2040, i.e. representing seven days in hourly resolution. The theoretical framework presented in Section **??** is the starting point for the estimation of future charging profiles. In the geographic aggregation of charging profiles, Austria is represented by nine nodes ($|\mathcal{N}_1| = 9$) mirroring the nine federal states of Austria.

Light-duty trucks TheLDT fleet classified as vehicles that have a maximum total weight of 3.5 tons [172]. The Austrian vehicle fleet of this type amounted to a size of around 510,000 vehicles by the end of 2023 and is expected to be 535,000 by 2040 in decarbonization scenarios by Angelini et al. [173]. According to [174], the average daily range of a light-duty vehicle operated in Germany is 70km on average. We adopted this value for Austria. Further assumptions for the vehicle are summarized in Table 4.15. Based on this, we assume the routes of vehicles of this fleet type to only within one Federal State and, therefore, the charging profile of the LDT fleet is modeled based on the operational pattern of *local transport*.

The core assumptions for the modeling of a charging profile of light-duty vehicles are the following:

- **Charging strategy:** Vehicles are charged during nighttime and weekends. Recharging is done at depots. During weekends, the vehicles have a substantially higher flexibility in the recharging processes. The charging process is performed at the lowest possible power level and is evenly distributed during the plug-in period. Departure times are evenly distributed between 7 am and 9 am, and arrival times are between 16 pm and 18 pm.

- **Geographic scaling:** The expected amount ofLDT is distributed proportionally to the population size.

- **Sizing of charging infrastructure:** The aggregated charging infrastructure for all depots within a federal state region is sized based on assuming that the charging power of 11kW and charging the vehicle every sixth night is sufficient.

Heavy-duty truck TheHDT has a total weight greater than 3.5 tons. Technical assumptions for this fleet are summarized in Table 4.16. Origin-destination flow data on the European level is used here to capture charging demand from national as well as transit transport. The ETISplus dataset provides estimations of the annual tons transported by trucks in the year 2030 [175]. We apply the charging profile generation for *interregional transport* here under the following considerations:

- **Charging strategy:** For this we consider that the European regulation dictates a maximum driving shift of 4.5 hours.HDTs are charged at depots and highway service areas [176]. Further, in Austria, a regulation prohibits driving between 10 pm and 5 am for multiple transport segments [177]. During Saturdays and Sundays, the operation of heavy-duty transport is also restricted. For trips that take longer than 4.5 hours, the 45-minute break takes place at a highway service area, and during this time, theHDT is connected to a megawatt charging station and is fully charging. The remaining energy demand is covered at the origin and the destination depots during nighttime. For trips shorter than 4.5 hours, charging is fully compensated at depots. Recharging happens also at depots during the day if the driving range is not sufficient to be covered with one charged battery. During trips that take multiple days, charging demand is also compensated at service areas during nighttime.

- **Sizing of charging infrastructure:** Charging in depots is assumed to be conducted at 100kW of peak power and megawatt charging is available at charging stations. As a reference point for the sizing of charging infrastructure, the goals for the installment of 5MW of grid connection at each charging station until 2035/2040 along the Austrian highway network by the highway infrastructure operator ASFINAG [178][5]. This is assumed to be available for the electrification rate of a heavy-duty fleet of 30%. A relative factor to the charging demand is deducted for each region based on the amount of existing service areas and applied to all charging location demands within the respective region.

- **Flexibility in the fast-charging along highway service areas:** Due to the short time

[5]The exact number of 5MW was stated by a representative of ASFINAG in the course of a stakeholder interview.

Table 4.15.: Parameter settings for modeling the charging demand of light-duty vehicles

Parameter	Value	Reference
Fleet size (Austria 2040)	535,000	[173]
Battery capacity	145 kWh	[179]
Specific energy consumption	27 kWh/100km	[173]
Daily range	70 km/day	[174]
Charging power at charging station depot	11 kW	[174]

frame of the break of 45 minutes and the used temporal resolution of the modeling being one hour, there is no flexibility given for highway charging processes. It may be foreseeable that the operational times and the timing of the break and, therefore, the charging process would be adapted to charge for lower expenses in the future. Therefore, to introduce some flexibility here, we created three distinct plug-in time frames of two hours each, which are allocated around noon.

Busses For the bus fleet, we consider the application case for public transport in both cities and rural regions. Similar to the modeling of light-duty vehicles, the routes of buses do not go through multiple regions. Table 4.17 comprises technical assumptions for an average bus in public transport.

The charging profiles for busses are estimated based on the following:

- **Charging strategy:** Most charging demand is covered during night time at the bus depot. If required, busses enter the depot for fast charging during daytime. The operation is assumed to be similar during each day of the week.

- **Geographic scaling:** City busses are considered for Austrian cities that surpass the population size of 100,000. For each city, the size of the bus fleet is scaled using the population size. The bus fleet in rural areas is sized using the total area of the rural region within a federal state.

- **Sizing of charging infrastructure:** During night time, each bus is connected to a 100kW charging station. The power connection of the depot is limited to two-thirds of the summed peak power of the charging stations which is sufficient to cover the charging demand during down times at night and is in line with the peak power achieved with coordinated and, therefore, cost-efficient charging [90].

4.4.1.1. Scenarios and case studies

In the application to the Austrian electricity system in 2040, we explore two different dimensions: On the one side, the variation in the share of electrification of the commercial fleet in 2040, and,

Table 4.16.: Parameter settings for modeling of charging profile of heavy-duty vehicles

Parameter	**Value**	**Reference**
Battery capacity	350 kWh	[180]
Specific energy consumption	1.3 kWh/km	[181]
Average load	10.63 tons	[181]
Average driving speed	50 km/h	[182]
Peak charging power at highway/depot charging station (fast charging during daytime)	1000 kW	
Peak charging power at highway/depot charging station (slow charging during nighttime)	100 kW	

Table 4.17.: Parameter settings for the modeling of charging profile by busses in the application in the city and rural areas

Parameter	**Value**	**Reference**
Fleet size (Austria 2040)	12,900	extrapolated based on [183], [184]
Battery capacity	350 kWh	[185]
Specific energy consumption	1.3 kWh/km	[181]
Daily range - city	236 km	[183]
Daily range - rural	138 km	[184]
Peak charging power at depot charging station (fast charging during daytime)	1000 kW	
Peak charging power at depot charging station (slow charging during nighttime)	100 kW	

Table 4.18.: Electrification scenarios considered in the analysis

	Share of electrification in fleet		
Scenario	Light-duty vehicles	Heavy-duty vehicles	Busses
LOW	100%	30%	100%
MEDIUM	100%	50%	100%
HIGH	100%	100%	100%

on the other side, the role of the commercial BEV fleet in the electricity market. Table 4.18 displays the shares of electrification for the three considered electrification scenarios, *LOW*, *MEDIUM*, and *HIGH*. The three case studies for market participation describe:

4.4.1.2. Electricity market 2040

All 13 countries (Austria, Germany, Netherlands, Belgium, Luxembourg, Czech Republic, Slovenia, Switzerland, Poland, Slovakia, Hungary, Italy, and France) within the optimized area are expected to fulfill the European national energy and climate plans [186] by 2030 and continue to compensate for the rising electricity demand by expanding generation capacities in 2040. According to Austria's national energy and climate plans, RESE compensates for 100% of the yearly electricity demand. The used *National Trends* dataset, provided by the European Network of Transmission System Operators for Electricity is based on the fulfillment of these plans [187]. It includes a power plant fleet with RESE, fossil fuel-fired power plants (coal, oil, lignite, and combined-cycle gas turbines, and other types of thermal power plants that do not emit CO_2 (nuclear, biogas, biomass, and fuel cells).

- **Baseline:** The charging demand of the commercial BEV fleet is inflexible.
- **Dispatch:** The flexibility of the charging of the commercial BEV fleet is optimized in the dispatch.
- **Redispatch:** The flexibility of the commercial BEV fleet charging is utilized in redispatch measures.

4.4.2. Spatio-temporal demand distribution

Table 4.19 summarizes the total values for annual load estimates for the commercialBEV fleet under different electrification scenarios in Austria for 2040. The annual charging demands are 7.4 TWh / 9.6 TWh / 15.2 TWh. This makes up for 8 % / 11 % / 16.9% of the estimated total electricity load in 2040. The highest charging load stems from the charging of heavy-duty trucks (HDT) which is mostly allocated to depot locations. In the electrification scenarios *MEDIUM* and *HIGH* which indicate a respective electrification rate of 50% and 100% of the HDT fleet, the fleet is the clear dominant demand segment with 5.6 TWh/ 11.2 TWh of annual charging demand.

Figure 4.20 displays peak power levels along the highway for the *HIGH* scenario. Positions of service areas along the Austrian highway network are shown together with the coloring indicating approximated peak power levels. The peak power level at a service area is deducted from the estimated accumulated load within the Federal state which is then equally distributed among all existing service areas. It is important to note here that this image is an estimation of the peak power distribution aimed to illustrate geographical variations and an estimate for peak power levels at service areas in 2040. The figure indicates a wide range in required power levels at highway service areas, between 7 and 29 MW. The illustration suggests distinctly higher power levels at the service areas located in the Northeast than in the Western regions of Austria.

The accumulated demand profile for the charging of the commercial BEV fleet is visible in Figure 4.21 for the whole of Austria. The figure displays an average week of charging load. During daytime time, most charging at highway service areas by the HDT fleet around noon is conducted. The load peaks at night time around midnight due to accumulated night charging of the LDT fleet, city, and regional busses, and the HDT fleet. During weekend days, the charging demand is substantially lower than during work days as vehicle operation is mostly performed during workdays. Fleet operators of vehicles that are not operated during the weekend recharge the vehicle for the next operation between Friday night and Monday morning and have, therefore, also higher flexibility in the recharging process while still having similar charging infrastructure as used during a workday available.

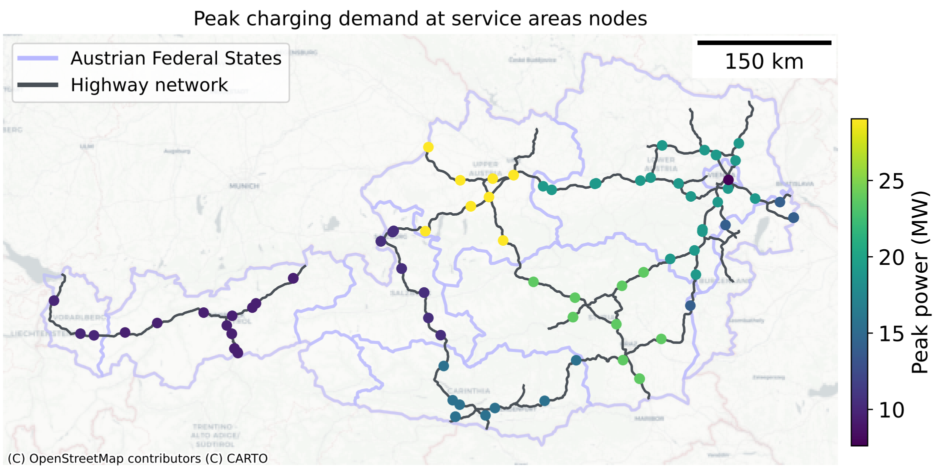

Figure 4.20.: Approximation of charging peaks at Austrian resting areas for an 100% electrified commercial vehicle fleet (positions retrieved from [188])

Table 4.19.: Annual charging demand of three different considered fleet types under different assumptions on electrification rate for the commercial vehicle fleet in Austria in 2040.

	Total annual demand of fleet types 2040 (TWh)						
	Total	LDV	Bus	HDV			
				Charging location			
Scenario				Depot	Highway (day time)	Highway (night time)	Total
LOW	**7.4**	2.7	1.3	1.8	1.1	0.5	3.4
MEDIUM	**9.6**	2.7	1.3	3.1	1.7	0.8	5.6
HIGH	**15.2**	2.7	1.3	6.2	3.5	1.5	11.2

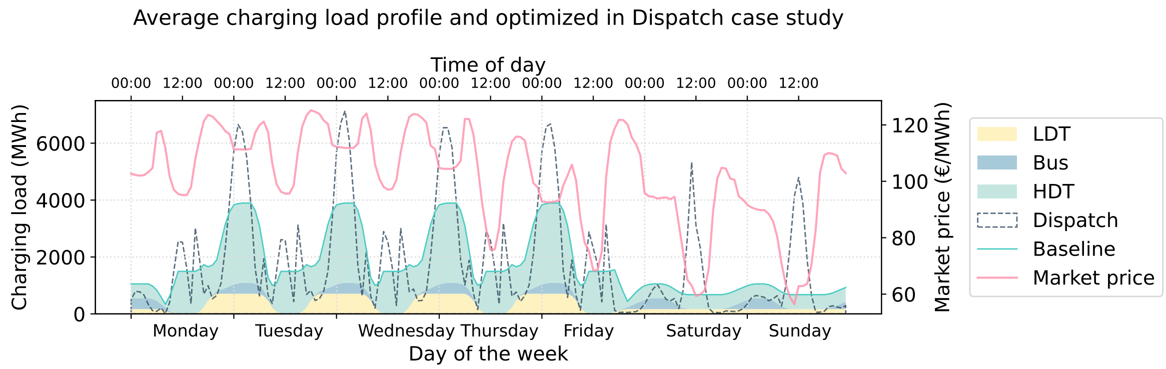

Figure 4.21.: Aggregated charging demand profile of the commercialBEV fleet in the *Baseline* and *Dispatch* case study with the average day-ahead market price.

4.4.2.1. Composition of heavy-duty truck charging demand by traffic type

The traffic-flow-based estimation of heavy-duty truck charging demand enables a decomposition by the type of traffic generating the demand within Austria. Table 4.20 classifies the origin-destination flows from the ETISplus dataset into three categories: inland transport between Austrian regions, transit flows that neither originate nor terminate in Austria, and import/export flows with one end in Austria. Of the 319,447 origin-destination pairs considered, the vast majority (239,209) are transit flows. The total transport volume amounts to 529.5 Mio. tons, of which transit and import/export contribute similar shares (175.4 and 185.0 Mio. tons, respectively). In terms of ton-kilometers traveled within Austria, transit traffic accounts for 32.6 Mrd. tkm (37.4%), import/export for 32.2 Mrd. tkm (36.9%), and inland transport for 22.5 Mrd. tkm (25.7%). The resulting daily charging demand within Austria follows this distribution, with transit traffic generating 15.3 GWh/d and import/export 15.1 GWh/d, together accounting for 74.3% of the total daily demand of 40.9 GWh/d.

Table 4.21 presents the annual charging demand for the *HIGH* scenario (100% HDT electrification), disaggregated by traffic type and charging location. The total annual HDT charging demand of 11.24 TWh is predominantly covered at depot locations during nighttime (10.97 TWh),

Table 4.20.: Classification of heavy-duty truck charging demand in Austria by traffic type, based on ETIS+ 2030 transport flow projections. AT tkm refers to ton-kilometers traveled within Austria. Daily demand represents electricity required for charging within Austria.

Traffic Type	**Flows** [-]	**Transport Volume** [Mio. t]	**AT tkm** [Mrd. tkm]	**Daily Demand** [GWh/d]	**Share** [%]
Inland (AT $\rightarrow$ AT)	1,160	169.0	22.5	10.5	25.7
Transit (non-AT $\rightarrow$ non-AT)	239,209	175.4	32.6	15.3	37.4
Import/Export (AT $\leftrightarrow$ non-AT)	79,078	185.0	32.2	15.1	36.9
Total	**319,447**	**529.5**	**87.3**	**40.9**	**100.0**

while highway charging accounts for 0.27 TWh. Highway charging is concentrated in transit and import/export traffic (0.12 TWh each), whereas inland transport generates negligible highway charging demand (0.02 TWh). This reflects the shorter route segments of inland transport within Austria, for which depot charging at origin and destination is sufficient without en-route stops.

Table 4.21.: Annual heavy-duty truck charging demand in Austria by traffic type under 100% electrification. Annual values are calculated with 261 workdays and 95% charging efficiency.

Traffic Type	**Highway** [TWh/a]	**Depot Night** [TWh/a]	**Total** [TWh/a]	**Share** [%]
Inland (AT $\rightarrow$ AT)	0.02	2.87	2.89	25.7
Transit (non-AT $\rightarrow$ non-AT)	0.12	4.08	4.20	37.4
Import/Export (AT $\leftrightarrow$ non-AT)	0.12	4.03	4.15	36.9
Total	**0.27**	**10.97**	**11.24**	**100.0**

Table 4.22 summarizes the relationship between transport volume, ton-kilometers within Austria, and resulting annual charging demand by traffic type. Despite contributing the lowest transport volume (169.0 Mio. t), inland traffic generates 2.9 TWh of annual demand due to shorter distances traveled entirely within the country. Transit traffic, with 175.4 Mio. t and the highest share of Austrian ton-kilometers (32.6 Mrd. tkm), results in 4.2 TWh. Import/export flows generate a comparable 4.1 TWh from 185.0 Mio. t.

Table 4.22.: Heavy-duty truck charging demand in Austria by traffic type: transport volume, ton-kilometers within Austria, and resulting annual charging demand under 100% electrification.

Traffic Type	**Transport [Mio. t]**	**AT tkm [Mrd. tkm]**	**Demand [TWh/a]**	**Share [%]**
Inland (AT $\rightarrow$ AT)	169.0	22.5	2.9	25.7
Transit	175.4	32.6	4.2	37.4
Import/Export	185.0	32.2	4.1	36.9
Total	**529.5**	**87.3**	**11.2**	**100.0**

4.4.2.2. Regional charging demand profiles

Figure 4.22 displays the weekly charging demand profiles disaggregated by Austrian federal state. The profiles reveal substantial regional differences in both the magnitude and temporal shape of charging demand. Regions with high shares of transit and long-haul truck traffic exhibit higher daytime peaks due to en-route highway charging, while regions with predominantly local commercial vehicle fleets show demand profiles dominated by nighttime depot charging. The weekday-weekend contrast is visible across all regions, with weekend charging demand being substantially lower. The relative magnitude of the charging demand varies across regions, reflecting the heterogeneous distribution of commercial vehicle fleets and transit traffic volumes across Austria.

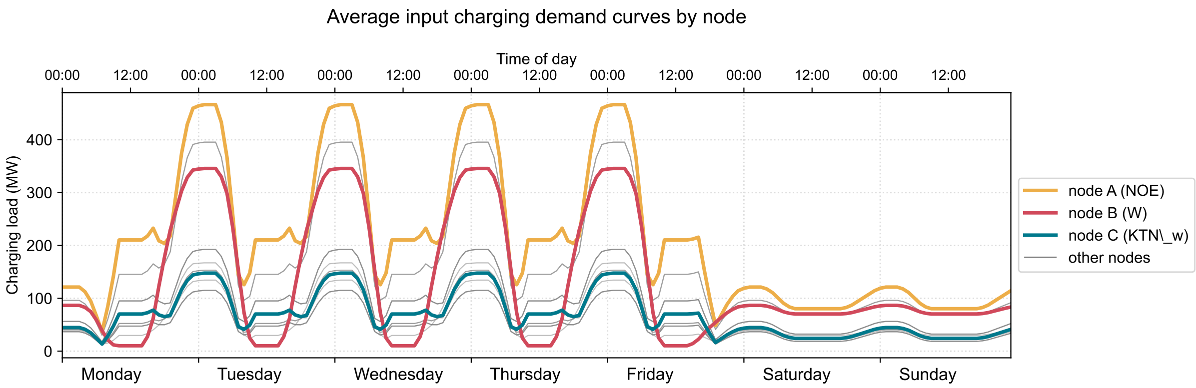

Figure 4.22.: Weekly charging demand profiles of the commercial BEV fleet by Austrian federal state.

Figure 4.23 shows the composition of the weekly charging demand by fleet type for three selected locations in Austria. The stacked profiles illustrate the relative contributions of light-duty trucks, heavy-duty trucks, and buses to the total charging load at each location. The composition varies across the selected regions: locations along major transit corridors show a higher share of heavy-duty truck charging, whereas regions with large urban centers exhibit a more balanced composition due to the presence of bus fleets and light-duty commercial vehicles. The temporal patterns also differ by fleet type, with bus charging showing a more uniform daily pattern compared to the pronounced weekday peaks of truck charging.

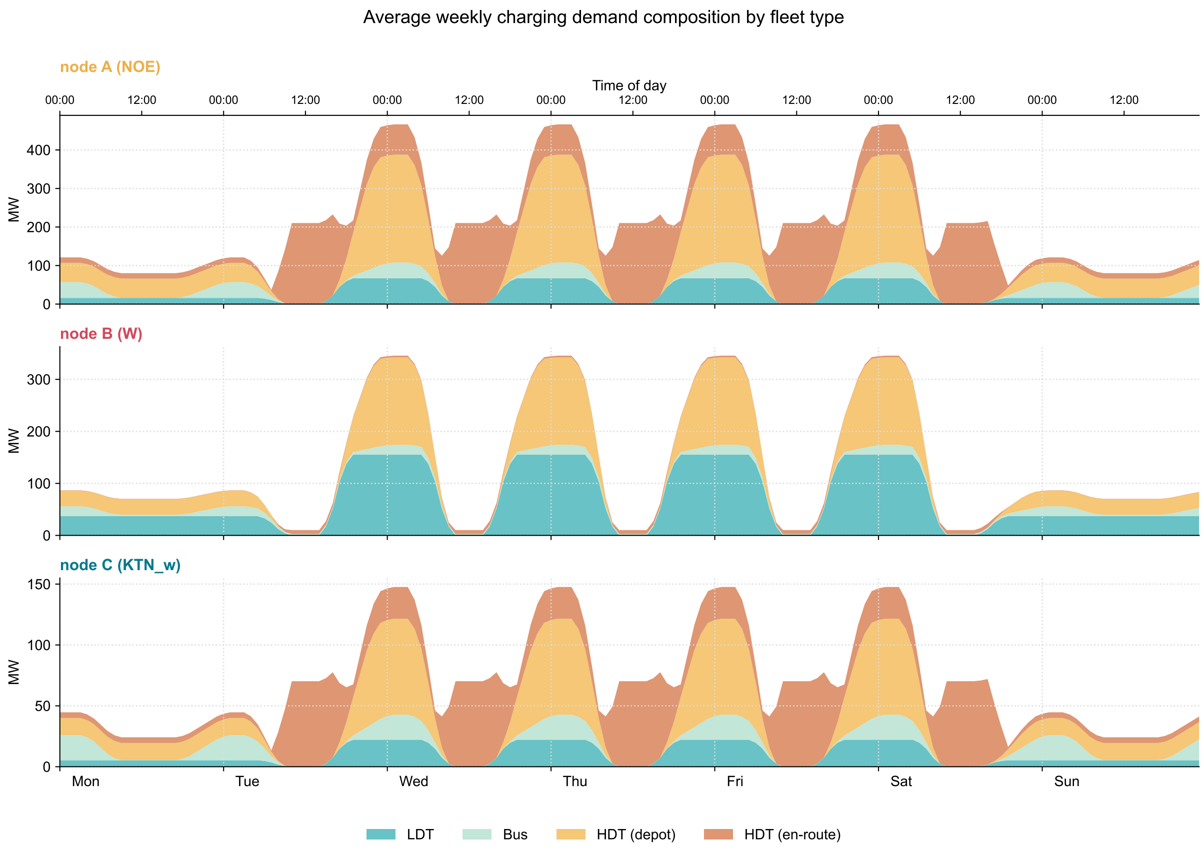

Figure 4.23.: Composition of the weekly charging demand by fleet type (light-duty trucks, heavy-duty trucks, buses) for three selected locations in Austria.

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5. Discussion & Synthesis of Results

This chapter discusses and synthesizes the findings of the four contributions presented in this thesis. Section 5.1 contextualizes the results of each research question within the existing literature, comparing estimates against policy targets and prior studies, and discussing the limitations of the applied methodologies. Section 5.2 synthesizes the findings across contributions, drawing out cross-cutting insights. Parts of this discussion have been adopted, reformulated, and extended from the respective journal publications.

5.1. Contextualizing findings in existing literature

5.1.1. Research question 1

RQ1: *What are the required fast-charging capacities along Austria's highway network for battery-electric passenger cars in 2030 under different decarbonization scenarios, and where are these allocated?*

Planned capacities Along the Austrian highway network, the results indicate 160–240 MW of fast-charger capacities distributed across 54–55 stations to serve passenger BEVs traveling in 2030.

The required values for the total capacity vary by decarbonization scenario. The following discussion first contextualizes the results within policy-based expansion goals for charging infrastructure, particularly the Alternative Fuels Infrastructure Regulation (AFIR), and the estimates from the study by Jochem et al. [99]. Subsequently, the discussion focuses on influential parameters of the decarbonization scenarios.

Table 5.1 provides a quantitative comparison of the results. Since the studies differ in their assumptions regarding passenger car fleet development, the adjustments applied to ensure comparability are summarized in the column "Key Assumptions".

Comparison with AFIR In the AFIR, there are two charger-capacity related specifications that are relevant for the present discussion. The first specifies required capacities along highways until 2027; the second defines national targets for public charging infrastructure.

Regarding the first specification, the AFIR mandates 600 kW of charging capacity every 60 km for light-duty vehicles along the European TEN-T core network by 2027 [11]. Given a 2,258 km highway network in Austria, this translates to a minimum requirement of 22.6 MW. This is substantially lower than the estimated range of 160–240 MW.

Regarding the national targets, the AFIR mandates that public charging infrastructure is continuously expanded to at least 1.3 kW of power output per BEV [11]. Since the AFIR establishes minimum deployment requirements, the resulting capacity targets represent a policy-mandated lower bound. Assuming 30% electrification of Austria's current passenger car fleet of 5.2 million vehicles, this translates to approximately 2 GW of required public charging infrastructure. However, only a small share of public charging is highway-based fast charging. Drawing on Norwegian deployment data[1], highway charging capacity is estimated to account for roughly 210 MW. That the optimization-based estimates of 160–240 MW are in the same order of magnitude as, and partially exceed, this policy-derived lower bound is consistent with the expectation that cost-optimal infrastructure sizing meets or surpasses minimum regulatory requirements.

Comparison with previous research To the best of the author's knowledge, the study by Jochem et al. [99] is, alongside Golab et al. [17], the only prior work that estimates concrete locations and capacities for fast chargers along Austria's highway network. It is important to note here that the focus of the study by Jochem et al. [99] is not specifically on Austrian highway charging but rather on European highway charging. Under 15% BEV electrification, Jochem et al. [99] identify 15 charging sites with approximately eight 150 kW points each (18 MW total, scaling to 36 MW at 30% electrification)—roughly one-tenth of the capacity estimated by the present approach. Several factors explain this divergence.

First, the two studies differ fundamentally in their modeling approach. Jochem et al. [99] employ a Flow Refueling Location Model, which minimizes the number of stations required to cover origin-destination flows, with station capacity determined ex post through queuing models. The proposed approach, by contrast, jointly optimizes both location and capacity within a single cost-minimization framework. This integrated formulation allows infrastructure costs to enter the objective directly, yielding capacity estimates that are driven by economic trade-offs rather than minimum-coverage requirements. However, this also means that the resulting capacity sizing is highly sensitive to the assumed peak demand, as station dimensions are directly determined by the traffic volumes at each node rather than based on an operational modeling of the station utilization.

Second, the studies differ in how charging demand is derived. Jochem et al. [99] estimate demand from Europe-wide origin-destination flows, applying two filtering thresholds that systematically exclude demand: only O/D pairs with a trip length of at least 100 km are considered, and flows below 5,000 vehicles per year are removed to reduce computational complexity. This excludes shorter highway trips that may still require en-route charging as well as low-volume routes that

[1]Norway is globally an exception for the relative share of fast chargers. According to IEA [189], fast chargers account for 5–6% of public charging points globally. In Norway, highway chargers represent 20% of total charging points but 35% of installed capacity, reflecting a 1.75× capacity-to-points ratio due to higher power ratings. Applying this ratio yields an estimated highway charging capacity of approximately 210 MW (6% × 1.75 × 2,000 MW).

collectively contribute to aggregate demand. The traffic count-based approach applied in this work, by contrast, observes aggregate volumes at each highway link and therefore captures a broader share of charging demand, including shorter-distance and domestic trips that fall below these thresholds. The assumption of a constant share of long-distance traffic along the highway network neglects the heterogeneously distributed long-distance traffic that is particularly captured by origin-destination data. Furthermore, the international scope of Jochem et al. [99] allows demand to be served at charging stations outside Austria's national borders, further reducing the capacity allocated domestically.

The authors themselves, Jochem et al. [99], acknowledge the tendency of their approach to underestimate, by noting that their results for Germany of 128 stations appear "absurdly small" compared to actual deployment of 315 stations from a single operator alone [99, p. 126].

A key advantage of the proposed approach is that it relies on traffic count data that are directly observed at actual service area locations along the highway network, rather than on origin-destination flows for which up-to-date data are difficult to obtain[2]. However, this comes at the cost of information on individual vehicle trajectories: aggregate link volumes do not reveal where a vehicle entered the network, its battery state of charge, or whether it has already charged upstream. Charging demand is therefore inferred from traffic volumes rather than from vehicle-level charging needs, which may lead to spatial misallocation of capacity across the network rather than a systematic over- or underestimation of total demand. The absence of explicit route information remains a crucial limitation relative to origin-destination-based formulations.

Regarding the overall magnitude, the assumed 30% BEV electrification by 2030 is optimistic relative to Austria's current share of 3.8%, suggesting that the total capacity estimates likely represent an upper bound. Under more conservative projections of approximately 15% electrification, required highway charging capacity would be 80–120 MW. The resulting range of 80–240 MW thus brackets realistic capacity requirements between conservative and ambitious electrification scenarios.

Geographic allocation of charging sites The optimized allocation distributes capacity across 54–55 charging sites. This aligns with observed deployment trends: 31 sites are operational at the time of model calibration, expanding to 47 sites at service areas by early 2026 [191], suggesting continued convergence toward the modeled 55-site network. The highest modeled capacity for a single passenger BEV charging site is 4.9 MW, equivalent to approximately 14 charging points at 350 kW each. This capacity level has precedent in Austrian highway infrastructure: one existing service area features 4.8 MW of installed capacity across twelve 400 kW charging points, primarily targeting battery-electric trucks but demonstrating the technical and operational feasibility of multi-megawatt charging sites.

[2]This statement exclusively applies to the Austrian case study for which ASFINAG provides traffic count data for free [190].

The modeled geographic allocation is primarily driven by existing charging site locations, highway network topology, and a uniform maximum capacity constraint applied across all service areas. This uniform constraint does not account for site-specific limitations, such as available parking spaces or grid connection capacity, which may limit deployment at individual locations. A notable spatial pattern emerges: charging capacity concentrates near Austria's borders, particularly at network entry and exit points. This edge effect likely constitutes a modeling artifact arising from the constraint that all charging demand must be satisfied within the Austrian network, thereby neglecting the option for cross-border charging in neighboring countries. The allocation ensures at least one charging site per highway segment.

Drivers in decarbonization scenarios Different scenarios are used to account for uncertainties in the possible developments in the transport sector. Further, sensitivity analyses are conducted on specific parameters. The focus is on uncertainties in the charging demand and technological parameters, reflecting diverging developments in vehicle battery technology, driving range, and charging power.

Investment costs scale nearly proportionally with charging demand. A 31% demand reduction decreases total investment from 100 million EUR to 68 million EUR (32% cost reduction), while a 17% demand reduction yields 18 million EUR savings (18% cost reduction). Higher charging power produces comparable cost benefits: increasing charging power beyond the baseline 350 kW per charging point reduces required capacity similarly to demand reduction or vehicle efficiency improvements, as fewer charging points are needed to serve the same traffic volume. By contrast, increasing driving range has a limited impact on required capacities. This is in line with the observations by Wang et al. [100] and Nugroho et al. [105]. Overall, the two defining parameters for the estimation are, on the one hand, the share of long-distance-traveling passenger BEVs and, on the other hand, charging efficiency.

Table 5.1.: Comparison of Fast-Charging Infrastructure Capacity Estimates for Austrian Highways

Approach	BEV Share	Capacity (MW)	Target Year	Key Assumptions
Golab et al. [17]	30%	**160–240**	2030	54–55 stations, 350kW points, traffic count-based, demand reduction 21%, cost optimization
Golab et al. [17] (adjusted)	15%[a]	**80–120**	2030	Adjusted for more realistic fleet growth trajectory, includes demand reduction
AFIR distance-based	—	**22.6**	2027	600kW per 60km on TEN-T core, 2,258km network, minimum connectivity requirement
AFIR fleet-based	30%	**∼210**	2030	1.3kW per BEV (1.56M BEVs), highway share 35% of total capacity[b], public charging total: 2.03 GW
Jochem et al. (2019)	15%	**18**	2030	15 stations, 8 points/station, 150kW, FRLM approach, O/D flows, excludes flows <5,000 cars/year
Jochem (scaled to 30%)	30%	**36**	2030	Linear scaling from 15% scenario, likely underestimate per authors' own assessment[c]

[a] Accounts for actual electrification trajectory (current: 3.8% in Austria).
[b] Based on IEA (2025) global average of 5–6% highway share of charging points and Norwegian capacity ratio (1.75×).
[c] Authors note: "our results [...] seem absurdly small. Today only a single platform provides already 315 FCS [for Germany vs. modeled 128]" [99, p. 126].

5.1.2. Research question 2

> **RQ2.a:** *To what extent does the spatial density of public charging infrastructure influence battery-electric vehicle adoption across consumers with different income levels?*

Impact of spatial density on BEV adoption across income levels Scenarios that differ in the spatial densification strategy of charging infrastructure rollout lead to distinct solutions for cost-optimal infrastructure expansion, BEV adoption across consumer groups, and the allocation of charging processes to different charger types. The following relation is evident: Higher spatial density of public charging infrastructure leads to higher adoption among lower-income groups.

This aligns with the objectives of existing charging planning tools that prioritize equitable charger distribution at the neighborhood level [118, 119]. The co-optimization of BEV adoption with charging capacities further emphasizes that, beyond enabling uniform charging access across neighborhoods, this equity consideration is also crucial for supporting the adoption of passenger cars across all consumer groups.

Charger utilization Despite facing both longer detour times to reach fast chargers and longer charging durations, cost-optimal solutions indicate BEV adoption in lower-income groups. This occurs because their lower valuation of time makes the time spent on fast charging less costly in monetary terms than for higher-income groups. From the system perspective of meeting passenger car travel demand at minimum cost, BEV adoption by lower-income consumers using fast charging remains economically efficient, even accounting for these time costs. Overall, in the proposed optimization model, the expansion of public fast charging infrastructure is mainly driven by the constraint that ensures the simultaneous expansion of both slow and fast charging capacities, and by the achievable reduction in average detour time, as modeled by the detour fueling time reduction function.

Throughout all scenarios, fast charger utilization only grows significantly after 2040. Realistically, this would lead to very high charging prices to cover installations, maintenance, and operation costs. The pricing and consumer-specific constraints to charging costs are not reflected in the modeling.

It should be noted that the cost-optimal allocation of fast-charger utilization to low-income classes is a system-level outcome that does not account for current market pricing structures. In practice, fast charging prices are significantly higher than slow charging prices per kilowatt-hour. Realizing the modeled allocation would therefore require a shift in pricing to be better aligned with system costs — specifically, lowering fast charging prices and raising slow charging prices — which would also increase the utilization of fast charger capacities.

Concerning the interpretation of the differences in scenarios, the following limitations should be noted. The approach to linking infrastructure capacity with access time operates at the NUTS-2 level, which introduces two important simplifications. First, the modeled regions are treated as spatially homogeneous: charger locations are implicitly distributed uniformly, and road network characteristics and residential patterns are held constant. This abstraction overlooks variation in urban density and ignores the strategic placement of charging infrastructure at trip attractors (shopping centers, workplaces, etc.), which in practice substantially reduces access times without requiring additional detours. Second, the formulation does not capture congestion dynamics at individual charging sites. The model assumes drivers possess perfect information about charger availability and can always access their nearest station without queuing or searching for an available plug. In reality, sites with limited capacity may force drivers to visit multiple locations before finding an open charging point—a phenomenon analyzed by [112]. This effectively assumes zero probability of encountering occupied chargers, implying that drivers never incur additional search time beyond the initial detour.

A further assumption with significant implications for the results is the limited availability of home charging for lower-income groups. In the model, lower-income households are assumed to have restricted access to home chargers, reflecting the empirical observation that these households are more likely to reside in multi-family housing without private parking or dedicated electrical infrastructure [20]. This assumption is deliberately imposed to isolate the role of public charging infrastructure in driving BEV adoption across income classes. However, it simultaneously suppresses adoption in lower-income groups, as it removes the most convenient and cost-effective charging option from their consideration set. In practice, the availability of home charging varies considerably within income groups depending on housing type, tenure, and local infrastructure investments. Relaxing this constraint — for instance, through subsidized installation of shared charging facilities in multi-family buildings or employer-provided workplace charging — likely yields higher BEV adoption numbers among lower-income consumers, as the dependence on public infrastructure for daily charging needs diminishes. The main finding that higher spatial density of public charging supports adoption in lower-income groups is expected to hold even with relaxed home charging constraints, as public infrastructure remains critical for consumers without home charging access and for trips exceeding daily commuting distances.

RQ2.b.: *How does a local rapid charging infrastructure expansion affect the adoption of battery-electric vehicles in neighboring regions with slower charging infrastructure expansion?*

Cross-border effects Enhanced charging deployment in one region, Bizkaia, generates spillover effects in neighbouring regions, Araba and Gipuzkoa, though the magnitude remains modest (electrification share increases by at most 2.1%) and varies substantially between the two neighboring regions. The asymmetric response across regions likely stems from differences in mobility patterns captured in the input data. When examining commuting-specific adoption versus fleet-wide electrification trends, a mechanism emerges: expanded destination-region capacity enables commuters to charge at their workplace or destination, thereby releasing origin-region infrastructure to serve local, non-commuting vehicle use. The elevated utilization rates observed for public charging infrastructure support this reallocation dynamic. However, fully characterizing these cross-regional dynamics and explaining the observed asymmetries would require addressing several methodological and data limitations. First, resolving the inconsistent spillover patterns demands methodological extensions in two directions: expanding the geographic scope to include many more regions would enable statistical analysis of spillover determinants, while isolating commuter flows from other trip purposes would clarify their specific contribution. The current framework optimizes system-wide infrastructure deployment and adoption trajectories subject to endogenous capacity decisions and fleet electrification rate constraints—assumptions that presume strong coordination between infrastructure availability and individual vehicle choices, potentially masking heterogeneous regional dynamics. Second, a critical data limitation concerns commuter trip volumes. The baseline draws on 2010 calibrations from the ETISplus project [152] and extrapolates all trip categories proportionally to GDP growth, as temporal trends distinguishing commuting from other purposes remain unavailable. Spillover magnitudes are highly sensitive to these assumed commuter flows. Additionally, regional trip distance distributions lack validation from Austrian-specific mobility surveys, introducing further uncertainty. Despite these limitations, the findings provide a complementary perspective to previous studies that report significant spillover effects in the early adoption phase based on empirical evidence [23], [24]. The purely techno-economic mechanisms captured by the model likely account for only a small share of the spillover effects observed empirically, where social and behavioral factors are expected to dominate.

5.1.3. Research Question 3

RQ3: *To what extent do geographically varying charging prices impact the cost-optimal spatial reallocation of long-haul battery-electric truck charging demand along transport corridors across multiple electricity market zones?*

Geographically varying electricity prices affect the distribution of charging demand at multiple levels. For the Scandinavian-Mediterranean corridor, the following is observed. First, energy system scenarios with different relative electricity price rankings across corridor countries produce substantially different spatial charging patterns over time, with allocated charging loads varying by up to 135% between the most divergent scenarios. Second, comparing the cost-optimized allocation against a deterministic benchmark with uniform charging distribution reveals that 39% of total demand shifts location under cost optimization. Third, localized surcharges on charging costs effectively redirect demand to surrounding regions, demonstrating that even modest price differentials can reshape the spatial concentration of charging load across the corridor.

Two effects drive these spatial differences in charging demand distribution: the spatially and temporally varying cost advantage of BETs over conventional trucks, and the geographic variation in charging costs.

Comparison with Pagnier et al. [122] These cross-scenario differences are consistent with findings by Pagnier et al. [122], who draw electricity-zone-dependent prices from a Norwegian energy system model and quantify their impact on regional charging infrastructure investment. They observe that, across different electricity pricing levels, investments in charging infrastructure differ by up to a factor of three at a given location. In the present analysis, the maximum cross-scenario spread in charging demand at the NUTS-2 level in a selected region reaches a factor of seven in 2032, though this reduces substantially over the optimization horizon to approximately two.

A key methodological difference is that Pagnier et al. [122] iterate between a freight transport model and an energy system model, thereby capturing the feedback effect of regionally developing charging demand on electricity prices. The present analysis does not account for this feedback, which may lead to some overestimation of the magnitude of demand reallocation and local concentration. However, the total charging loads in the case study remain small relative to national electricity demand (less than 0.35% across all countries), suggesting that price feedback effects would be limited in practice. A further simplification in the analysis is the use of country-level electricity prices, which is reasonable given that most countries along the corridor correspond to a single electricity market zone. Exceptions are Norway and Italy, which comprise multiple price zones — a distinction the energy system model input does not capture. Incorporating more granular price zones, including locational signals from balancing markets, could refine the spatial allocation results and represents an avenue for future work.

Modal shift to rail When modal shift to rail is enabled, approximately 15.4% of total charging demand along the corridor is spatially redistributed across regions. This shift predominantly reduces charging demand in the corridor's peripheral regions — Scandinavia and Italy — where long-distance routes are most cost-effectively shifted to rail. As a result, charging demand concentrates in the centrally located countries, particularly Germany and Austria, while overall electrification levels remain comparable. On a per-highway-km basis, Austrian highways face the highest demand concentration, reflecting the country's short transit corridor and high freight volumes.

These findings have direct implications for infrastructure planning. Cost-optimal decarbonization of the corridor requires not only consistent expansion of rail freight capacity along its full length, but also accelerated deployment of charging infrastructure in Denmark, Austria, and Germany to accommodate the concentrated road freight electrification.

These results should be interpreted in light of implementation constraints. The optimization model includes constraints based on the assumption that the pace of vehicle electrification is independent of the pace of modal shift — effectively treating rail as an accelerator for road freight decarbonization. In practice, both transitions face capacity and regulatory constraints that may slow their parallel deployment. Consequently, while the spatial concentration of charging demand in central regions is a robust finding, the required pace of charging infrastructure expansion may be slower than indicated here.

5.1.4. Research Question 4

> **RQ4:** *What is the expected spatio-temporal distribution of charging demand from the commercial battery electric vehicle fleet in Austria in 2040?*

In 2040, the estimated annual charging demand of the commercial BEV fleet in Austria ranges from 7.4 TWh (*LOW* scenario, 30% HDT electrification) to 15.2 TWh (*HIGH* scenario, 100% HDT electrification), corresponding to 8–17% of the projected total electricity demand in Austria. Heavy-duty trucks account for the bulk of commercial charging demand. In the *HIGH* scenario, HDT charging accounts for 11.2 TWh of the total 15.2 TWh (74%). The decomposition by traffic type reveals that 74.3% of the HDT charging demand within Austria stems from transit and import/export traffic (Table 4.20). This underscores the importance of the traffic-flow-based methodology.

Regarding temporal distribution, the findings indicate substantial daytime charging activity across most Austrian regions, though peak demand predominantly occurs during nighttime hours, with lower intensity on weekends. Daytime charging demand may be overestimated, as the model assumes that highway charging infrastructure is the preferred option for heavy-duty trucks during mandatory rest periods, potentially introducing an upward bias in projected daytime loads at motorway service areas. Conversely, the model excludes operators without access to depot charging, whose vehicles would rely entirely on public infrastructure during daytime hours. These opposing effects may partially offset each other.

In the absence of empirical data on the spatio-temporal distribution of commercial charging demand across Austrian regions, the estimates are contextualized by comparing peak loads and charging demand against policy targets and scientific literature. Table 5.2 summarizes these comparisons, which are discussed in detail below.

Comparison of charging load estimations The estimated increase in electricity demand from the commercial fleet is close to estimates by Li et al. [123]. Li et al. [123] estimate that EV charging in China would increase national peak demand by 5.2–12.5% under unmanaged charging scenarios. While a direct comparison of demand between China and Austria is limited by the vastly different fleet sizes and electricity systems, the share of total electricity demand attributable to commercial vehicle charging is comparable. In the present analysis, the commercial fleet alone—excluding private passenger BEVs—accounts for 8–17% of total demand, highlighting the significance of commercial vehicle electrification for electricity system planning in a European transit country.

For heavy-duty vehicles specifically, the Transition Mobility 2040 scenario implies an energy demand of approximately 25,000 TJ ($\approx$6.9 TWh) for SNF in 2040 (Figure 16 in [173]). The HIGH scenario estimates 11.2 TWh for HDT charging — approximately 1.6 times higher. This difference can be attributed to several factors: the traffic-flow-based methodology captures transit and international freight, which constitutes 74.3% of HDT charging demand on Austrian highways; the Transition Mobility 2040 scenario assumes a modal shift in freight transport from 70% to 62% road share (by tonne-kilometres), with rail increasing from 27% to 34%; and it includes a 15% FCEV share for heavy-duty vehicles, whose energy demand would not appear as stationary charging.

Comparison with European truck charging infrastructure studies Shoman et al. [192] estimate daily charged energy for battery-electric long-haul trucks across European countries at 15% fleet electrification. For Austria, they report a daily charged energy of 3.5 GWh and identify 71 charging areas, with approximately 1,169 CCS and 240 MCS charging points required. Linearly scaling the *HIGH* scenario (40.9 GWh/d at 100% electrification) to 15% yields approximately 6.1 GWh/d — noting that linear scaling is an approximation, as charging patterns and infrastructure utilization may not scale proportionally with fleet size. This is higher than the estimate by Shoman et al. [192] by a factor of approximately 1.7. The difference is primarily attributable to the broader scope of the estimation: the methodology includes all HDT traffic types—inland, transit, and import/export—and encompasses the full range of heavy-duty vehicle weights, whereas Shoman et al. [192] focus specifically on long-haul truck operations. Additionally, differences in assumptions on specific energy consumption, average loads, and the allocation between en-route and depot charging contribute to the divergence.

Comparison with planned infrastructure in Austria The modeled peak power levels at highway service areas range between 7 and 29 MW for the *HIGH* scenario. As a reference, the Austrian highway infrastructure operator ASFINAG has communicated plans to provide 5 MW of grid connection capacity at each service area by 2035/2040 [178]. This planned capacity corresponds to a commercial fleet electrification rate of approximately 30% in the modeling framework. At higher electrification rates, the modeled peak power levels substantially exceed the planned grid connection capacities, particularly in the northeastern regions of Austria where transit traffic is concentrated. This indicates that grid reinforcement beyond current plans will be required to support full electrification of the commercial fleet.

The AFIR defines progressively increasing HDV recharging infrastructure targets along the TEN-T network [11]. By 2030, recharging pools must be deployed every 60 km on the TEN-T core network, each providing a minimum of 3,600 kW. For Austria, this translates to approximately 135 MW across the network (assuming roughly 38 pools over 2,258 km). The modeled peak power levels of 7–29 MW per service area substantially exceed these per-pool minimums, reflecting the high concentration of freight traffic on Austrian transit corridors. Even in aggregate, the AFIR minimum is an order of magnitude below the modeled total peak demand.

Table 5.2.: Quantitative comparison of commercial fleet charging demand and infrastructure estimates for Austria

Source	**Electrification**	**Metric**	**Key Assumptions & Notes**
Charging demand estimates			
This work (*LOW*)	30% HDT	7.4 TWh/a	All commercial fleet segments (HDT, LDT, bus); traffic-flow-based; Austria 2040
This work (*HIGH*)	100% HDT	15.2 TWh/a (40.9 GWh/d)	74% from HDT; 74.3% of HDT demand from transit + import/export traffic
This work (scaled)	15% HDT	∼6.1 GWh/d	Linear scaling of *HIGH* scenario for comparability with Shoman et al. [192]
Shoman et al. [192]	15% BET	3.5 GWh/d	Long-haul trucks only; 71 charging areas; 1,169 CCS + 240 MCS points
Transition Mobility 2040[a]	100%[b]	127 PJ/a (≈35 TWh/a)	Total transport energy; 85% BEV/ERS + 15% FCEV for SNF
Infrastructure capacity per service area			
This work (*HIGH*)	100% HDT	7–29 MW	Peak power at highway service areas; highest in NE Austria (transit corridors)
ASFINAG (planned)[c]	—	5 MW	Grid connection capacity per service area by 2035/2040
AFIR HDV (2030)[d]	—	3.6 MW/pool	Minimum per pool every 60 km on TEN-T core; ≈135 MW aggregate for Austria

[a] Angelini et al. [173]. Projects truck freight modal share declining from 70% to 62%, rail increasing from 27% to 34%.
[b] 100% zero-emission fleet, but only 85% require stationary charging (BEV/ERS); 15% are FCEV.
[c] [178]. Corresponds to ≈30% electrification in the modeling framework.
[d] [11]. Minimum connectivity requirement; 2025 interim target: 1,400 kW/pool along 15% of TEN-T.

5.2. Synthesis of Results

The four contributions address different vehicle segments, geographic scopes, and planning dimensions. Yet they share a common analytical perspective: charging infrastructure planning at scales that extend beyond individual cities or regions — spanning national highway networks, international transport corridors, and cross-regional adoption dynamics. This section draws out three cross-cutting insights that emerge from considering the contributions jointly rather than in isolation.

Driving range and spatial density As framed by the review paper by Metais et al. [96], the overcoming of *range anxiety* is a primary design goal for most charging infrastructure capacity planning algorithms. Throughout the proposed model, the range of battery-electric vehicles is a core parameter determining the maximum distance between charging opportunities. The objective in most graph-based optimization algorithms applied for planning charging infrastructure at large scale focuses on establishing a minimum required set of charging sites (as in Contribution 1), while range dictates the maximum distance between them. Although battery capacities have grown to cover the vast majority of daily travel needs — largely rendering this anxiety no longer technically justified — and substantial charging infrastructure is deployed across European countries, the modeling tools applied in large-scale analyses remain largely unchanged, for both passenger and commercial vehicle fleets. In Contribution 1, range has no significant influence on siting decisions. In the regional consideration of both passenger and commercial BEVs addressed in Contributions 2 and 4, the impact of range on the relative siting of charging stations is inherently negligible by design. Results of Contribution 3 demonstrate that the expected range of long-haul trucks allows for substantial flexibility in the allocation of charging demand, suggesting that local electricity prices are more important determinants for infrastructure planning than range. Contribution 2 analytically demonstrates the benefit of increased spatial density of charging sites.

With charging infrastructure for passenger cars along European highway networks now largely established, the central problem in geographic charging capacity allocation is no longer the siting of initial charging sites, but rather the densification of the charger network and the expansion of on-site capacity, accounting for the spatial configuration of chargers. For trucks, while more foundational infrastructure efforts are still required, future tools should increasingly target the densification of charging sites, expansion of on-site capacity, and integration of local pricing signals.

Spatial flexibility Spatial flexibility in the allocation of charging demand is a key feature across the methodologies of all four contributions. Contribution 4 demonstrates that substantial charging loads arise from interregional transport flows, underscoring the cross-border dimension of spatial demand allocation. Contribution 3 quantifies significant spatial flexibility in charging demand allocation along transport corridors. Contribution 1, by contrast, finds that the siting of charging stations is not significantly affected by extending the driving range of vehicles — that is, by increasing the spatial extent of this flexibility. While the study area in Contribution 1 does not extend across borders, nor does it draw on origin-destination flows rather than traffic-count-derived demand, the findings of Contribution 3 — albeit for a different vehicle segment and geographic scope — point in the same direction, suggesting that the limited impact of range on spatial allocation may generalize beyond the specific setting of Contribution 1. Contribution 2 shows that spatial flexibility does not significantly affect BEV adoption from a techno-economic perspective, but reveals flexibility in the utilization of different charging infrastructure types within a region, depending on the spatial density of the network.

Taken together, these findings challenge the prevailing exogenous treatment of charging demand in large-scale analyses, which neglects flexibility in vehicle operation, the relative configuration of infrastructure, and the favorable spatial reallocation of demand that this flexibility enables. While endogenous modeling of charging demand and vehicle operation flexibility is explored at smaller scopes in the literature, i.e., for smaller geographic regions and for individual routes, its application to large-scale contexts remains limited and warrants further development.

Value of time in consumer segmentation In the commercial freight sector, the monetization of time is relatively straightforward, expressed directly through labor costs, as in Contribution 3, and delivery schedules. This is reflected in the vehicle-type segmentation applied in the charging load estimation of Contribution 4, where demand is characterized by energy consumption, installed battery capacity, and mobility patterns. For passenger vehicles, however, the trade-off between time and money is individual and closely tied to income level. The results of Contribution 2 illustrate this concretely: income-specific behavioral differences produce divergent preferences in charging infrastructure utilization, suggesting that a homogeneous representation of passenger BEV users is an oversimplification with practical planning consequences. Contribution 1 further demonstrates the high sensitivity of highway charging infrastructure efficiency to installed charging power levels.

Incorporating income-class-dependent value of time into charging infrastructure planning could therefore translate into two distinct implications for highway network operators. First, higher-income drivers may systematically prefer faster, higher-cost charging, shifting the cost-optimal configuration of charging power levels at highway sites. Second, lower-income drivers may prefer cheaper off-highway charging alternatives, which could reduce utilization of highway charging infrastructure. Neither behavioral pattern is captured by current aggregate, homogeneous modeling approaches.

6. Conclusion

The findings of the four contributions presented in the previous chapters address the three interconnected dimensions of charging infrastructure planning outlined in the Introduction (Section 1), with implications for transport policy makers, transport and charging infrastructure planners, as well as, electricity grid and system operators:

(i) **The identification of locations and required capacities for charging infrastructure:** The optimization of charging infrastructure siting should account endogenously for the interactions between capacity sizing, station utilization, and vehicle operation. Doing so shifts the basis for infrastructure siting from capital expenditures alone toward a fuller consideration of operational expenditures. Current large-scale methodologies predominantly optimize for coverage: sites are selected to serve maximum demand, with utilization implicitly assumed rather than explicitly modeled. While smaller-scope studies of individual vehicle movements do capture these operational dynamics, they are rarely incorporated at the geographic scales relevant for international corridor planning.

(ii) **The influence of spatial configuration on charging demand and driver behavior**: Spatially dense charging infrastructure supports BEV adoption across all income groups, making geographic reach the decisive factor for large-scale uptake. This finding implies that expansion targets framed solely in terms of charging points per BEV or installed capacity are insufficient — such metrics do not capture whether infrastructure is actually accessible to drivers where and when they need it. Instead, definitions should be grounded in maximum spacing requirements within the road transport network. The AFIR already reflects this logic for highway charging through distance-based deployment mandates, but country-level definitions remain count-based — aligning these with a spacing-based approach would better support equitable and effective fleet electrification.

(iii) **The integration of charging operations with the electricity system**: Charging infrastructure sits at the intersection of electrified vehicle fleets and the electricity grid, making it a natural lever for demand-side flexibility. The flexibility perspective should extend beyond timing-based demand-response to encompass the spatial dimension of charging demand allocation. Grid operators and regulators should leverage this potential, and the findings of this work demonstrate that it is substantial along international corridors.

(iv) Overall, cost-optimal BEV adoption for long-haul and international transport requires coordinated action from transport policy makers, infrastructure planners, and electricity grid operators beyond the national level. International transport flows along the TEN-T corridors encompass substantial freight and passenger movements, but the resulting charging demand does not follow transport activity directly — it is shaped by the relative configuration of charging infrastructure and costs across national boundaries.

7. Future Work

Future work should address the limitations identified in Section 5. Beyond these, the spatial allocation of charging infrastructure for mobile consumers — the central problem of this work — opens several directions for further research that extend its scope. In the following, four concrete research directions are discussed that address broader questions at the intersection of transport electrification and energy system planning.

Logistical flexibility In the aggregated representation of vehicle movements, individual decision-making is strongly simplified and system-optimal charging behavior is assumed. However, such behavior can only be initiated through significant pricing signals or incentives. For example, cost-optimal charging allocation along a TEN-T corridor only emerges when charging costs differ substantially across locations. Moreover, reaching the system optimum requires flexibility in mobility and logistics patterns, which in practice are constrained by commercial and operational processes. Further research is needed to understand what incentives are required to drive cost-optimal charging behavior, particularly in the context of rigid logistics schedules and mobility patterns, and to what extent individual behavioral responses translate to the aggregated representation adopted in this work.

Charging infrastructure as a lever for second-hand vehicle markets This work has made efforts to extend the role of charging infrastructure beyond covering existing demand, towards serving as a lever for the adoption of battery-electric vehicles. A particularly relevant extension concerns the second-hand vehicle market. In Europe, there are substantial cross-border flows of used vehicles between certain countries. In this context, the question arises what role the deployment of charging infrastructure in these destination countries plays in enabling the adoption of second-hand battery-electric vehicles — by ensuring availability and contributing to reasonable total costs of ownership. This is especially relevant for trucks, where second-hand markets are a key pathway for fleet renewal. Further research should examine this relationship under consideration of challenges associated with battery degradation and state of health, which directly affect the economic viability of used battery-electric vehicles.

Cross-modal perspective While this work focuses on the electrification of road transport, electrification is increasingly relevant across other transport modes, including rail, shipping, and aviation. Together with other renewable fuels, these modes are becoming increasingly intertwined through sector coupling. This calls for a cross-modal perspective on charging and fueling infrastructure planning that identifies synergies in both investment and operation across modes. A key aspect is the spatial allocation of infrastructure at intermodal nodes — such as ports, airports, and rail terminals — where energy demand from multiple transport modes converges. Analyzing the transformation of such nodes into *energy hubs*, with consideration of their geographic allocation and connected network infrastructure, offers potential for cost-optimal solutions in both multi-modal transport and energy supply. This is further motivated by policy efforts to shift freight transport towards rail and inland waterways, which increase the relevance of intermodal transfer points as shared infrastructure investment sites.

Along with the research directions outlined above, there are several *modeling challenges* to address and overcome for the analytical tools. The co-optimization framework for charging infrastructure allocation, as extended in scope towards broader system interactions, requires methodological improvements in several areas:

- **Sub-annual temporal resolution in demand data:** Origin-destination data is predominantly available for aggregated annual flows, and both up-to-date data and forward-looking projections are scarce. Sub-annual fluctuations in transport demand — such as seasonal variations in freight and passenger volumes — are typically neglected, yet become increasingly important when considering operational interactions with the electricity grid.

- **Uncertainty in origin-destination flow projections:** Current approaches place high trust in origin-destination data, with little consideration for the inherent uncertainty in these projections. However, future transport flows are shaped by spatial planning, infrastructure development, socio-economic dynamics, and labor market evolution — involving multiple layers of decision-making that are difficult to predict. Rather than extending the modeling scope towards macroeconomic representations, which would result in prohibitively large systems, a more tractable approach is to incorporate demand uncertainty directly into the optimization, enabling more robust infrastructure investment decisions.

- **Spatio-temporal complexity in optimization:** Jointly resolving spatial and temporal dimensions introduces substantial computational challenges, well known from energy system modeling. These are further amplified by vehicle mobility, where the discretization of time steps must account for moving consumers whose behaviors may interact, for example through queuing at charging sites. While such dynamics are typically addressed through agent-based approaches, their integration into system-optimization frameworks requires careful identification of which spatio-temporal elements are essential to represent at the aggregated level and which can be simplified. Two complementary strategies emerge: adopting aggregation techniques from other optimization domains, and identifying the minimal set of dynamic representations required — for instance, to capture intra-day demand response of long-haul trucks to volatile charging prices.

Appendix A.

RQ2

A.0.1. all the relevant snestivity analysis and data methodology stuff

think hard about including additionally the break down on objective function values + their distribution and differences among the three main densification scenarios maybe i also need to show the development of the fast charging **time** with the years to justify the sudden development

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